THIRSTY'S CODEX

STANDARDS EDITION

VOLUME IV / 04 OF 20

ELECTRICAL ENGINEERING THROUGH USER VALIDATION

Lifecycle Sections 061-106

NO PROOF, NO CLAIM

CONTINUOUS MASTER FOLIO / SOURCE-PRESERVING EDITION

Edition Statement

THIRSTY'S CODEX / STANDARDS EDITION / SOURCE-PRESERVING SET

This volume is one part of a continuous 20-volume edition. Canonical lifecycle section numbers remain stable across the set; no volume restarts or renumbers them.

The visual treatment has been rebuilt as a unified standards library: coordinated covers, volume architecture, source dividers, running headers, continuous master folios, local page numbers, and source-identity marks.

The source corpus remains authoritative over the presentation layer. Status terms are preserved exactly. CONTENT-COMPLETE does not silently become source-reconciled, hostile-reviewed, owner-accepted, ratified, or published.

Every registered source witness is identified by path, byte count, and SHA-256 digest. The complete source archive is preserved as an embedded attachment in Volume XX, while the two supplied foundational PDFs are preserved as attachments in Volume I.

CONTINUITY RULE

Read the master folio and canonical section sequence as one uninterrupted work. Each succeeding volume begins where the preceding volume ends.

Standards Edition Set Map

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Doctrine, prompts, front matter, shared governance, roles, and output contracts

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Research, Strategy, and Risk
Lifecycle Sections 001-018

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Requirements through Hardware Architecture
Lifecycle Sections 019-060

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Reference Registers and Working Documentation
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VOLUME XIV
Standards Source Library
Preserved standards, command editions, blueprints, and source artifacts

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Canonical Authority and Assurance
Authority, constitutions, policies, schemas, provenance, evaluation, and verifica...

VOLUME XVI
Applied Corpus and V3Q
Agent-output corpus, V3Q implementation, fixtures, tests, and governed examples

VOLUME XVII
Training Views and Indexes
Continuous-pretraining, evaluation, retrieval, and training views with canonical ...

VOLUME XVIII
Agents and Owner Command Standards
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061

SECTION 061 - ELECTRICAL ENGINEERING

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8,512 BYTES / PRESERVED SOURCE IDENTITY

SECTION 061 — Electrical Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Define the governed engineering of electrical power, distribution, conversion, protection, grounding, wiring, energy storage, electromagnetic behavior, and electrical safety for a bounded system and lifecycle. This section addresses physical electrical behavior and its evidence without treating a schematic, simulation, component rating, or certification claim as proof of safe installed operation.

REQUIRED OUTCOME

An approved electrical design with explicit loads, sources, protection, isolation, grounding, thermal and environmental limits, interfaces, safety controls, maintainability, test methods, operating modes, and lifecycle responsibilities.

WHY THIS MATTERS

Electrical faults can cause fire, shock, loss of control, equipment damage, unsafe energy release, data loss, and service interruption. Correct voltage or nominal load is not sufficient evidence of safe behavior during transients, faults, maintenance, environmental change, or degraded operation.

APPLICABILITY

Applies to generation, distribution, conversion, wiring, batteries, motors, power supplies, grounding, protection, electrical interfaces, facilities, vehicles, devices, and systems where electrical energy affects function, safety, reliability, or compliance.

NON-APPLICABILITY CONDITIONS

May be recorded as non-applicable when the bounded change has no electrical source, load, interface, energy storage, protection, grounding, electromagnetic, environmental, safety, or lifecycle effect. The decision must name the baseline, reviewer, evidence date, assumptions, and reopening trigger.

PREREQUISITES AND DEPENDENCIES

Requirements, operational modes, load profiles, environments, hazards, applicable jurisdictional rules, component data, schematics, protection philosophy, installation constraints, maintenance needs, and qualified human review. Sections 019-060 provide requirements, system, architecture, safety, reliability, resilience, and hardware context.

REQUIRED INPUTS

Authorized objective and boundary; source and load data; nominal and worst-case conditions; fault assumptions; insulation and clearance requirements; grounding and bonding; protection coordination; thermal, mechanical, EMC, accessibility, service, disposal, and evidence constraints.

HUMAN RESPONSIBILITIES AND AUTHORITY

The accountable human engineer and owner approve safety assumptions, hazard acceptance, design basis, component substitutions, protection settings, energization, test limits, field changes, and final acceptance. A license, model, or vendor recommendation does not transfer authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may calculate bounded scenarios, check rules, identify omissions, compare designs, draft schematics or test cases, and summarize evidence. It must preserve units, assumptions, source versions, uncertainty, and calculation provenance; it must not approve energization, invent ratings, accept hazards, or conceal unresolved field conditions.

APPLICABLE EXISTING AGENT ROLES

Architect, Researcher, Analyst, Critic, Security Auditor, Tester, Verifier, Implementer, Refactorer, Planner, Coordinator, Observer, Documentation Writer, Governance Reviewer, Red Team, Optimizer, and Memory Curator, selected according to consequence and existing role contracts.

REQUIRED ACTIVITIES

1. Confirm boundary, load, source, operating modes, authority, jurisdiction, and lifecycle.
2. Establish design basis, single-line and wiring views, grounding, bonding, protection, isolation, and energy-release assumptions.
3. Calculate steady-state, transient, short-circuit, fault-current, coordination, voltage-drop, thermal, derating, and energy-storage cases.
4. Allocate electrical interfaces and protections to equipment, controls, people, facilities, and maintenance procedures.
5. Assess shock, arc, fire, overcurrent, overvoltage, backfeed, battery, EMC, lightning, environmental, and human-factors hazards.
6. Define inspection, test, commissioning, maintenance, lockout, emergency, degraded, repair, replacement, and retirement activities.
7. Review with qualified human specialists; baseline decisions, deviations, evidence, and open gaps.

HUMAN-AI INTERACTION REQUIREMENTS

AI-generated calculations, diagrams, component choices, and recommendations remain proposals. The cockpit must show the exact model, units, assumptions, source data, limitations, reviewer, and approval state. AI may not direct a person to energize or bypass protection. Human stop, isolation, inspection, and emergency controls remain available during model or network failure.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Represent sources, loads, conductors, protective devices, ground paths, energy stores, ratings, clearances, interfaces, hazards, controls, tests, and lifecycle states as versioned records. Preserve separation between design, installed configuration, energized state, measured result, and certification. Changes to loads, components, firmware, enclosure, environment, or protection require impact analysis.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show design baseline, energized state, source and load status, protection state, alarms, isolation authority, test evidence, hazards, deviations, maintenance, and recovery actions. Permit authorized stop, isolate, quarantine, inspect, restore, substitute, and retire actions, with identity and evidence for each consequential action.

REQUIRED OUTPUTS AND ARTIFACTS

Design basis; load and fault studies; schematics and wiring schedules; grounding and protection plan; hazard analysis; component and derating register; interface contracts; installation and commissioning plan; test records; maintenance and emergency procedures; deviation, decision, evidence, and handoff registers.

EVIDENCE REQUIREMENTS

Evidence identifies exact design, revision, source data, units, environment, assumptions, instrument calibration, installation state, test method, result, reviewer, and limitation. Simulation, nameplate rating, supplier statement, visual inspection, and field measurement remain distinct evidence types.

VERIFICATION AND ACCEPTANCE TESTS

Verify calculations, polarity, continuity, insulation, grounding, protection operation, isolation, test limits, thermal limits, EMC, battery behavior, emergency stop, maintenance access, and recovery under nominal, boundary, degraded, and fault conditions. Independently reproduce material safety claims.

FAILURE STATES AND ANTI-PATTERNS

Unsafe energization; undocumented field modification; missing grounding; uncoordinated protection; ignored inrush or backfeed; overstated component rating; hidden stored energy; uncalibrated measurement; bypassed interlock; and treating a passing simulation as installed safety.

RECOVERY AND RECONSIDERATION

Stop or isolate the affected energy path, preserve evidence, identify the hazard and authority, and use approved de-energized inspection, containment, repair, substitution, restoration, and re-test. Reconsider after changed loads, environment, components, protection, incident, or evidence invalidate the baseline.

COMPLETION CRITERIA

Scope, design basis, hazards, protections, interfaces, operating modes, maintenance, tests, evidence, owner, and unresolved gaps are explicit. No readiness or safety claim exceeds its verified environment and installation state.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve revision, installed configuration, settings, measurements, deviations, incidents, maintenance, decisions, and next safe action. Hand off to Electronics, Hardware, Embedded, Safety, Reliability, Resilience, Testing, Operations, Maintenance, Compliance, and Retirement sections.

RELATED CODEX SECTIONS

019 Requirements Engineering; 020 Systems Engineering; 045 Security Architecture; 050 Safety Architecture; 051 Reliability Architecture; 052 Resilience Architecture; 060 Hardware Architecture; 062 Electronics Engineering; 064 Mechanical Engineering; 232 Hardware Testing; 243 Reliability Testing; 252 Safety Testing; 351 Manufacturing; 399 Hardware Disposal.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance files, Sections 019-060, Owner Cockpit Requirements, Status and Evidence Lexicon, and the integration roadmap. Jurisdiction-specific electrical codes, licensing, utility rules, product standards, site conditions, qualified-person requirements, and owner acceptance remain unresolved until supplied and verified.

062

SECTION 062 - ELECTRONICS ENGINEERING

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7,249 BYTES / PRESERVED SOURCE IDENTITY

SECTION 062 — Electronics Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Define the governed design of electronic circuits and assemblies that sense, condition, convert, communicate, compute, control, or protect signals and energy within a bounded system. Preserve analog, digital, mixed-signal, power, timing, EMC, thermal, safety, manufacturing, service, and lifecycle concerns.

REQUIRED OUTCOME

A traceable electronic design with schematics, constraints, interfaces, component identity, tolerances, power integrity, signal integrity, EMC behavior, thermal limits, test access, manufacturing controls, failure handling, and evidence for the declared environment.

WHY THIS MATTERS

Small component, timing, noise, thermal, or layout errors can produce intermittent faults, unsafe outputs, data corruption, field failures, and impossible diagnosis. Simulation and nominal bench success do not establish production robustness.

APPLICABILITY

Applies to analog, digital, mixed-signal, power, sensing, actuation, communications, control, instrumentation, and electronic assemblies, including prototypes and production transitions.

NON-APPLICABILITY CONDITIONS

Non-applicability requires no material circuit, signal, power, timing, EMC, component, assembly, firmware interface, test, manufacturing, or lifecycle effect, with baseline, evidence, authority, and reopening trigger recorded.

PREREQUISITES AND DEPENDENCIES

Functional requirements, electrical environment, schematics, timing and signal budgets, component data, tolerances, power and thermal limits, EMC targets, safety constraints, manufacturing process, firmware and mechanical interfaces, and service requirements. Sections 019-061 provide upstream context.

REQUIRED INPUTS

Circuit boundary, net intent, part identity, approved alternatives, power domains, clocks, signal levels, tolerances, derating, environmental profile, EMC limits, safety controls, assembly process, test access, calibration, service, and evidence constraints.

HUMAN RESPONSIBILITIES AND AUTHORITY

The human design authority approves critical parts, substitutions, tolerances, safety and EMC margins, prototype-to-production transition, deviations, destructive tests, and acceptance. Human experts decide whether evidence represents the intended field population.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze schematics, datasheets, simulations, layouts, bills of material, and test results; draft checks and alternatives; and identify missing margins. It must not invent pinouts, assume equivalent parts, approve a design, hide uncertainty, or authorize field use.

APPLICABLE EXISTING AGENT ROLES

Architect, Researcher, Analyst, Critic, Implementer, Tester, Verifier, Security Auditor, Red Team, Optimizer, Planner, Coordinator, Observer, Documentation Writer, Governance Reviewer, and Memory Curator under the existing role contracts.

REQUIRED ACTIVITIES

1. Establish circuit boundary, functions, modes, interfaces, environment, criticality, and lifecycle.
2. Capture schematics, net intent, component identity, tolerances, derating, power, timing, signal, thermal, and EMC budgets.
3. Analyze operating, transient, fault, startup, shutdown, calibration, sensor, actuator, communication, and recovery behavior.
4. Review layout, grounding, return paths, shielding, isolation, test points, manufacturability, serviceability, and obsolete or counterfeit parts.
5. Define prototype, qualification, production, incoming, in-process, end-of-line, field, and regression tests.
6. Baseline evidence, deviations, substitutions, hazards, open gaps, and approval state.

HUMAN-AI INTERACTION REQUIREMENTS

AI outputs remain proposed and retain source part numbers, revisions, units, simulation configuration, data, uncertainty, and reviewer. Human approval is required for safety, security, EMC, production, substitution, and release decisions. Stop and diagnostic paths must not depend on AI availability.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Model circuits, components, nets, signals, power domains, clocks, interfaces, constraints, tolerances, layouts, assembly versions, test points, configuration, firmware boundaries, and lifecycle state as controlled artifacts. Separate simulation, prototype, production, and field evidence.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show assembly revision, powered state, rails, signals, temperatures, faults, test status, component substitutions, calibration, evidence freshness, and authorized isolate or repair actions. Preserve human inspection, safe shutdown, and non-AI diagnosis.

REQUIRED OUTPUTS AND ARTIFACTS

Schematics; net and interface definitions; bill of materials; approved parts and substitutions; budgets; simulations; layout and EMC review; hazard analysis; assembly drawings; manufacturing and test plans; qualification results; service and lifecycle records.

EVIDENCE REQUIREMENTS

Evidence binds part and assembly revision, instruments, fixtures, calibration, firmware, environment, stimulus, limits, result, operator, and limitations. Supplier data and simulation are not production evidence without correlation.

VERIFICATION AND ACCEPTANCE TESTS

Verify connectivity, polarity, timing, power-up, thermal, signal integrity, noise, EMC, tolerance, fault insertion, protection, calibration, production variation, aging, vibration interface, and recovery behavior. Reproduce failures and confirm corrective evidence.

FAILURE STATES AND ANTI-PATTERNS

Uncontrolled substitutions; stale datasheets; floating inputs; missing return paths; ignored tolerances; layout afterthought; untested startup or brownout; hidden counterfeit risk; inadequate test access; and treating nominal simulation as field reliability.

RECOVERY AND RECONSIDERATION

Quarantine affected assemblies or functions, preserve configuration and test evidence, identify safe operation, and use approved repair, substitution, rework, rollback, recalibration, or redesign. Reopen after component, layout, firmware, environment, manufacturing, or failure changes.

COMPLETION CRITERIA

The electronic baseline, constraints, interfaces, parts, margins, tests, evidence, service, production, and unresolved gaps are explicit for the declared environment; no production or safety claim exceeds evidence.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve design and assembly revisions, BOM, substitutions, test fixtures, calibration, failures, decisions, and next action. Hand off to Electrical, Semiconductor, PCB, Firmware, Embedded Software, Hardware Testing, Manufacturing, Security, Reliability, and Retirement sections.

RELATED CODEX SECTIONS

061 Electrical Engineering; 063 Computer Engineering; 066 Semiconductor Engineering; 067 PCB Design; 068 Firmware Engineering; 069 Embedded Software Engineering; 232 Hardware Testing; 246 Security Testing; 252 Safety Testing; 351 Manufacturing; 354 Hardware Certification.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance files, Sections 019-060, and the integration roadmap. Part availability, EMC jurisdiction, certification, manufacturing capability, environmental profile, counterfeit controls, and owner acceptance remain unresolved until supplied and verified.

063

SECTION 063 - COMPUTER ENGINEERING

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7,245 BYTES / PRESERVED SOURCE IDENTITY

SECTION 063 — Computer Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Define the governed co-design of computing hardware, firmware boundaries, memory, buses, timing, processing, storage, I/O, accelerators, and software-visible behavior for a bounded system. Address hardware-software contracts, performance, safety, security, reliability, manufacturability, service, and lifecycle.

REQUIRED OUTCOME

A traceable computer design with allocated functions, hardware-software interfaces, resource budgets, timing, boot and update behavior, identity, protection, diagnostics, testability, manufacturing, recovery, and evidence for the declared workloads and environment.

WHY THIS MATTERS

Computing failures often arise at boundaries: reset, boot, memory, timing, buses, power, concurrency, updates, drivers, and thermal limits. A benchmark or successful boot does not prove safe behavior under faults, aging, malicious input, or resource exhaustion.

APPLICABILITY

Applies to processors, boards, memory, storage, buses, accelerators, controllers, computers, embedded compute, firmware interfaces, and hardware-software co-design.

NON-APPLICABILITY CONDITIONS

Non-applicability requires no material compute, memory, storage, timing, boot, bus, I/O, hardware-software contract, security, update, test, or lifecycle effect, with a recorded baseline and reopening trigger.

PREREQUISITES AND DEPENDENCIES

Functional allocation, workload and timing, hardware and firmware constraints, memory and storage maps, bus and I/O contracts, power and thermal envelope, boot, update, identity, security, safety, diagnostics, manufacturing, service, and retirement requirements. Sections 019-062 provide upstream context.

REQUIRED INPUTS

System boundary, workloads, resource budgets, memory map, storage model, bus and I/O contracts, reset and boot conditions, update authority, trust boundaries, environmental limits, diagnostics, manufacturing, service, and evidence constraints.

HUMAN RESPONSIBILITIES AND AUTHORITY

Human engineering authority approves architecture, criticality, resource margins, trusted boot or equivalent controls, update and recovery behavior, field changes, risk, and acceptance. Human operators retain authority to stop, isolate, recover, and retire equipment.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze architecture, resource traces, interface contracts, fault logs, and tests; propose allocations and test cases; and detect inconsistencies. It must not invent hardware behavior, approve trust claims, change boot or update controls, or treat benchmark performance as readiness.

APPLICABLE EXISTING AGENT ROLES

Architect, Analyst, Researcher, Implementer, Refactorer, Tester, Verifier, Security Auditor, Red Team, Critic, Planner, Coordinator, Observer, Governance Reviewer, Documentation Writer, Optimizer, and Memory Curator under the existing contracts.

REQUIRED ACTIVITIES

1. Establish system boundary, workloads, modes, criticality, users, operators, and lifecycle.
2. Allocate functions across hardware, firmware, operating environment, software, people, and external services.
3. Define memory, storage, bus, I/O, timing, interrupt, reset, boot, update, identity, and resource contracts.
4. Analyze performance, capacity, concurrency, fault containment, thermal, power, security, safety, service, and manufacturing conditions.
5. Define diagnostics, test access, logging, recovery, rollback, field update, replacement, and retirement.
6. Validate hardware-software integration with qualified specialists and baseline deviations and evidence.

HUMAN-AI INTERACTION REQUIREMENTS

AI recommendations must expose traces, workloads, configuration, assumptions, uncertainty, and alternatives. Human review is required for allocation, boot trust, privileged update, safety, security, resource margins, and release. Essential stop, recovery, and diagnosis must function without AI.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Represent compute elements, resource maps, buses, memories, boot states, update states, identities, trust boundaries, hardware-software contracts, configurations, evidence, and lifecycle states as versioned artifacts. Separate design, manufactured unit, deployed configuration, runtime state, and test evidence.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show unit identity, hardware and firmware revisions, boot and update state, resource use, thermal and power condition, faults, trust status, evidence, and authorized isolation or recovery actions. Expose rollback, safe mode, replacement, and manual diagnosis.

REQUIRED OUTPUTS AND ARTIFACTS

Computer architecture; allocation matrix; memory and bus maps; hardware-software interface contracts; timing and resource budgets; threat and hazard analysis; boot and update design; diagnostics; manufacturing and service plans; integration and qualification evidence.

EVIDENCE REQUIREMENTS

Evidence identifies unit, part and firmware revisions, workload, configuration, environment, instruments, stimulus, timing, result, operator, and limitations. Benchmarks, emulation, simulation, and field observations remain distinct.

VERIFICATION AND ACCEPTANCE TESTS

Verify boot, reset, memory protection, storage integrity, bus behavior, timing, resource exhaustion, concurrency, update interruption, rollback, fault containment, thermal and power limits, security controls, diagnostics, and recovery.

FAILURE STATES AND ANTI-PATTERNS

Hidden hardware-software assumptions; unbounded resource use; insecure boot or update; untested reset; shared fault domains; stale firmware; opaque diagnostics; benchmark-only claims; and recovery that requires the failed control path.

RECOVERY AND RECONSIDERATION

Quarantine the unit or function, preserve image, configuration, logs, and evidence, identify safe mode, and use approved rollback, replacement, reflash, restore, or manual recovery. Reopen after hardware, firmware, workload, security, supplier, or environment change.

COMPLETION CRITERIA

The co-design boundary, allocations, contracts, resources, trust, update, diagnostics, tests, recovery, evidence, and unresolved gaps are explicit; status is limited to the tested unit and environment.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve unit identity, hardware and firmware revisions, maps, contracts, traces, faults, updates, decisions, evidence, and next action. Hand off to Electronics, Computer Science, Firmware, Embedded Software, Operating Systems, Security, Testing, Operations, and Retirement.

RELATED CODEX SECTIONS

020 Systems Engineering; 033 Software Architecture; 035 Infrastructure Architecture; 045 Security Architecture; 051 Reliability Architecture; 052 Resilience Architecture; 060 Hardware Architecture; 062 Electronics Engineering; 068 Firmware Engineering; 069 Embedded Software Engineering; 113 Operating Systems Engineering; 232 Hardware Testing; 264 Build Engineering; 379 Firmware Updates.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance files, Sections 019-060, and the integration roadmap. Processor, platform, certification, supply-chain, threat, workload, field-service, and owner-acceptance assumptions remain unresolved until supplied and verified.

064

SECTION 064 - MECHANICAL ENGINEERING

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7,570 BYTES / PRESERVED SOURCE IDENTITY

SECTION 064 — Mechanical Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Define the governed engineering of structures, mechanisms, materials, motion, loads, interfaces, enclosures, thermal paths, manufacturability, serviceability, and physical safety for a bounded system across its lifecycle.

REQUIRED OUTCOME

A traceable mechanical design with loads, materials, tolerances, joints, motion limits, interfaces, environmental ratings, human access, hazard controls, manufacturing, inspection, maintenance, recovery, and evidence for the declared use conditions.

WHY THIS MATTERS

Mechanical defects can cause collapse, pinch, impact, leakage, overheating, loss of alignment, unsafe motion, or inaccessible maintenance. A CAD model or static calculation does not prove behavior under fatigue, abuse, tolerance, environment, assembly variation, or field repair.

APPLICABILITY

Applies to structures, mechanisms, enclosures, mounts, rotating or moving parts, thermal and fluid interfaces, materials, packaging, tooling, facilities, devices, vehicles, and physical service environments.

NON-APPLICABILITY CONDITIONS

Non-applicability requires no material load, motion, material, enclosure, thermal, fluid, vibration, human-access, manufacturing, service, safety, or lifecycle effect, with baseline, authority, evidence, and reopening trigger recorded.

PREREQUISITES AND DEPENDENCIES

Functional and load requirements; operating modes; materials; geometry; tolerances; environments; interfaces; human factors; hazards; thermal, vibration, corrosion, fluid, acoustic, manufacturing, assembly, maintenance, accessibility, transport, and disposal constraints. Sections 019-063 provide upstream context.

REQUIRED INPUTS

Boundary, loads, motion, materials, geometry, interfaces, tolerances, environmental profile, human access, hazards, thermal and fluid conditions, manufacturing process, inspection, transport, maintenance, recovery, and disposal constraints.

HUMAN RESPONSIBILITIES AND AUTHORITY

The human mechanical authority approves design basis, materials, safety factors, critical tolerances, prototypes, substitutions, field modifications, hazard acceptance, manufacturing release, and final acceptance. Human operators decide safe use, isolation, repair, and retirement.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze CAD metadata, load cases, tolerances, simulations, inspection records, and alternatives; draft calculations and tests; and identify inconsistencies. It must preserve units and assumptions and must not certify structural safety, invent material properties, or authorize operation.

APPLICABLE EXISTING AGENT ROLES

Architect, Researcher, Analyst, Critic, Implementer, Refactorer, Tester, Verifier, Security Auditor, Red Team, Planner, Coordinator, Observer, Governance Reviewer, Documentation Writer, Optimizer, User Experience Reviewer, and Memory Curator under existing role contracts.

REQUIRED ACTIVITIES

1. Establish boundary, loads, motions, environments, users, operators, hazards, production, service, and lifecycle.
2. Define geometry, materials, interfaces, joints, tolerances, fits, fasteners, seals, thermal paths, and motion limits.
3. Analyze static, dynamic, fatigue, fracture, buckling, vibration, shock, wear, corrosion, thermal, fluid, acoustic, and abuse cases as applicable.
4. Allocate physical interfaces and controls to electrical, electronic, software, human, facility, and maintenance boundaries.
5. Review manufacturability, assembly, inspection, transport, accessibility, maintenance, replacement, recovery, and disposal.
6. Validate with qualified specialists; baseline calculations, drawings, prototypes, tests, deviations, and evidence.

HUMAN-AI INTERACTION REQUIREMENTS

AI-generated geometry, material selections, simulations, and optimization remain proposals. Human review must inspect loads, assumptions, margins, manufacturing and field conditions, and affected users. AI must not approve lifting, motion, pressure, structural, or safety operation; stop and manual inspection remain available.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Represent geometry, materials, loads, joints, interfaces, tolerances, motions, hazards, environmental ratings, manufacturing revisions, installed configuration, inspection, maintenance, and lifecycle as controlled artifacts. Separate CAD intent, manufactured part, assembled unit, measured state, and field evidence.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show asset identity, configuration, load or motion state, guards, temperature, pressure, alarms, inspection, maintenance, deviations, evidence, and authorized stop, isolate, repair, recover, and retire actions. Keep manual release, emergency stop, and physical access controls visible.

REQUIRED OUTPUTS AND ARTIFACTS

Design basis; drawings and CAD; material and process register; load and tolerance analyses; hazard assessment; interface control; prototype and inspection plans; manufacturing and assembly instructions; maintenance and recovery procedures; test and deviation records.

EVIDENCE REQUIREMENTS

Evidence identifies part or assembly revision, material batch, manufacturing process, inspection instrument, calibration, load, environment, setup, result, reviewer, and limitations. Simulation, supplier data, inspection, and destructive testing are distinct.

VERIFICATION AND ACCEPTANCE TESTS

Verify dimensions, materials, interfaces, fit, motion, load, fatigue, vibration, thermal, wear, corrosion, guards, pinch and impact hazards, pressure or fluid containment, assembly variation, maintainability, and recovery under declared conditions.

FAILURE STATES AND ANTI-PATTERNS

Uncontrolled material or fastener substitution; missing fatigue or tolerance analysis; unguarded motion; inaccessible maintenance; hidden corrosion; CAD-only acceptance; ignored assembly variation; and recovery dependent on unsafe access or failed controls.

RECOVERY AND RECONSIDERATION

Stop motion or isolate energy, secure the area, preserve configuration and evidence, identify the hazard and authority, and use approved inspection, repair, replacement, containment, rollback, or redesign. Reopen after load, material, geometry, manufacturing, environment, incident, or evidence change.

COMPLETION CRITERIA

Loads, materials, geometry, interfaces, hazards, manufacturing, inspection, service, lifecycle, tests, evidence, owner, and gaps are explicit for the declared use; no safety or durability claim exceeds verified conditions.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve drawings, CAD revision, material and batch data, as-built state, inspections, failures, maintenance, decisions, evidence, and next action. Hand off to Electrical, Electronics, Robotics, Embedded, Hardware Testing, Manufacturing, Safety, Reliability, Operations, and Disposal.

RELATED CODEX SECTIONS

020 Systems Engineering; 050 Safety Architecture; 051 Reliability Architecture; 052 Resilience Architecture; 058 Robotics Architecture; 060 Hardware Architecture; 061 Electrical Engineering; 062 Electronics Engineering; 067 PCB Design; 073 Control Systems Engineering; 074 Robotics Engineering; 232 Hardware Testing; 351 Manufacturing; 399 Hardware Disposal.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance files, Sections 019-060, and the integration roadmap. Sector-specific structural, pressure, machinery, material, environmental, accessibility, manufacturing, licensing, certification, and owner-acceptance requirements remain unresolved until supplied and verified.

065

SECTION 065 - INDUSTRIAL DESIGN

CODEX/03_SECTIONS/065_INDUSTRIAL_DESIGN.txt

SHA-256 f487e95ff93a81194445c31c70e47bbd14ef49eed21c2451cb1989f4200d9840

6,039 BYTES / PRESERVED SOURCE IDENTITY

SECTION 065 — Industrial Design

STATUS: CONTENT-COMPLETE

PURPOSE

Define the governed design of products and physical experiences so form, function, usability, manufacture, service, accessibility, safety, identity, and lifecycle reinforce the intended outcome.

REQUIRED OUTCOME

A documented industrial design baseline linking user needs, form, ergonomics, materials, interfaces, production, quality, service, evidence, and lifecycle decisions.

WHY THIS MATTERS

Appearance alone does not establish usability, safety, accessibility, manufacturability, durability, or responsible disposal. Early form decisions can create lasting cost and harm.

APPLICABILITY

Applies to physical products, devices, enclosures, controls, packaging, tools, public-facing artifacts, and connected products where human use or physical realization matters.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when no material effect exists on form, ergonomics, physical interaction, accessibility, manufacture, service, identity, safety, or lifecycle, naming baseline and reopening trigger.

PREREQUISITES AND DEPENDENCIES

Requirements, user research, domain and capability outcomes, functional and mechanical boundaries, manufacturing constraints, materials, accessibility, safety, sustainability, cost, service, and lifecycle obligations.

REQUIRED INPUTS

Audience, use context, journeys, environments, anthropometric range, task evidence, brand or identity constraints, materials, production process, tolerances, hazards, maintenance, packaging, and disposal constraints.

HUMAN RESPONSIBILITIES AND AUTHORITY

Human owners approve intended users, protected outcomes, aesthetic and functional tradeoffs, accessibility, safety, cultural claims, production release, exceptions, and acceptance. Design capability does not create authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may synthesize research, generate alternatives, check dimensions, compare materials, and draft evaluation plans. It must label invention, preserve source evidence, avoid stereotyping, and never approve safety, accessibility, production, or user acceptance.

APPLICABLE EXISTING AGENT ROLES

Architect, Researcher, Analyst, Critic, User Experience Reviewer, Implementer, Tester, Verifier, Planner, Documentation Writer, Governance Reviewer, Red Team, Optimizer, and Memory Curator under existing role contracts.

REQUIRED ACTIVITIES

1. Define users, context, outcomes, constraints, and lifecycle. 2. Translate research into form, interaction, material, ergonomic, accessibility, and service requirements. 3. Develop alternatives and record tradeoffs. 4. Review prototype, production, quality, safety, repair, and disposal implications. 5. Test with representative users and qualified specialists. 6. Baseline evidence, dissent, decisions, and gaps.

HUMAN-AI INTERACTION REQUIREMENTS

Show source research, assumptions, affected populations, alternatives, uncertainty, and evaluation status. Human review is required for identity, inclusion, safety, accessibility, cultural meaning, and production decisions. Preserve manual and accessible paths.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Represent form, dimensions, materials, interfaces, ergonomics, controls, accessibility, manufacturing revision, as-built state, and lifecycle as controlled artifacts. Changes trigger impact analysis across mechanical, electronics, software, packaging, service, and testing.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show design revision, intended users, constraints, prototype or production state, evidence, unresolved risks, approvals, and authorized revise, quarantine, recall, repair, or retire actions.

REQUIRED OUTPUTS AND ARTIFACTS

Design brief; user and context model; sketches and CAD; material and finish register; ergonomic and accessibility analysis; prototype records; manufacturing and service review; test plan; decision and evidence register.

EVIDENCE REQUIREMENTS

Bind claims to user population, context, prototype or production revision, method, environment, sample limits, reviewer, result, and unresolved interpretation. Renderings and preference surveys do not prove safe or effective use.

VERIFICATION AND ACCEPTANCE TESTS

Verify fit, reach, comprehension, accessibility, control use, hazards, durability, cleaning, assembly, service, production variation, packaging, and disposal with representative conditions and recorded results.

FAILURE STATES AND ANTI-PATTERNS

Aesthetic-first design; token user testing; inaccessible controls; hidden maintenance burden; unsafe edges or force; prototype-only claims; unapproved material substitution; and treating brand consistency as user success.

RECOVERY AND RECONSIDERATION

Stop release or isolate affected products, preserve samples and evidence, identify affected users, and use approved redesign, repair, recall, retrofit, or withdrawal. Reopen after user, material, production, hazard, or evidence change.

COMPLETION CRITERIA

Design intent, users, form, function, accessibility, safety, manufacture, service, evidence, acceptance limits, and open gaps are explicit; status remains CONTENT-COMPLETE until reconciliation and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve revision, rationale, research, prototypes, materials, approvals, tests, deviations, and next action. Hand off to Mechanical, Electronics, UX, UI, Visual, Manufacturing, Testing, Operations, and Retirement.

RELATED CODEX SECTIONS

015 User Research; 019 Requirements Engineering; 064 Mechanical Engineering; 075 Human-Computer Interaction; 076 Interaction Design; 082 UX Design; 083 UI Design; 103 Accessibility Design; 104 Design Systems; 232 Hardware Testing; 351 Manufacturing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance files, Sections 015, 019, 020, 060-064, and the roadmap. Product standards, user populations, cultural context, manufacturing capability, sustainability rules, and owner acceptance remain unresolved until supplied and verified.

066

SECTION 066 - SEMICONDUCTOR ENGINEERING

CODEX/03_SECTIONS/066_SEMICONDUCTOR_ENGINEERING.txt

SHA-256 6d53edd15d8002722144ec30bb738e28f04779278fbe15f5e54284ebc5260922

5,799 BYTES / PRESERVED SOURCE IDENTITY

SECTION 066 — Semiconductor Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Define the governed engineering of semiconductor devices, process technology, fabrication assumptions, packaging, yield, reliability, security, and lifecycle evidence.

REQUIRED OUTCOME

A traceable device or process baseline with device behavior, process limits, masks or equivalent design artifacts, packaging, test, yield, qualification, supply, and retirement controls.

WHY THIS MATTERS

Small process variation, aging, contamination, defect, or supply change can alter safety, timing, leakage, reliability, and trust while remaining invisible to nominal functional tests.

APPLICABILITY

Applies to integrated circuits, dies, process nodes, memory, sensors, power devices, accelerators, packaging, wafer or device test, and semiconductor supply decisions.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when no device, process, packaging, yield, reliability, supply, security, test, or lifecycle property is affected, with baseline and reopening trigger.

PREREQUISITES AND DEPENDENCIES

Functional and electrical requirements, process design rules, device models, packaging, thermal and reliability targets, fab and supplier constraints, security, qualification, and lifecycle plans.

REQUIRED INPUTS

Device intent, masks or layout, process assumptions, libraries, models, tolerances, operating envelope, test structures, packaging, environmental profile, yield targets, provenance, and evidence requirements.

HUMAN RESPONSIBILITIES AND AUTHORITY

Qualified human engineers approve process assumptions, critical device behavior, foundry or supplier choices, substitutions, qualification, security, export or compliance constraints, and release. Human owners accept residual risk.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze models, layouts, yield data, defects, and test results; propose checks; and identify anomalies. It must preserve provenance and units and must not certify silicon, infer yield from small samples, or approve release.

APPLICABLE EXISTING AGENT ROLES

Architect, Researcher, Analyst, Critic, Security Auditor, Implementer, Tester, Verifier, Red Team, Optimizer, Planner, Coordinator, Observer, Governance Reviewer, Documentation Writer, and Memory Curator.

REQUIRED ACTIVITIES

1. Define device, process, package, environment, criticality, and lifecycle. 2. Capture design rules, models, masks, libraries, and dependencies. 3. Analyze variation, power, timing, thermal, aging, defect, yield, and security cases. 4. Define wafer, package, production, incoming, field, and failure tests. 5. Qualify process and supply changes. 6. Baseline evidence and gaps.

HUMAN-AI INTERACTION REQUIREMENTS

AI suggestions expose source model, process revision, sample, uncertainty, and alternatives. Human review is required for tape-out, qualification, security, supplier change, and release. Non-AI diagnostic and safe fallback paths remain available.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Control device identity, process revision, package, masks, models, libraries, test data, provenance, configuration, and lifecycle. Changes trigger electrical, computer, firmware, thermal, security, supply, and reliability impact analysis.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show device and process revision, lot, package, qualification, yield, defect, reliability, supply, security, evidence, and authorized quarantine, substitution, recall, or retire actions.

REQUIRED OUTPUTS AND ARTIFACTS

Device specification; process and design-rule record; models and libraries; layout or mask set; packaging record; test and yield plan; qualification report; supplier and change register; failure and evidence register.

EVIDENCE REQUIREMENTS

Bind results to lot, wafer or device, process, package, test setup, calibration, workload, environment, sample, method, reviewer, and limitations. Simulation, characterization, qualification, and field evidence remain distinct.

VERIFICATION AND ACCEPTANCE TESTS

Verify design rules, electrical behavior, timing, power, thermal, variation, aging, packaging, defect screening, yield, security, environmental qualification, traceability, and controlled change response.

FAILURE STATES AND ANTI-PATTERNS

Uncontrolled process change; stale models; insufficient sample; hidden lot dependence; yield-only acceptance; counterfeit or opaque supply; ignored aging; and treating simulation or vendor data as field qualification.

RECOVERY AND RECONSIDERATION

Quarantine affected lots or devices, preserve traceability, identify safe scope, and use approved re-test, binning, substitution, recall, redesign, or withdrawal. Reopen after process, foundry, package, defect, security, or evidence change.

COMPLETION CRITERIA

Device, process, package, tests, yield, qualification, supply, reliability, security, evidence, owner, and open gaps are explicit for the declared population and environment.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve masks, models, libraries, lots, package, tests, qualification, deviations, incidents, decisions, evidence, and next action. Hand off to Electronics, Computer, PCB, Hardware Testing, Manufacturing, Security, Supply Chain, and Retirement.

RELATED CODEX SECTIONS

062 Electronics Engineering; 063 Computer Engineering; 067 PCB Design; 188 Human-in-the-Loop Operations; 199 Hardware Security; 203 Cryptography; 232 Hardware Testing; 246 Security Testing; 351 Manufacturing; 354 Hardware Certification.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance files, Sections 019-065, and the roadmap. Process node, foundry, packaging, qualification, export, supply-chain, security, and owner-acceptance requirements remain unresolved until supplied and verified.

067

SECTION 067 - PCB DESIGN

CODEX/03_SECTIONS/067_PCB_DESIGN.txt

SHA-256 9d1800f0c9d719f17a2f4e9d7df296219a0b67f3e39e1bc4a5014415a08c9816

5,542 BYTES / PRESERVED SOURCE IDENTITY

SECTION 067 — PCB Design

STATUS: CONTENT-COMPLETE

PURPOSE

Define the governed printed-circuit-board design from schematic intent through layout, fabrication, assembly, inspection, test, service, and retirement.

REQUIRED OUTCOME

A controlled PCB baseline with stackup, placement, routing, power and signal integrity, thermal and EMC constraints, manufacturability, test access, revision identity, and evidence.

WHY THIS MATTERS

Layout, fabrication, and assembly variation can create failures absent from schematics or simulation. An untraceable board revision can invalidate safety, security, and field evidence.

APPLICABILITY

Applies to rigid, flexible, rigid-flex, high-speed, power, RF, mixed-signal, prototype, production, and assembled PCB designs.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when no board geometry, stackup, routing, component placement, fabrication, assembly, test, or lifecycle property changes.

PREREQUISITES AND DEPENDENCIES

Schematics, net classes, component library, electrical and mechanical interfaces, stackup, manufacturing capability, signal and power budgets, EMC, thermal, safety, test, and service constraints.

REQUIRED INPUTS

Controlled schematic and BOM, approved footprints, stackup, impedance targets, current and thermal limits, keepouts, enclosure, test points, assembly process, fab rules, revision, and evidence requirements.

HUMAN RESPONSIBILITIES AND AUTHORITY

Human PCB authority approves stackup, critical routing, substitutions, design-rule exceptions, fabrication release, field deviation, and acceptance. Human owners accept residual production and reliability risk.

AI RESPONSIBILITIES AND BOUNDARIES

AI may check rules, inspect layout, compare libraries, estimate routing or thermal issues, and draft tests. It must preserve revisions and units and must not approve release, infer equivalent footprints, or hide violations.

APPLICABLE EXISTING AGENT ROLES

Architect, Analyst, Researcher, Critic, Implementer, Tester, Verifier, Security Auditor, Planner, Coordinator, Observer, Governance Reviewer, Red Team, Documentation Writer, Optimizer, and Memory Curator.

REQUIRED ACTIVITIES

1. Establish board boundary, interfaces, environment, criticality, and lifecycle. 2. Control libraries, footprints, stackup, placement, routing, power, return paths, and keepouts. 3. Review SI, PI, thermal, EMC, safety, manufacturability, assembly, inspection, and test access. 4. Release fab and assembly data. 5. Verify first article and production variation. 6. Baseline deviations and evidence.

HUMAN-AI INTERACTION REQUIREMENTS

AI findings show rule, revision, geometry, source, confidence, and limitation. Human review is required for exceptions, critical nets, isolation, security features, production release, and substitutions. Manual inspection and safe shutdown remain available.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Version stackup, layers, nets, footprints, constraints, placement, routing, fab files, assembly files, test points, hardware revision, and as-built identity. Changes trigger electronics, mechanical, firmware, thermal, EMC, security, and reliability impact analysis.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show board revision, lot, fab and assembly status, rule violations, critical nets, test status, substitutions, evidence, and authorized hold, quarantine, rework, recall, or retire actions.

REQUIRED OUTPUTS AND ARTIFACTS

Schematic link; controlled libraries; stackup; layout; design-rule report; fab and assembly outputs; BOM; inspection and test plan; first-article report; deviation and evidence register.

EVIDENCE REQUIREMENTS

Bind evidence to board revision, library, fab lot, assembly lot, fixture, instrument, calibration, environment, test, result, and reviewer. Layout review and simulation do not prove as-built behavior alone.

VERIFICATION AND ACCEPTANCE TESTS

Verify net connectivity, clearances, impedance, power, thermal, EMC, assembly, solder, polarity, test access, rework, environmental, security, and production variation behavior.

FAILURE STATES AND ANTI-PATTERNS

Uncontrolled libraries; wrong footprint; missing return path; ignored impedance; unreviewed rule exception; stale fab files; counterfeit part; inaccessible test point; and schematic-only acceptance.

RECOVERY AND RECONSIDERATION

Hold or quarantine boards, preserve lot and revision evidence, identify safe population, and use approved rework, substitution, respin, recall, or withdrawal. Reopen after library, fab, assembly, component, or environment change.

COMPLETION CRITERIA

Board intent, revision, constraints, production data, tests, evidence, deviations, and lifecycle owner are explicit for the declared population.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve design files, libraries, fab outputs, lots, tests, rework, failures, decisions, evidence, and next action. Hand off to Electronics, Semiconductor, Firmware, Hardware Testing, Manufacturing, Supply Chain, Security, and Retirement.

RELATED CODEX SECTIONS

062 Electronics Engineering; 066 Semiconductor Engineering; 068 Firmware Engineering; 069 Embedded Software Engineering; 232 Hardware Testing; 246 Security Testing; 351 Manufacturing; 354 Hardware Certification.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance files, Sections 019-066, and the roadmap. Fab capability, stackup, EMC, qualification, supplier, certification, and owner-acceptance requirements remain unresolved until supplied and verified.

068

SECTION 068 - FIRMWARE ENGINEERING

CODEX/03_SECTIONS/068_FIRMWARE_ENGINEERING.txt

SHA-256 46c9b8febde65d44e70774093dcce251be33afa87a8ba1ef31eec32a2f3befdb

5,755 BYTES / PRESERVED SOURCE IDENTITY

SECTION 068 — Firmware Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Define the governed engineering of low-level software that initializes, controls, updates, diagnoses, and protects hardware within declared timing, resource, safety, security, and lifecycle boundaries.

REQUIRED OUTCOME

A traceable firmware baseline with hardware contracts, boot, timing, interrupts, drivers, state, update, diagnostics, security, recovery, test, and evidence controls.

WHY THIS MATTERS

Firmware controls physical behavior and often survives longer than its tooling. A hidden reset, unsafe update, stale driver, or unbounded interrupt path can defeat higher-level safeguards.

APPLICABILITY

Applies to boot code, device firmware, microcontroller software, drivers, board support, bootloaders, update agents, diagnostics, and safety or security monitors.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when no low-level code, hardware contract, boot, timing, update, diagnostic, security, safety, or lifecycle behavior changes.

PREREQUISITES AND DEPENDENCIES

Functional and computer architecture, hardware and register contracts, toolchain, memory and timing budgets, threat and hazard analysis, update authority, test hardware, and field-service plan.

REQUIRED INPUTS

Hardware revision, register map, interrupts, peripherals, timing, power, memory, boot trust, identity, update, fault, diagnostic, compiler, dependency, coding, and evidence requirements.

HUMAN RESPONSIBILITIES AND AUTHORITY

Human owners approve privileged behavior, safety and security boundaries, toolchains, release, signing, field update, recovery, exceptions, and acceptance. Operators retain authority to stop, isolate, reflash, or replace equipment.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inspect code, traces, specifications, and tests; draft implementations and reviews; and identify defects. It must preserve versions and assumptions and must not sign, release, bypass review, or claim hardware behavior without evidence.

APPLICABLE EXISTING AGENT ROLES

Architect, Researcher, Analyst, Implementer, Refactorer, Tester, Verifier, Security Auditor, Critic, Red Team, Planner, Coordinator, Observer, Governance Reviewer, Documentation Writer, Optimizer, and Memory Curator.

REQUIRED ACTIVITIES

1. Define hardware boundary, lifecycle, privilege, timing, resource, and failure requirements. 2. Implement boot, drivers, state, interrupts, diagnostics, and update contracts. 3. Review concurrency, memory, reset, power, fault, security, and safety behavior. 4. Build reproducibly and control signing and release. 5. Test on representative hardware. 6. Baseline evidence and deviations.

HUMAN-AI INTERACTION REQUIREMENTS

AI-generated code and review remain proposals. Show source, toolchain, dependencies, hardware revision, test evidence, and uncertainty. Human review is required for privileged, safety, security, boot, signing, update, and release actions.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Version firmware, source, toolchain, dependencies, configuration, hardware target, image, signing, update state, logs, diagnostics, and lifecycle. Preserve rollback, recovery, anti-rollback, identity, and offline service paths.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show device, firmware, boot, signing, update, health, fault, diagnostic, authority, evidence, and recovery state. Permit authorized pause, rollback, quarantine, reflash, restore, replace, and retire actions.

REQUIRED OUTPUTS AND ARTIFACTS

Source and build record; hardware contract; configuration; reproducible image; SBOM; signing and release record; test and static-analysis results; update and recovery runbook; evidence and decision register.

EVIDENCE REQUIREMENTS

Bind claims to source, commit, toolchain, dependencies, hardware, configuration, image hash, test, environment, result, reviewer, and limitations. Host tests and simulations do not prove field behavior alone.

VERIFICATION AND ACCEPTANCE TESTS

Verify boot, reset, timing, interrupts, memory, drivers, power, faults, privilege, secure update, interrupted update, rollback, diagnostics, watchdog, recovery, and hardware variation.

FAILURE STATES AND ANTI-PATTERNS

Unsigned or untraceable image; unsafe boot default; hidden privilege; stale dependency; unbounded ISR; ignored brownout; update without rollback; disabled watchdog; and benchmark or unit-test-only acceptance.

RECOVERY AND RECONSIDERATION

Stop release, quarantine images or devices, preserve logs and hashes, identify safe image, and use approved rollback, recovery boot, reflash, replacement, or withdrawal. Reopen after hardware, toolchain, dependency, threat, or update change.

COMPLETION CRITERIA

Firmware identity, contracts, build, signing, update, diagnostics, tests, evidence, recovery, owner, and open gaps are explicit for the tested hardware and environment.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve source, toolchain, dependencies, images, keys or signing references, hardware, logs, tests, incidents, decisions, evidence, and next action. Hand off to Embedded Software, Computer, Security, Testing, Operations, Updates, and Retirement.

RELATED CODEX SECTIONS

063 Computer Engineering; 069 Embedded Software Engineering; 107 Software Engineering; 113 Operating Systems Engineering; 198 Embedded Security; 203 Cryptography; 232 Hardware Testing; 246 Security Testing; 264 Build Engineering; 379 Firmware Updates.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance files, Sections 019-067, and the roadmap. Hardware targets, toolchain, signing authority, update infrastructure, safety regime, field population, and owner acceptance remain unresolved until supplied and verified.

069

SECTION 069 - EMBEDDED SOFTWARE ENGINEERING

CODEX/03_SECTIONS/069_EMBEDDED_SOFTWARE_ENGINEERING.txt

SHA-256 2857dac63d65a4d9121ffd12e7f57111240b61f7c3fe16c9c0b0b3396335c166

9,114 BYTES / PRESERVED SOURCE IDENTITY

SECTION 069 — Embedded Software Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Define the disciplined engineering of software that executes within or directly controls embedded hardware, including real-time behavior, resource limits, device interfaces, updateability, safety, security, and lifecycle constraints. It preserves the distinction between a software design, a tested build, a flashed image, and verified behavior in the target device.

REQUIRED OUTCOME

- A bounded embedded-software scope, hardware boundary, operating modes, timing model, resource budget, ownership, and lifecycle.
- Requirements, architecture, interfaces, state behavior, build provenance, configuration, tests, evidence, update, recovery, and retirement records.
- Explicit human authority for safety, security, release, field update, degraded operation, and acceptance.

WHY THIS MATTERS

Embedded software can control physical effects while operating with limited memory, power, bandwidth, timing margin, observability, and recovery options. A defect may persist in deployed devices, create unsafe state, corrupt data, or make later repair impossible. Engineering discipline makes timing, resource use, hardware assumptions, update authority, and evidence visible.

APPLICABILITY

Applies when software runs on a device, controller, appliance, instrument, vehicle, industrial asset, sensor, actuator, or constrained compute target, or when software depends materially on hardware timing, interrupts, peripherals, boot state, power, or physical environment.

NON-APPLICABILITY CONDITIONS

- Purely conceptual software with no claim about a target device, timing, resource use, interface, deployment, or physical effect.
- A covered embedded baseline is unchanged by a reversible documentation correction and impact analysis confirms no material effect.

Record the exact baseline, target, evidence date, owner, assumptions, and reopening trigger.

PREREQUISITES AND DEPENDENCIES

- Sections 019-020, 033, 045, 050-052, 059-068, and 231-252 as applicable.
- Hardware specifications, board support package, compiler and toolchain, device interfaces, timing and resource budgets, threat model, safety constraints, update mechanism, test fixtures, and accountable release authority.

REQUIRED INPUTS

- Approved requirements, hazards, operating modes, hardware revisions, schematics or interface descriptions, memory and power budgets, timing constraints, data classifications, and lifecycle horizon.
- Source, dependencies, toolchain, build configuration, calibration, test equipment, field population, update inventory, support, and recovery constraints.

HUMAN RESPONSIBILITIES AND AUTHORITY

The human owner approves device purpose, hazards, acceptable risk, interface contracts, timing and resource margins, security posture, field deployment, update authority, exceptions, and acceptance. Humans assign competent reviewers, approve release and rollback plans, decide when evidence is sufficient, and retain stop, quarantine, recall, downgrade, and retirement authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inspect source and specifications, propose implementation or tests, analyze traces, identify timing and resource risks, and draft evidence records. AI must label inference and simulation, preserve build and hardware identity, and escalate unresolved safety, security, timing, licensing, or authority questions. It must not flash devices, approve release, change safety behavior, hide warnings, or infer target compatibility from compilation alone.

APPLICABLE EXISTING AGENT ROLES

Architect, Implementer, Tester, Verifier, Security Auditor, Governance Reviewer, Analyst, Observer, Planner, Critic, Red Team, Documentation Writer, Memory Curator, Optimizer, and Decision Arbiter Advisory. Selection is proportional to physical consequence and must preserve independent verification.

REQUIRED ACTIVITIES

1. Establish target hardware, software boundary, operating modes, hazards, users, environments, and lifecycle.
2. Derive and trace timing, resource, interface, data, safety, security, update, and recovery requirements.
3. Define tasks, interrupts, scheduling, state machines, drivers, middleware, diagnostics, initialization, shutdown, and degraded modes.
4. Control source, dependencies, compiler, flags, generated code, configuration, calibration, reproducible builds, signing, and artifact identity.
5. Test normal, boundary, overload, power loss, reset, clock fault, communication loss, malformed input, update interruption, and recovery behavior.
6. Baseline evidence and obtain authorized release and field-deployment decisions.

HUMAN-AI INTERACTION REQUIREMENTS

- Identify human, AI, automation, and device authority at each consequential action.
- AI-generated code, tests, timing claims, and diagnostics remain proposals until review and target evidence exist.
- The owner must see assumptions, model or tool identity, source version, build, target hardware, test result, uncertainty, and unresolved defects.
- No conversational path may bypass signing, review, safety interlocks, update authorization, audit, or recovery controls.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use deterministic or bounded scheduling where required; define interrupt and concurrency rules, memory ownership, watchdog and reset semantics, hardware abstraction, versioned interfaces, diagnostic channels, secure boot and update boundaries, fail-safe states, and power-loss behavior. Preserve traceability from requirement to source, binary, device, test, field event, and retirement.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show device and image identity, hardware revision, boot state, software version, configuration, authority, health, resource margin, active update, alarms, evidence freshness, and available stop, isolate, downgrade, recover, and retire actions. Explain whether status is simulated, lab-tested, field-observed, or verified.

REQUIRED OUTPUTS AND ARTIFACTS

- Embedded software requirements and architecture baseline.
- Interface, timing, memory, power, safety, security, diagnostic, update, and recovery contracts.

- Reproducible build manifest, signed artifact record, test results, defect and waiver register, deployment inventory, and field handoff.

EVIDENCE REQUIREMENTS

Evidence identifies source, commit, dependency, toolchain, flags, binary hash, hardware revision, configuration, fixture, environment, test method, operator, date, result, and limitation. Compilation, simulation, unit tests, and one device demonstration prove only their bounded claims.

VERIFICATION AND ACCEPTANCE TESTS

Verify traceability, deterministic or bounded timing, resource margin, reset and watchdog behavior, peripheral and protocol handling, fault injection, power interruption, secure update, rollback, diagnostics, signed artifacts, hardware compatibility, and independent reproduction on representative targets.

FAILURE STATES AND ANTI-PATTERNS

- Treating a successful build, simulator, unit test, or nominal bench run as field acceptance.
- Unbounded timing, hidden shared state, unchecked integer or memory behavior, unsafe defaults, missing watchdog, undocumented hardware assumptions, unsigned updates, and recovery depending on the failed component.

RECOVERY AND RECONSIDERATION

On fault, uncertainty, resource exhaustion, corrupted image, lost communication, or authority conflict, enter the approved safe state, preserve diagnostics, isolate affected devices, and stop propagation. Reconsider only after hardware, image, configuration, evidence, user impact, security, safety, and rollback conditions are reviewed by the accountable authority.

COMPLETION CRITERIA

Scope, requirements, architecture, interfaces, resource and timing budgets, source and build identity, tests, evidence, field inventory, update and recovery procedures, open risks, and owner acceptance authority are recorded. Status remains CONTENT-COMPLETE until separate source reconciliation, hostile review, and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve target, hardware revision, image, configuration, toolchain, dependencies, decisions, evidence, defects, waivers, field population, update state, and next safe action. Hand off to Application, Software, Hardware, Firmware, Security, Safety, Testing, Operations, Maintenance, Updates, and Retirement sections as applicable.

RELATED CODEX SECTIONS

059 Embedded Systems Architecture; 060 Hardware Architecture; 062 Electronics Engineering; 068 Firmware Engineering; 070 Internet of Things Engineering; 115 Driver Engineering; 231-252 Testing sections; 378-380 Updates and Security Patching; 406 System Retirement.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance files, Sections 019-068, Owner Cockpit Requirements, Status and Evidence Lexicon, and the Remaining-Section Integration Roadmap. Target hardware, regulated safety regime, timing class, field population, toolchain, update authority, and domain-specific standards remain unresolved until supplied and verified.

070

SECTION 070 - INTERNET OF THINGS ENGINEERING

CODEX/03_SECTIONS/070_INTERNET_OF_THINGS_ENGINEERING.txt

SHA-256 ffae0cfade8526dd2fdef19db661d7aa719a73418a983f530801989113e7762a

7,999 BYTES / PRESERVED SOURCE IDENTITY

SECTION 070 — Internet of Things Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Define the engineering of connected physical things, gateways, device identities, telemetry, commands, lifecycle operations, and human control across intermittent, heterogeneous, and distributed environments.

REQUIRED OUTCOME

- A bounded device, network, cloud, edge, data, user, operator, and lifecycle scope.
- Governed identity, provisioning, telemetry, command, firmware, privacy, security, resilience, interoperability, and retirement contracts.
- Traceable evidence from device behavior through service outcome and owner control.

WHY THIS MATTERS

IoT joins physical consequences with unreliable networks, fleet scale, long-lived devices, supplier dependencies, and sensitive telemetry. A locally correct device can create system-wide harm through identity failure, stale commands, insecure provisioning, or unmanageable updates.

APPLICABILITY

Applies to networked sensors, actuators, appliances, vehicles, facilities, industrial assets, wearables, gateways, edge devices, device fleets, and services that provision, monitor, command, or retire them.

NON-APPLICABILITY CONDITIONS

- A disconnected prototype with no claim of production identity, fleet behavior, physical control, privacy, security, or lifecycle readiness.
- An approved IoT baseline is unchanged by a reversible documentation-only change with recorded impact analysis. Record scope, baseline, evidence, authority, assumptions, and reopening trigger.

PREREQUISITES AND DEPENDENCIES

- Device and service architecture, embedded software, network, identity, privacy, security, data, operations, safety, and lifecycle baselines.
- Device inventory, protocol and connectivity constraints, provisioning authority, fleet population, telemetry purpose, command policy, update mechanism, and recovery capacity.

REQUIRED INPUTS

- Mission, device classes, users, operators, physical effects, locations, jurisdictions, connectivity, data classes, service levels, and retirement horizon.
- Schemas, protocols, credentials, certificates, firmware, gateway and cloud dependencies, support model, threat and hazard analysis, and test environments.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve purpose, collection and command authority, device identity policy, safety and privacy boundaries, fleet release, emergency action, vendor commitments, acceptable risk, and retirement. They assign owners for devices, data, security, operations, field support, and affected-party response, and retain pause, revoke, isolate, recall, and decommission authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze fleet telemetry, draft schemas and provisioning plans, detect anomalies, recommend maintenance, and simulate bounded rollout. It must preserve device, firmware, location, time, data, and authority identity and expose uncertainty. It must not issue consequential commands, enroll devices, widen collection, or deploy fleet changes without explicit delegation and approval.

APPLICABLE EXISTING AGENT ROLES

Architect, Implementer, Analyst, Observer, Security Auditor, Safety specialist through the existing role contract, Privacy and Governance Reviewers, Tester, Verifier, Planner, Coordinator, Critic, Red Team, Documentation Writer, Memory Curator, Optimizer, and Decision Arbiter Advisory.

REQUIRED ACTIVITIES

1. Define device classes, fleet boundaries, physical effects, users, operators, environments, jurisdictions, and modes.
2. Establish identity, enrollment, ownership, trust, key, provisioning, inventory, and revocation models.
3. Define telemetry, command, event, protocol, schema, timing, ordering, retry, idempotency, and offline behavior.
4. Design fleet rollout, segmentation, monitoring, diagnostics, update, support, incident, recovery, and retirement paths.
5. Test spoofing, duplication, replay, stale commands, partition, loss, scale, power failure, unsafe actuation, and partial rollout.
6. Baseline evidence and secure owner approval for each consequential fleet transition.

HUMAN-AI INTERACTION REQUIREMENTS

Show acting identity, device and fleet scope, command authority, data purpose, firmware, connectivity, freshness, health, rollout state, physical consequence, and stop or revoke controls. AI proposals must show source telemetry, uncertainty, alternatives, and human approval. Non-AI fallback and manual safe operation remain available.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use zero-trust device identity, least privilege, secure provisioning, certificate rotation, command authorization, replay protection, device twins only as bounded representations, store-and-forward semantics, fleet segmentation, safe offline behavior, signed updates, audit, and physical fail-safe states. Keep device, edge, network, cloud, data, and owner boundaries explicit.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Expose fleet inventory, identity health, connectivity, telemetry freshness, command queue, firmware, configuration, risk, affected locations, rollout, alerts, evidence, and authorized actions to pause, quarantine, revoke, isolate, roll back, recall, restore, or retire.

REQUIRED OUTPUTS AND ARTIFACTS

Device and fleet architecture; identity and provisioning records; telemetry and command contracts; data and privacy map; threat and hazard analysis; rollout and update plan; monitoring and incident runbook; test and field evidence; inventory, decision, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify device, firmware, configuration, certificate, protocol, gateway, network, cloud, location, time, data purpose, actor, command, test method, result, and limitation. Fleet averages, dashboards, lab devices, and simulated twins do not prove every device or physical outcome.

VERIFICATION AND ACCEPTANCE TESTS

Verify enrollment and revocation, identity isolation, authorization, telemetry integrity and freshness, command ordering and replay resistance, offline behavior, partition recovery, fleet scale, signed update, rollback, physical safe state, privacy deletion, and independent reproduction on representative device classes.

FAILURE STATES AND ANTI-PATTERNS

Shared credentials, universal device assumptions, unrestricted commands, hidden telemetry, stale twins, unbounded retries, insecure default provisioning, fleet-wide rollout without canary evidence, and dashboards that hide missing or disconnected devices.

RECOVERY AND RECONSIDERATION

Quarantine affected devices or commands, preserve logs and physical evidence, revoke authority or credentials, enter safe mode, and contain rollout. Reconsider after fleet scope, identity, firmware, data, network, safety, security, privacy, and owner authority are revalidated.

COMPLETION CRITERIA

All device classes, boundaries, identities, contracts, modes, controls, evidence, rollout, recovery, support, retirement, open risks, and acceptance authority are recorded. Remain CONTENT-COMPLETE pending source reconciliation, hostile review, and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve fleet and device identities, firmware, configuration, locations, owners, certificates, decisions, evidence, incidents, rollouts, exceptions, and next safe action. Hand off to Embedded, Network, Cloud, Edge, Data, Security, Privacy, Operations, Updates, Support, and Retirement sections.

RELATED CODEX SECTIONS

059 Embedded Systems Architecture; 069 Embedded Software Engineering; 278 Edge Computing Engineering; 284 Network Engineering; 355 Device Provisioning; 379 Firmware Updates; 198 Embedded Security; 218 Privacy Engineering; 327-330 Recovery sections.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance files, Sections 019-069, and the Owner Cockpit and evidence conventions. Device classes, physical hazard regime, connectivity, fleet size, jurisdiction, vendor contracts, data purposes, and update authority remain unresolved until supplied.

071

SECTION 071 - TELECOMMUNICATIONS ENGINEERING

CODEX/03_SECTIONS/071_TELECOMMUNICATIONS_ENGINEERING.txt

SHA-256 4aceaefd7bfc7f92d8a4d037290408f3c32cddb6dd99ec0d20454081d0d7ced1

7,880 BYTES / PRESERVED SOURCE IDENTITY

SECTION 071 — Telecommunications Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Define the engineering of telecommunications systems that convey authorized information, signaling, voice, video, or control across shared or dedicated transmission resources, including spectrum, timing, interoperability, quality, resilience, and lawful operation.

REQUIRED OUTCOME

- A bounded service, endpoint, medium, protocol, spectrum, traffic, user, operator, jurisdiction, and lifecycle scope.
- Explicit interfaces, identity, authorization, quality, capacity, availability, privacy, security, emergency, and recovery obligations.
- Evidence linking network behavior to user and service outcomes.

WHY THIS MATTERS

Communications failures can prevent emergency action, corrupt control, disclose sensitive content, or create cascading outages. Nominal throughput is not sufficient: timing, loss, jitter, contention, coverage, interoperability, lawful use, and recovery determine real service.

APPLICABILITY

Applies to wired, wireless, optical, satellite, radio, cellular, voice, messaging, media, signaling, control, and carrier or private communications systems with material service or physical consequences.

NON-APPLICABILITY CONDITIONS

- A local conceptual diagram with no operational, interoperability, spectrum, privacy, security, service, or physical-effect claim.
- An approved telecommunications baseline is unaffected by a reversible documentation correction and impact is recorded.

Record the baseline, scope, evidence date, authority, assumptions, and reopening trigger.

PREREQUISITES AND DEPENDENCIES

- Requirements, network and communications architecture, spectrum or regulatory constraints, endpoints, protocols, traffic models, service objectives, security, privacy, safety, and continuity needs.
- Coverage, capacity, interference, latency, jitter, loss, redundancy, carrier, equipment, configuration, monitoring, emergency, and restoration evidence.

REQUIRED INPUTS

- Approved objective, service scope, endpoints, media, spectrum, traffic, users, operators, environments, jurisdictions, and lifecycle horizon.
- Requirements, protocols, quality, capacity, security, privacy, safety, regulatory, continuity, monitoring, vendor, and restoration constraints.
- Coverage, capacity, interference, latency, jitter, loss, redundancy, equipment, configuration, emergency, and service evidence.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans authorize service purpose, spectrum or regulatory use, endpoints, lawful intercept or privacy boundaries, service commitments, emergency priority, security, capacity, acceptable risk, and release. They decide outages, traffic restrictions, emergency operation, vendor escalation, and retirement.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze traces, propose topology or capacity options, classify alarms, model traffic, and draft test plans. It must preserve measurement context, configuration, endpoint identity, and uncertainty. It must not reconfigure routes, spectrum, priority, lawful controls, or service commitments without delegated authority.

APPLICABLE EXISTING AGENT ROLES

Architect, Analyst, Researcher, Observer, Implementer, Tester, Verifier, Security Auditor, Governance Reviewer, Planner, Coordinator, Critic, Red Team, Documentation Writer, Memory Curator, Optimizer, and Decision Arbiter Advisory.

REQUIRED ACTIVITIES

1. Establish service scope, endpoints, media, spectrum, traffic, users, operators, environments, and modes.
2. Define addressing, identity, authorization, protocol, timing, ordering, quality, capacity, priority, and interface contracts.
3. Model normal, congested, partitioned, degraded, emergency, maintenance, roaming, failover, and restoration behavior.
4. Assess interoperability, interference, supply chain, privacy, security, resilience, accessibility, and regulatory constraints.
5. Exercise load, loss, jitter, delay, replay, spoofing, failover, power loss, route changes, and emergency priority.
6. Baseline evidence and obtain authorized service acceptance.

HUMAN-AI INTERACTION REQUIREMENTS

Show endpoint identity, route, service mode, traffic priority, quality, capacity, freshness, authority, privacy and security state, active changes, and stop or failover actions. AI recommendations expose measurements and uncertainty; operators can override safely without bypassing audit or authorization.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Define layered interfaces, timing and synchronization, addressing, routing, redundancy, queueing, admission, congestion, encryption, key and identity boundaries, observability, service isolation, emergency traffic, graceful degradation, store-and-forward, failover, and restoration. Separate service definition from live network state.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Expose topology, endpoints, protocols, spectrum or medium, quality, capacity, alarms, dependencies, configuration, authority, evidence freshness, and authorized actions to approve, pause, isolate, reroute, fail over, restore, downgrade, or retire.

REQUIRED OUTPUTS AND ARTIFACTS

Communications architecture and interface-control records; traffic and capacity model; quality and service objectives; identity, privacy, security, emergency, resilience, and recovery design; configuration and change records; test, measurement, incident, and continuity package.

EVIDENCE REQUIREMENTS

Identify endpoint, path, medium, configuration, workload, time, location, instrument, method, actor, result, uncertainty, and limitation. A lab link, average latency, carrier statement, or dashboard proves only bounded conditions.

VERIFICATION AND ACCEPTANCE TESTS

Verify addressing and identity, protocol interoperability, timing, capacity, loss, jitter, congestion, priority, encryption, failover, emergency traffic, partition, spoofing, replay, accessibility, monitoring, restoration, and independent reproduction under representative load and geography.

FAILURE STATES AND ANTI-PATTERNS

Planning for bandwidth only; hidden shared failure domains; unowned spectrum or endpoints; unauthenticated control; indefinite queues and retries; treating carrier availability as end-to-end service; suppressing alarms; and failing over without evidence or privacy review.

RECOVERY AND RECONSIDERATION

Protect affected users and services, preserve traffic and configuration evidence, isolate faults, apply approved degraded or emergency mode, reroute or fail over, and communicate limits. Reconsider after quality, authority, security, privacy, regulatory, safety, and recovery conditions are reverified.

COMPLETION CRITERIA

Scope, service contracts, endpoints, protocols, traffic, quality, capacity, controls, modes, evidence, recovery, support, regulatory assumptions, open risks, and acceptance authority are explicit. Status remains CONTENT-COMPLETE pending reconciliation, hostile review, and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve topology, endpoint and configuration identity, measurements, changes, incidents, decisions, vendor contacts, spectrum or regulatory records, evidence, and continuation point. Hand off to Network, Communications, Integration, API, Security, Operations, Reliability, Resilience, and Support sections.

RELATED CODEX SECTIONS

039 Network Architecture; 040 Communications Architecture; 042 API Architecture; 043 Event-Driven Architecture; 071 Telecommunications Engineering; 284 Network Engineering; 305 Network Operations; 318 Alerting; 322 Availability Management; 327-330 Recovery.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance files, Sections 019-069, and owner-control and evidence conventions. Specific carrier, spectrum, regulatory, geographic, accessibility, emergency, and service-level requirements remain unresolved until supplied.

072

SECTION 072 - SIGNAL PROCESSING

CODEX/03_SECTIONS/072_SIGNAL_PROCESSING.txt

SHA-256 80696630f46dcd7141280b4a32ebda72a5aa2c1d8a41b28ae631f435fa8120b5

7,887 BYTES / PRESERVED SOURCE IDENTITY

SECTION 072 — Signal Processing

STATUS: CONTENT-COMPLETE

PURPOSE

Define the disciplined acquisition, conditioning, transformation, interpretation, and verification of signals from physical or digital sources while preserving timing, provenance, uncertainty, fidelity, and safe downstream use.

REQUIRED OUTCOME

- A declared signal, sensor, sampling, processing, output, environment, latency, resource, and lifecycle scope.
- Versioned algorithms, calibration, transforms, models, thresholds, interfaces, evidence, failure, and recovery records.
- Human authority over interpretation, thresholds, physical action, and acceptance of signal-derived outcomes.

WHY THIS MATTERS

Signal processing can turn noise, aliasing, drift, bias, compression, or a misplaced threshold into a confident but wrong decision. Evidence must show the acquisition conditions and processing chain, not merely a plausible plot or model score.

APPLICABILITY

Applies to sensing, communications, audio, image, radar, biomedical, industrial, control, scientific, or machine-learning pipelines that acquire or transform signals into information or action.

NON-APPLICABILITY CONDITIONS

- A conceptual mathematical example with no claim of calibrated acquisition, production inference, control, safety, or operational performance.
- A covered processing baseline is unchanged by a reversible documentation correction and impact is recorded. Record exact signal scope, baseline, evidence, authority, assumptions, and reopening trigger.

PREREQUISITES AND DEPENDENCIES

- Sensor or source specifications, sampling and timing, calibration, noise and environment, domain requirements, algorithms, models, thresholds, compute, interfaces, privacy, safety, security, and acceptance criteria.
- Representative labeled and unlabeled data, ground truth, fixtures, instrumentation, configuration, artifacts, and failure or drift history.

REQUIRED INPUTS

- Approved signal purpose, source and sensor specifications, sampling and timing, calibration, noise and environment, domain requirements, algorithms, models, thresholds, compute, interfaces, privacy, safety, security, and acceptance criteria.
- Representative labeled and unlabeled data, ground truth, fixtures, instrumentation, configuration, artifacts, failure history, drift history, and lifecycle constraints.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve signal purpose, collection, calibration, transformations, thresholds, acceptable uncertainty, privacy, safety, security, downstream action, and release. They decide ambiguous or conflicting interpretations, retain stop and quarantine authority, and approve changes to material processing or model behavior.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inspect traces, propose filters or features, compare algorithms, detect anomalies, generate test vectors, and summarize performance. It must preserve source, time, calibration, preprocessing, model, and uncertainty identity. It must not change thresholds, label ground truth, claim detection, or trigger physical or rights-affecting action without authority.

APPLICABLE EXISTING AGENT ROLES

Researcher, Analyst, Architect, Implementer, Tester, Verifier, Observer, Security Auditor, Governance Reviewer, User Experience Reviewer, Critic, Red Team, Planner, Documentation Writer, Memory Curator, Optimizer, and Decision Arbiter Advisory.

REQUIRED ACTIVITIES

1. Define source, signal semantics, sampling, timing, units, environment, target outputs, and consequence.
2. Characterize calibration, noise, drift, aliasing, missingness, saturation, quantization, latency, and resource limits.
3. Specify acquisition, filtering, transforms, feature extraction, inference, thresholds, outputs, provenance, and safe handling.
4. Validate on representative, boundary, adversarial, shifted, missing, corrupted, and low-quality signals.
5. Test timing, numerical stability, resource exhaustion, calibration loss, source failure, replay, and downstream recovery.
6. Baseline algorithm, configuration, evidence, and owner decision.

HUMAN-AI INTERACTION REQUIREMENTS

Show source and sensor identity, calibration, timestamp, sampling, processing version, model or filter, confidence limits, missingness, threshold, downstream consequence, and human approval. Generated or AI-derived interpretations remain proposals until verified; raw and non-AI inspection paths remain available.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Preserve acquisition-to-output lineage, units, clock behavior, calibration, numerical precision, windowing, state, resource budgets, real-time limits, versioned algorithms, model and feature identity, uncertainty, and safe output states. Do not conflate processed signal with ground truth.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Provide signal catalog, processing chain, calibration and quality records, algorithm and configuration manifest, interface and timing contract, test vectors, performance and uncertainty report, failure and recovery design, and decision and evidence register.

REQUIRED OUTPUTS AND ARTIFACTS

Identify source, sensor, calibration, time, location, sample rate, units, preprocessing, algorithm, model, threshold, ground truth, workload, method, result, uncertainty, and limitation. Plots, benchmark scores, and synthetic signals do not prove field validity.

EVIDENCE REQUIREMENTS

Verify calibration, sampling and alias control, timing, numerical behavior, noise and drift handling, missing and saturated inputs, adversarial signals, threshold sensitivity, representative generalization, resource and latency limits, safe outputs, and independent reproduction.

VERIFICATION AND ACCEPTANCE TESTS

Filtering away inconvenient cases; treating correlation as detection; leakage from labels or future data; unreported preprocessing; uncalibrated sensors; threshold tuning on test data; synthetic-only evidence; hidden drift; and confidence treated as truth.

FAILURE STATES AND ANTI-PATTERNS

On source, calibration, timing, quality, model, or provenance failure, mark outputs invalid, stop consequential use, preserve raw and processed evidence, and switch to approved fallback or human review. Reconsider after recalibration, reprocessing, validation, and downstream impact review.

RECOVERY AND RECONSIDERATION

Signal purpose, chain, assumptions, algorithms, calibration, interfaces, limits, tests, evidence, safe output, recovery, lifecycle, open risks, and acceptance authority are explicit. Remain CONTENT-COMPLETE until source reconciliation, hostile review, and owner acceptance.

COMPLETION CRITERIA

Preserve raw and processed artifacts, timestamps, calibration, source and algorithm versions, configurations, decisions, evidence, failures, affected outputs, and next safe action. Hand off to Embedded, Telecommunications, Control, Robotics, Data, AI, Testing, Operations, Safety, Security, and Retirement sections.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve raw and processed artifacts, timestamps, calibration, source and algorithm versions, configurations, decisions, evidence, failures, affected outputs, and next safe action. Hand off to Embedded, Telecommunications, Control, Robotics, Data, AI, Testing, Operations, Safety, Security, and Retirement sections.

RELATED CODEX SECTIONS

020 Systems Engineering; 040 Communications Architecture; 056-057 AI and Machine Learning Architecture; 061-069 Engineering sections; 073 Control Systems Engineering; 172 Computer Vision; 173 Speech Engineering; 254 Data Testing; 323 Performance Management.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance files, Sections 019-069, and evidence conventions. Signal modality, calibration regime, ground truth, acceptable error, operating environment, regulatory constraints, and downstream consequence remain unresolved until supplied.

073

SECTION 073 - CONTROL SYSTEMS ENGINEERING

CODEX/03_SECTIONS/073_CONTROL_SYSTEMS_ENGINEERING.txt

SHA-256 a0665d79141f9873ffb921519844db9e1225effad5dc8b96b9aaad1e22cda891

5,923 BYTES / PRESERVED SOURCE IDENTITY

SECTION 073 — Control Systems Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Define engineering of systems that sense, decide, and influence physical or computational processes through feedback, supervision, estimation, actuation, and bounded operating modes.

REQUIRED OUTCOME

- A scoped plant, controller, sensor, actuator, timing, safety, stability, authority, and lifecycle model.
- Traceable control laws, interfaces, simulations, tests, margins, failure modes, recovery, and acceptance evidence.

WHY THIS MATTERS

Feedback errors can amplify, destabilize, or physically harm. Stability in a model does not prove robustness on a noisy, delayed, saturated, changing, or adversarial plant.

APPLICABILITY

Applies to automatic, supervisory, adaptive, robotic, industrial, vehicle, energy, medical, environmental, and software control loops.

NON-APPLICABILITY CONDITIONS

Conceptual equations with no claim about a plant, actuation, stability, safety, or deployed behavior; or an unchanged approved baseline with documented impact analysis. Record scope, baseline, owner, evidence, and reopening trigger.

PREREQUISITES AND DEPENDENCIES

Plant and actuator specifications, operating modes, requirements, hazards, sensor characteristics, sampling, delays, control objectives, constraints, models, interfaces, test fixtures, and authority.

REQUIRED INPUTS

Plant and actuator specifications, operating modes, requirements, hazards, sensor characteristics, sampling, delays, control objectives, constraints, models, interfaces, test fixtures, and authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Approved outcomes, plant limits, safety and security boundaries, control authority, operating envelopes, release, overrides, emergency stop, and acceptance remain human decisions.

AI RESPONSIBILITIES AND BOUNDARIES

AI may model, simulate, tune proposals, detect drift, and draft tests. It must expose assumptions, model mismatch, uncertainty, and evidence; it must not tune live control, change limits, or command actuation without delegated authority.

APPLICABLE EXISTING AGENT ROLES

Architect, Analyst, Implementer, Tester, Verifier, Observer, Safety and Security Auditors, Governance Reviewer, Critic, Red Team, Planner, Coordinator, Documentation Writer, Memory Curator, Optimizer, and Decision Arbiter Advisory.

REQUIRED ACTIVITIES

Establish plant and loop boundaries; define states, sensors, actuators, sampling, timing, control law, constraints, authority, safe modes, and recovery; analyze stability, robustness, saturation, delay, faults, and uncertainty; simulate and test representative, boundary, degraded, and emergency cases; baseline evidence and obtain approval.

HUMAN-AI INTERACTION REQUIREMENTS

Show controller, plant, mode, setpoint, measured state, authority, limits, latency, uncertainty, alarms, override, and stop status. AI recommendations remain proposals; manual and independent safe-stop paths remain available.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Separate plant, controller, estimator, supervisor, safety monitor, and operator authority. Define sampling, clocks, delays, saturation, anti-windup, state transitions, fail-safe output, watchdog, observability, and controlled model updates.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show loop identity, mode, stability margin, sensor freshness, actuator state, constraints, authority, active tuning, alarms, evidence, and controls to pause, limit, transfer, or stop.

REQUIRED OUTPUTS AND ARTIFACTS

Control requirements and architecture; plant and loop models; control-law and interface contracts; stability and robustness analysis; simulation and test evidence; hazard, fault, recovery, calibration, release, and continuity records.

EVIDENCE REQUIREMENTS

Identify plant, controller, model, parameters, hardware, sampling, workload, environment, test method, result, uncertainty, and limitation. Simulation or nominal stability analysis proves only bounded conditions.

VERIFICATION AND ACCEPTANCE TESTS

Verify stability, robustness, delay, noise, saturation, sensor loss, actuator limits, mode transitions, disturbances, faults, adversarial input, manual override, emergency stop, recovery, and independent reproduction.

FAILURE STATES AND ANTI-PATTERNS

Ignoring delay and saturation; tuning on one scenario; hidden integrator state; unsafe loss of signal; unbounded adaptive behavior; conflating setpoint tracking with safe outcome; and relying on the controller to recover the failed safety monitor.

RECOVERY AND RECONSIDERATION

Enter approved safe mode, limit actuation, preserve telemetry, notify the owner, isolate faulty sensors or actuators, and require human review before resuming. Reassess plant, model, parameters, hazards, evidence, and downstream impact.

COMPLETION CRITERIA

Scope, models, requirements, authority, control behavior, margins, tests, evidence, safe modes, recovery, lifecycle, open risks, and acceptance authority are explicit. Status remains CONTENT-COMPLETE pending reconciliation, hostile review, and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve model, parameters, hardware, calibration, configurations, tests, incidents, decisions, evidence, and next action. Hand off to Embedded, Robotics, Signal Processing, Safety, Security, Testing, Operations, Maintenance, and Retirement.

RELATED CODEX SECTIONS

020 Systems Engineering; 050 Safety Architecture; 059 Embedded Systems Architecture; 069 Embedded Software Engineering; 072 Signal Processing; 074 Robotics Engineering; 132 Real-Time Systems; 252 Safety Testing; 322 Availability Management.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance files, Sections 019-072, and owner-control and evidence conventions. Plant class, stability criteria, hazard regime, actuator authority, sampling environment, and regulatory obligations remain unresolved until supplied.

074

SECTION 074 - ROBOTICS ENGINEERING

CODEX/03_SECTIONS/074_ROBOTICS_ENGINEERING.txt

SHA-256 e5f83441134913c6b65e0a11a0fa9d8f04dec27237cc3a068bb8e9d1536b3c30

6,140 BYTES / PRESERVED SOURCE IDENTITY

SECTION 074 — Robotics Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Define engineering of robots and autonomous physical agents that perceive, plan, decide, move, manipulate, communicate, and interact with people under bounded authority and safe operating conditions.

REQUIRED OUTCOME

- A scoped robot, environment, mission, human, sensor, actuator, autonomy, safety, security, and lifecycle model.
- Traceable behavior, motion, task, interaction, update, recovery, verification, field, and acceptance evidence.

WHY THIS MATTERS

Robots couple uncertain perception and planning to physical action near people and assets. A successful demonstration can hide collision, localization, battery, communication, manipulation, recovery, or authority failures.

APPLICABILITY

Applies to mobile, stationary, collaborative, industrial, service, medical, aerial, marine, space, and simulated robots with material autonomy or physical effect.

NON-APPLICABILITY CONDITIONS

Concept art or simulation with no claim about physical embodiment, autonomy, safety, task success, or deployment; or an unchanged approved robot baseline with impact recorded. Record reopening conditions.

PREREQUISITES AND DEPENDENCIES

Robot and environment models, mission and task requirements, hardware, control, perception, maps, safety zones, human factors, communications, power, security, test facilities, field rules, and authority.

REQUIRED INPUTS

Robot and environment models, mission and task requirements, hardware, control, perception, maps, safety zones, human factors, communications, power, security, test facilities, field rules, and authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve mission, autonomy, workspace, safety envelope, interaction, data, emergency stop, remote control, fleet release, acceptable risk, and field acceptance. They retain override, isolation, recovery, recall, and retirement authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may generate plans, analyze logs, detect obstacles or drift, and propose tests. It must reveal uncertainty and authority boundaries; it must not select unapproved missions, enter unsafe space, alter limits, or command physical action without explicit delegation.

APPLICABLE EXISTING AGENT ROLES

Architect, Analyst, Implementer, Tester, Verifier, Observer, Safety Auditor, Security Auditor, Governance Reviewer, User Experience Reviewer, Critic, Red Team, Planner, Coordinator, Documentation Writer, Memory Curator, Optimizer, and Decision Arbiter Advisory.

REQUIRED ACTIVITIES

Define mission, environment, embodiment, task, human interaction, autonomy modes, sensors, actuators, maps, localization, planning, control, communication, power, safety, security, diagnostics, update, and recovery. Test collision, loss, ambiguity, adversarial conditions, interruption, and recovery.

HUMAN-AI INTERACTION REQUIREMENTS

Show robot identity, mission, location, map freshness, autonomy, operator authority, sensor confidence, plan, physical effect, battery, communications, alarms, and stop or teleoperation controls. AI plans remain proposals until authorized and tested.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Separate mission, perception, localization, planning, control, safety monitor, human interface, and physical actuation. Define bounded autonomy, geofencing, collision avoidance, safe stop, degraded navigation, loss of link, power failure, audit, and recovery.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Expose fleet or robot identity, mode, workspace, task, plan, human authority, safety zones, telemetry, evidence, and controls to pause, stop, isolate, teleoperate, recover, recall, or retire.

REQUIRED OUTPUTS AND ARTIFACTS

Robot architecture and mission model; behavior and autonomy contracts; environment and safety model; perception, planning, control, interaction, update, security, recovery, test, field, incident, and continuity records.

EVIDENCE REQUIREMENTS

Identify robot, hardware, software, model, map, environment, task, operator, time, sensors, configuration, method, result, uncertainty, and limitation. A demo, simulation, or average success rate proves only bounded conditions.

VERIFICATION AND ACCEPTANCE TESTS

Verify safe stop, geofence, collision response, localization, obstacle and human detection, plan bounds, link loss, power loss, sensor faults, adversarial input, manual control, update rollback, recovery, and independent field reproduction.

FAILURE STATES AND ANTI-PATTERNS

Treating autonomy as authority; testing only open space; hiding recovery labor; unbounded exploration; stale maps; no independent safety monitor; remote control without latency limits; and measuring task completion while ignoring human or environmental harm.

RECOVERY AND RECONSIDERATION

Stop and isolate the robot, preserve physical and telemetry evidence, notify affected people, use approved teleoperation or recovery, and quarantine affected software or maps. Resume only after safety, security, authority, environment, and evidence review.

COMPLETION CRITERIA

Mission, environment, embodiment, autonomy, human authority, safety, behavior, evidence, tests, update, recovery, field limits, open risks, and acceptance are explicit. Remain CONTENT-COMPLETE pending reconciliation, hostile review, and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve robot and mission identity, maps, models, firmware, logs, plans, decisions, incidents, field population, evidence, and next safe action. Hand off to Embedded, Control, Signal, IoT, AI, Safety, Security, Operations, Testing, and Retirement.

RELATED CODEX SECTIONS

058 Robotics Architecture; 073 Control Systems Engineering; 074 Robotics Engineering; 090 Spatial Interface Design; 091 AR Design; 132 Real-Time Systems; 182 AI Safety; 252 Safety Testing; 355 Device Provisioning.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance files, Sections 019-073, and owner-control and evidence conventions. Robot class, workspace, autonomy level, human proximity, safety regime, mission, and field authority remain unresolved until supplied.

075

SECTION 075 - HUMAN COMPUTER INTERACTION

CODEX/03_SECTIONS/075_HUMAN_COMPUTER_INTERACTION.txt

SHA-256 79d7af361584a3ef6f8e0b0ddd7d6b7f79ce4624400501cdf134ba50a9085b50

6,045 BYTES / PRESERVED SOURCE IDENTITY

SECTION 075 — Human–Computer Interaction

STATUS: CONTENT-COMPLETE

PURPOSE

Define how people perceive, understand, control, correct, and recover from interaction with computational systems across users, operators, maintainers, and affected non-users.

REQUIRED OUTCOME

- A declared population, context, task, device, modality, accessibility, risk, and lifecycle scope.
- Evidence-backed interaction requirements, flows, feedback, error recovery, authority, privacy, and acceptance criteria.

WHY THIS MATTERS

Interaction is where system state becomes human action. Ambiguous feedback, inaccessible controls, hidden automation, or poor recovery can turn technically correct behavior into user harm or loss of control.

APPLICABILITY

Applies when people interact with software, devices, services, agents, physical systems, dashboards, tools, or automation, especially where comprehension, access, decision, or recovery matters.

NON-APPLICABILITY CONDITIONS

Purely internal implementation with no user, operator, maintainer, accessibility, explanation, or control effect; or an unchanged approved interaction baseline with impact recorded.

PREREQUISITES AND DEPENDENCIES

Users and tasks, domain and system context, requirements, journeys, accessibility needs, devices, modalities, content, policies, risks, research, telemetry, and acceptance authority.

REQUIRED INPUTS

Users and tasks, domain and system context, requirements, journeys, accessibility needs, devices, modalities, content, policies, risks, research, telemetry, and acceptance authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans define desired outcomes, populations, dignity, consent, accessibility, authority, acceptable burden, risk, and acceptance. They resolve conflicting needs, approve consequential interaction, and retain correction, appeal, stop, and override authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may synthesize research, generate alternatives, inspect interaction traces, and suggest improvements. It must distinguish observed behavior from inference, preserve participant context, and not infer consent, accessibility, preference, or comprehension from sparse signals.

APPLICABLE EXISTING AGENT ROLES

User Experience Reviewer, Researcher, Analyst, Architect, Interaction Designer through existing contract, Tester, Verifier, Accessibility specialist, Governance Reviewer, Critic, Observer, Documentation Writer, Memory Curator, Implementer, and Decision Arbiter Advisory.

REQUIRED ACTIVITIES

Define users, tasks, context, goals, states, actions, feedback, errors, recovery, accessibility, privacy, and authority. Research and model journeys; prototype; test normal, ambiguous, denied, interrupted, assistive, and recovery paths; baseline findings and acceptance.

HUMAN-AI INTERACTION REQUIREMENTS

Make system state, acting identity, automation, consequence, uncertainty, permission, and recovery visible. AI-generated interaction remains proposed; users can correct intent, stop actions, obtain non-AI help, and recover without chat dependence.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Connect user goals to system state and enforceable actions. Support perceivable feedback, understandable status, input validation, reversible actions, error recovery, assistive technology, localization, privacy, authorization, latency budgets, and audit.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show population, task, context, baseline, evidence, accessibility status, active automation, permissions, errors, open findings, and controls to stop, correct, undo, appeal, report, and recover.

REQUIRED OUTPUTS AND ARTIFACTS

Interaction requirements, personas or population records, journey and task models, prototypes, accessibility and risk findings, content and feedback specifications, test plans, research evidence, decisions, and handoff package.

EVIDENCE REQUIREMENTS

Identify population, task, context, device, modality, participant protection, method, sample, result, limitation, and evidence date. A prototype, preference, completion rate, or AI-generated persona is bounded evidence, not universal truth.

VERIFICATION AND ACCEPTANCE TESTS

Verify task success, comprehension, error prevention, recovery, accessibility, privacy, authorization, latency, modality, localization, cognitive load, and user control with representative and edge populations.

FAILURE STATES AND ANTI-PATTERNS

Designing for an imagined average user; hiding system state; dark patterns; inaccessible critical actions; irreversible errors; automation without explanation; measuring clicks instead of outcomes; and treating silence as consent.

RECOVERY AND RECONSIDERATION

Preserve participant safety and evidence, disable harmful paths, provide human support and non-AI fallback, correct or withdraw misleading content, and retest affected populations. Reconsider when context, population, policy, system state, or evidence changes.

COMPLETION CRITERIA

Population, tasks, context, interaction model, accessibility, authority, risks, evidence, tests, recovery, open findings, and acceptance owner are explicit. Status remains CONTENT-COMPLETE pending reconciliation, hostile review, and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve research, findings, decisions, prototypes, versions, participant limits, accessibility evidence, open risks, and next action. Hand off to Interaction, Experience, UI, Accessibility, Content, Application, Testing, Operations, and Governance.

RELATED CODEX SECTIONS

014 Stakeholder Management; 015 User Research; 053 Accessibility Architecture; 075 Human-Computer Interaction; 076 Interaction Design; 079-083 UX and UI sections; 247 Accessibility Testing; 249 Usability Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance files, Sections 001-074, Owner Cockpit Requirements, and evidence conventions. Actual populations, accessibility standards, domain risks, devices, modalities, and acceptance thresholds remain unresolved until supplied.

076

SECTION 076 - INTERACTION DESIGN

CODEX/03_SECTIONS/076_INTERACTION_DESIGN.txt

SHA-256 93609be54550af812ca44e645f14cee7683ca7dd3d207926ee4291769f416e6a

5,492 BYTES / PRESERVED SOURCE IDENTITY

SECTION 076 — Interaction Design

STATUS: CONTENT-COMPLETE

PURPOSE

Define the intentional design of actions, feedback, states, flows, controls, and transitions that let people and systems achieve authorized goals with comprehension, accessibility, and recovery.

REQUIRED OUTCOME

- A scoped interaction model tied to users, tasks, system states, modalities, content, permissions, and outcomes.
- Prototypes, specifications, evidence, accessibility, failure, recovery, and acceptance records.

WHY THIS MATTERS

Interaction design determines whether control is understandable and exercisable. A polished path can still conceal authority, error, exclusion, or irreversible consequence.

APPLICABILITY

Applies to user, operator, admin, assistive, conversational, physical, spatial, or multimodal controls and transitions.

NON-APPLICABILITY CONDITIONS

No material user, operator, accessibility, state, content, permission, or recovery effect; or an unchanged approved interaction specification with impact recorded.

PREREQUISITES AND DEPENDENCIES

User and task research, system state and contracts, content, domain rules, accessibility, privacy, security, devices, modalities, prototypes, and acceptance authority.

REQUIRED INPUTS

User and task research, system state and contracts, content, domain rules, accessibility, privacy, security, devices, modalities, prototypes, and acceptance authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve goals, populations, interaction authority, burden, inclusion, content meaning, risk, and acceptance. They decide tradeoffs and retain control over consequential and irreversible actions.

AI RESPONSIBILITIES AND BOUNDARIES

AI may propose flows, inspect session evidence, generate prototypes, and identify ambiguity. It must not claim user intent or comprehension, invent policy, or publish consequential behavior without human review and evidence.

APPLICABLE EXISTING AGENT ROLES

Interaction Designer, User Experience Reviewer, Researcher, Analyst, Architect, Implementer, Tester, Verifier, Accessibility specialist, Governance Reviewer, Critic, Observer, Documentation Writer, Memory Curator, and Decision Arbiter Advisory.

REQUIRED ACTIVITIES

Map goals and tasks; define states, actions, feedback, constraints, permissions, error and recovery; prototype alternatives; test with representative and assistive users; document behavior, content, evidence, and unresolved decisions.

HUMAN-AI INTERACTION REQUIREMENTS

Expose current state, pending action, actor, automation, consequence, permission, uncertainty, and recovery. AI proposals remain drafts; users can revise, cancel, undo, appeal, and use an accessible non-AI path.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use explicit state machines, predictable transitions, feedback timing, reversible actions, focus and input rules, permission enforcement, error semantics, audit, localization, assistive technology, and graceful degradation.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show flow and state baseline, user population, evidence, accessibility status, content, permissions, pending actions, errors, and controls for cancel, undo, correct, report, appeal, and recover.

REQUIRED OUTPUTS AND ARTIFACTS

Interaction specification, task and state models, wireframes or prototypes, content and feedback rules, accessibility annotations, behavior contracts, research findings, test evidence, decisions, and handoff.

EVIDENCE REQUIREMENTS

Identify user, task, context, device, version, prototype or implementation, method, participants, result, and limitation. A prototype or click path proves neither comprehension nor production accessibility.

VERIFICATION AND ACCEPTANCE TESTS

Verify state clarity, action feedback, focus, keyboard or alternative input, error prevention, undo, authorization, timing, localization, assistive support, interruption, and recovery with representative users.

FAILURE STATES AND ANTI-PATTERNS

Hidden state; ambiguous labels; disabled controls without explanation; irreversible defaults; confirmation theater; inaccessible focus; automation that looks manual; and optimizing completion while increasing errors or burden.

RECOVERY AND RECONSIDERATION

Stop or quarantine harmful interaction, preserve user intent and evidence, offer human support and alternative path, correct content or state, and retest. Reopen when system behavior, population, policy, modality, or evidence changes.

COMPLETION CRITERIA

States, actions, feedback, authority, accessibility, evidence, tests, failure, recovery, open findings, and acceptance owner are recorded. Remain CONTENT-COMPLETE pending source reconciliation, hostile review, and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve interaction identity, state model, prototypes, content, research, participant limitations, decisions, evidence, accessibility findings, and next safe action. Hand off to HCI, Experience, UI, Content, Application, Testing, and Operations.

RELATED CODEX SECTIONS

075 Human-Computer Interaction; 077 Experience Design; 082 UX Design; 083 UI Design; 088 Voice Interface Design; 089 Conversational Design; 103 Accessibility Design; 249 Usability Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance files, Sections 014-075, and evidence conventions. Actual user populations, devices, modalities, accessibility requirements, domain constraints, and acceptance thresholds remain unresolved.

077

SECTION 077 - EXPERIENCE DESIGN

CODEX/03_SECTIONS/077_EXPERIENCE_DESIGN.txt

SHA-256 5dbcb2cc8bf748ce88cf87e5b03ecddd051dbfea924b66f8e19f548124f69c88

5,495 BYTES / PRESERVED SOURCE IDENTITY

SECTION 077 — Experience Design

STATUS: CONTENT-COMPLETE

PURPOSE

Define the end-to-end quality of a person's experience across touchpoints, channels, states, emotions, tasks, service interactions, and recovery conditions.

REQUIRED OUTCOME

- A scoped experience journey, population, context, touchpoints, outcomes, risks, and lifecycle.
- Evidence-backed experience principles, journey models, prototypes, content, controls, measurement, and acceptance.

WHY THIS MATTERS

Experience quality is produced across systems and handoffs, not by a single screen. Breaks in trust, accessibility, continuity, or recovery can outweigh local usability gains.

APPLICABILITY

Applies to products, services, workplaces, public systems, physical environments, digital channels, support, onboarding, and any multi-touchpoint journey.

NON-APPLICABILITY CONDITIONS

No material effect on user, operator, support, accessibility, trust, journey, or recovery outcome; or an unchanged approved experience baseline with impact recorded.

PREREQUISITES AND DEPENDENCIES

Population and stakeholder research, journey evidence, service and process models, content, brand or domain constraints, accessibility, privacy, security, operational dependencies, and acceptance authority.

REQUIRED INPUTS

Population and stakeholder research, journey evidence, service and process models, content, brand or domain constraints, accessibility, privacy, security, operational dependencies, and acceptance authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans define desired experience outcomes, inclusion, dignity, consent, acceptable burden, tradeoffs, and acceptance. They resolve conflicts and retain authority over commitments, disclosure, access, and redress.

AI RESPONSIBILITIES AND BOUNDARIES

AI may synthesize research and propose journeys or language, but must preserve participant context, uncertainty, and dissent. It must not infer emotion, consent, satisfaction, or universal preference from proxy data.

APPLICABLE EXISTING AGENT ROLES

User Experience Reviewer, Researcher, Analyst, Architect, Interaction Designer, Service Designer, Content Designer, Tester, Verifier, Accessibility specialist, Critic, Observer, Governance Reviewer, Documentation Writer, and Memory Curator.

REQUIRED ACTIVITIES

Map populations and journeys; identify touchpoints, backstage dependencies, expectations, emotions, accessibility, failure, support, and recovery; prototype alternatives; test with representative people; baseline findings and decisions.

HUMAN-AI INTERACTION REQUIREMENTS

Show journey stage, actor, system state, automation, expectation, consequence, uncertainty, permission, and recovery. AI suggestions remain proposals, and human support and non-AI paths remain available.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Connect frontstage and backstage behavior, service contracts, content, identity, data, operations, accessibility, localization, error, escalation, continuity, and feedback loops. Preserve experience across channels and transitions.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show population, journey, touchpoint, evidence, sentiment data limits, accessibility, open findings, active incidents, and controls to correct, stop, escalate, appeal, and recover.

REQUIRED OUTPUTS AND ARTIFACTS

Experience principles, journey maps, service blueprints, personas or population records, prototypes, content and interaction specifications, research evidence, measurement plan, decision register, and handoff.

EVIDENCE REQUIREMENTS

Identify population, context, journey, touchpoint, method, sample, date, result, limitation, and affected-party protection. Ratings and analytics are bounded signals, not universal experience truth.

VERIFICATION AND ACCEPTANCE TESTS

Verify end-to-end task completion, continuity, comprehension, accessibility, trust, privacy, support, failure, recovery, channel consistency, and outcomes with representative and edge populations.

FAILURE STATES AND ANTI-PATTERNS

Optimizing a single touchpoint; hiding backstage labor; treating satisfaction as consent; designing for an average user; breaking continuity; inaccessible recovery; and counting engagement as benefit.

RECOVERY AND RECONSIDERATION

Protect participants, pause harmful journeys, preserve evidence, offer human support and alternate channels, correct misleading content, and retest affected populations. Reopen when service, population, policy, channel, or evidence changes.

COMPLETION CRITERIA

Journey scope, outcomes, population, touchpoints, dependencies, inclusion, risks, evidence, tests, recovery, open findings, and acceptance owner are explicit. Remain CONTENT-COMPLETE pending reconciliation, hostile review, and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve research, journey versions, blueprints, decisions, participant limits, evidence, unresolved gaps, and next action. Hand off to HCI, Interaction, Service, UX, Content, Operations, Accessibility, and Testing.

RELATED CODEX SECTIONS

075 Human-Computer Interaction; 076 Interaction Design; 078 Service Design; 079 UX Strategy; 080 UX Research; 014 Stakeholder Management; 015 User Research; 249 Usability Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance files, Sections 014-076, Owner Cockpit Requirements, and evidence conventions. Actual populations, channels, service commitments, cultural context, and acceptance thresholds remain unresolved.

078

SECTION 078 - SERVICE DESIGN

CODEX/03_SECTIONS/078_SERVICE_DESIGN.txt

SHA-256 a155af808c2d6a80dfada37e52778f823f31d9bd58471ab91c40e55961ec0d7e

5,562 BYTES / PRESERVED SOURCE IDENTITY

SECTION 078 — Service Design

STATUS: CONTENT-COMPLETE

PURPOSE

Define the coordinated design of services, people, processes, technology, policy, suppliers, channels, and evidence that produce an outcome for users and the organization.

REQUIRED OUTCOME

- A bounded service ecosystem, actors, value exchange, frontstage and backstage, dependencies, outcomes, and lifecycle.
- Service blueprint, ownership, measures, controls, accessibility, failure, continuity, and acceptance records.

WHY THIS MATTERS

Services fail at boundaries: policy to practice, channel to channel, provider to owner, and user to support. Service Design exposes those dependencies and assigns accountability for the whole outcome.

APPLICABILITY

Applies to public, commercial, internal, digital, physical, assisted, subscription, support, and multi-party services.

NON-APPLICABILITY CONDITIONS

No material service outcome, actor, process, channel, dependency, commitment, accessibility, or recovery effect; or an unchanged approved service baseline with impact recorded.

PREREQUISITES AND DEPENDENCIES

Service strategy, stakeholders, user research, journeys, processes, capabilities, applications, suppliers, policies, contracts, accessibility, privacy, security, operations, and acceptance authority.

REQUIRED INPUTS

Service strategy, stakeholders, user research, journeys, processes, capabilities, applications, suppliers, policies, contracts, accessibility, privacy, security, operations, and acceptance authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans own service purpose, commitments, eligibility, fairness, staffing, supplier choices, risk, exceptions, and acceptance. They decide conflicts and retain stop, escalation, redress, and retirement authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may map service evidence, identify friction, simulate demand, and propose blueprint changes. It must preserve source and affected-party limits and must not assign obligations, deny service, or change commitments without authority.

APPLICABLE EXISTING AGENT ROLES

Service Designer, Architect, Researcher, Analyst, User Experience Reviewer, Process and Operations owners, Governance Reviewer, Security Auditor, Accessibility specialist, Tester, Verifier, Critic, Coordinator, Documentation Writer, and Memory Curator.

REQUIRED ACTIVITIES

Define service outcome and eligibility; map actors, channels, frontstage, backstage, processes, systems, suppliers, evidence, controls, failure, support, and recovery; prototype and test; baseline ownership and handoff.

HUMAN-AI INTERACTION REQUIREMENTS

Show service state, actor, eligibility, authority, channel, dependency, automation, consequence, wait, explanation, and recovery. Keep human assistance and accessible alternatives available when AI or a provider fails.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Represent service ecosystem, contracts, queues, handoffs, data, policy, identity, suppliers, support, observability, continuity, and lifecycle. Distinguish service definition from actual delivery and outcome.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show service owner, current mode, journey, demand, capacity, dependencies, incidents, commitments, evidence, open findings, and controls to pause, narrow, escalate, compensate, restore, or retire.

REQUIRED OUTPUTS AND ARTIFACTS

Service blueprint, ecosystem map, journey and channel models, service standards, ownership matrix, supplier and support contracts, evidence plan, measures, risk register, recovery model, and handoff package.

EVIDENCE REQUIREMENTS

Identify population, service, channel, actor, period, contract or requirement, method, result, limitation, and owner. A blueprint, SLA, queue metric, or survey does not prove equitable or effective service.

VERIFICATION AND ACCEPTANCE TESTS

Verify end-to-end outcome, eligibility, accessibility, channel continuity, handoffs, supplier failure, demand peaks, privacy, security, support, complaint, compensation, and recovery.

FAILURE STATES AND ANTI-PATTERNS

Optimizing internal throughput; hiding eligibility or denial; outsourcing accountability; fragmented channels; inaccessible support; and treating an SLA or dashboard as the user outcome.

RECOVERY AND RECONSIDERATION

Contain affected service, preserve records, communicate limits, use approved alternate channel or manual support, escalate to the owner, and compensate where authorized. Reconsider after supplier, policy, capacity, accessibility, and evidence review.

COMPLETION CRITERIA

Service outcome, ecosystem, actors, channels, dependencies, commitments, controls, evidence, recovery, open risks, and acceptance authority are explicit. Remain CONTENT-COMPLETE pending reconciliation, hostile review, and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve blueprint, contracts, service owners, supplier and support contacts, measures, decisions, incidents, evidence, dissent, and next safe action. Hand off to Process, Operations, UX, Content, Architecture, Support, Testing, and Governance.

RELATED CODEX SECTIONS

CODEX/03_SECTIONS/078_SERVICE_DESIGN.txt

004 Mission Strategy; 014 Stakeholder Management; 015 User Research; 027 Process Architecture; 029 Operational Architecture; 077 Experience Design; 079 UX Strategy; 311 Service Management; 334 Customer Support.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance files, Sections 001-077, and evidence conventions. Actual service owners, commitments, suppliers, eligibility, populations, jurisdictions, and compensation rules remain unresolved.

079

SECTION 079 - UX STRATEGY

CODEX/03_SECTIONS/079_UX_STRATEGY.txt

SHA-256 ee347bd09853d23f98f5f4fe5baf18be2c11bb4c552adf0d90c11a710ca641ee

5,490 BYTES / PRESERVED SOURCE IDENTITY

SECTION 079 — UX Strategy

STATUS: CONTENT-COMPLETE

PURPOSE

Define how user experience contributes to approved outcomes, strategic choices, differentiated value, inclusion, risk treatment, and measurable long-term product or service direction.

REQUIRED OUTCOME

- A declared experience ambition, users, business or mission outcomes, constraints, portfolio choices, measures, and horizon.
- Prioritized experience principles, evidence, investment decisions, risks, governance, and acceptance criteria.

WHY THIS MATTERS

Experience strategy prevents local design activity from outrunning mission, user need, operational capacity, or ethical obligations. It makes tradeoffs and omissions visible.

APPLICABILITY

Applies when experience is a material source of value, adoption, safety, trust, inclusion, service quality, differentiation, or change priority.

NON-APPLICABILITY CONDITIONS

No material experience outcome, population, prioritization, investment, policy, journey, or measure effect; or an unchanged approved UX strategy with impact recorded.

PREREQUISITES AND DEPENDENCIES

Mission and product strategy, stakeholder and user evidence, market or domain context, journeys, accessibility, brand, architecture, operations, portfolio constraints, risks, and decision authority.

REQUIRED INPUTS

Mission and product strategy, stakeholder and user evidence, market or domain context, journeys, accessibility, brand, architecture, operations, portfolio constraints, risks, and decision authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans choose strategic outcomes, target populations, inclusion commitments, investment, tradeoffs, acceptable burden, and acceptance. They own commitments and decide when evidence supports reprioritization.

AI RESPONSIBILITIES AND BOUNDARIES

AI may synthesize evidence, identify patterns, compare strategic options, and draft hypotheses. It must expose evidence quality and opportunity cost and must not define user values or make portfolio commitments.

APPLICABLE EXISTING AGENT ROLES

UX Strategist, Architect, Researcher, Analyst, User Experience Reviewer, Product and Service owners, Governance Reviewer, Critic, Planner, Coordinator, Verifier, Documentation Writer, Memory Curator, and Decision Arbiter Advisory.

REQUIRED ACTIVITIES

Define experience outcomes and principles; segment populations; map journeys and pain; assess current capability; generate options; prioritize by evidence, impact, risk, and cost; define measures; secure decisions; review and replan.

HUMAN-AI INTERACTION REQUIREMENTS

Show evidence, population, goal, assumption, confidence limit, tradeoff, owner, investment, and expected consequence. AI recommendations remain advisory and do not bypass product, governance, accessibility, or user authority.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Link strategy to journeys, capabilities, processes, architecture, design system, content, operations, measures, experiments, and lifecycle. Define leading and outcome measures and invalidation conditions.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show experience goals, target populations, evidence freshness, roadmap choices, risks, measures, investment, dependencies, open dissent, and controls to approve, defer, stop, narrow, or reopen work.

REQUIRED OUTPUTS AND ARTIFACTS

Experience strategy, principles, population and journey map, opportunity register, evidence assessment, prioritization rationale, investment and dependency map, measures, decision record, and handoff.

EVIDENCE REQUIREMENTS

Identify source, population, context, method, date, sample, result, uncertainty, and limitation. Market signals, analytics, interviews, or AI synthesis are bounded evidence.

VERIFICATION AND ACCEPTANCE TESTS

Verify strategic traceability, population inclusion, outcome measures, accessibility, feasibility, opportunity cost, dependency, ethical risk, and whether decisions change under plausible evidence updates.

FAILURE STATES AND ANTI-PATTERNS

Chasing engagement over outcome; treating personas as facts; copying competitors; ignoring non-users; confusing aesthetic preference with value; and making commitments without owner, capacity, or evidence.

RECOVERY AND RECONSIDERATION

Record dissent, pause unsupported investment, preserve prior decisions, commission missing research, and re-evaluate affected populations and commitments. Reopen when mission, market, evidence, technology, policy, or user needs change.

COMPLETION CRITERIA

Outcomes, populations, principles, evidence, priorities, measures, tradeoffs, owners, risks, decisions, and reopening rules are explicit. Remain CONTENT-COMPLETE pending reconciliation, hostile review, and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve strategy version, evidence, assumptions, decisions, dissent, portfolio links, dependencies, measures, and next action. Hand off to Product, Service, UX, Research, Design Systems, Analytics, Operations, and Governance.

RELATED CODEX SECTIONS

003 Innovation; 006 Product Strategy; 014 Stakeholder Management; 015 User Research; 077 Experience Design; 078 Service Design; 080 UX Research; 373 Roadmap Management; 369 Experimentation.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance files, Sections 001-078, and evidence conventions. Actual strategic objectives, populations, investment constraints, market evidence, accessibility commitments, and decision authority remain unresolved.

080

SECTION 080 - UX RESEARCH

CODEX/03_SECTIONS/080_UX_RESEARCH.txt

SHA-256 3e1b6c6d0dba902c855ac40c465ebd1e719648a228eae41bec22f3ed9b6343d3

5,438 BYTES / PRESERVED SOURCE IDENTITY

SECTION 080 — UX Research

STATUS: CONTENT-COMPLETE

PURPOSE

Define ethical, rigorous research that establishes what people need, do, understand, experience, and cannot do in a declared context.

REQUIRED OUTCOME

- A research question, population, context, method, consent and protection plan, evidence record, analysis, limitation, and decision link.
- Reproducible findings that distinguish observation, interpretation, hypothesis, and recommendation.

WHY THIS MATTERS

Research reduces uncertainty only when its methods fit the question and its limitations are visible. Weak sampling or leading interpretation can institutionalize the wrong problem.

APPLICABILITY

Applies to discovery, evaluation, accessibility, service, product, operational, safety, and post-release questions involving people or affected communities.

NON-APPLICABILITY CONDITIONS

No human or affected-party claim, decision, behavior, need, burden, comprehension, or acceptance inference; or an unchanged research plan with impact recorded.

PREREQUISITES AND DEPENDENCIES

Research question, domain context, population, recruitment, consent, privacy, safety, accessibility, method, instruments, incentives, analysis plan, and decision owner.

REQUIRED INPUTS

Research question, domain context, population, recruitment, consent, privacy, safety, accessibility, method, instruments, incentives, analysis plan, and decision owner.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve research purpose, participant protection, consent, inclusion, compensation, interpretation, disclosure, and use of findings. They decide when evidence is sufficient and retain authority over decisions.

AI RESPONSIBILITIES AND BOUNDARIES

AI may help recruit within approved criteria, transcribe, code, summarize, compare, and identify gaps. It must preserve participant privacy, quotes, context, uncertainty, and dissent; it must not fabricate participants, consent, findings, or universal claims.

APPLICABLE EXISTING AGENT ROLES

Researcher, User Experience Reviewer, Analyst, Observer, Critic, Governance Reviewer, Accessibility specialist, Security and Privacy Auditors, Tester, Verifier, Documentation Writer, Memory Curator, and Decision Arbiter Advisory.

REQUIRED ACTIVITIES

Frame question; define population and method; protect and recruit participants; collect and preserve evidence; analyze with alternative interpretations; validate findings; state limits; link decisions; archive and hand off.

HUMAN-AI INTERACTION REQUIREMENTS

Show participant protection, method, source, quote or observation, interpretation, uncertainty, and decision use. AI-generated coding remains reviewable; participants can withdraw where promised and human contact remains available.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Separate participant evidence, researcher interpretation, hypothesis, recommendation, and decision. Protect identity, consent, access, retention, provenance, translation, accessibility, and reproducibility.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show question, sample, method, status, findings, confidence limits, dissent, risks, consent, evidence freshness, and controls to restrict, correct, withdraw, or reopen research.

REQUIRED OUTPUTS AND ARTIFACTS

Research protocol, participant and consent records, instruments, raw and derived evidence, analysis, findings, limitations, accessibility and risk records, decisions, and handoff package.

EVIDENCE REQUIREMENTS

Identify participant or population, context, method, sample, date, researcher, source, consent, result, uncertainty, and limitation. A small sample, quote, survey, or AI summary cannot support universal claims.

VERIFICATION AND ACCEPTANCE TESTS

Verify recruitment and consent, method fit, inclusion, leading effects, data protection, transcription, coding, rival interpretations, member checking where appropriate, and decision traceability.

FAILURE STATES AND ANTI-PATTERNS

Leading questions; convenience samples presented as representative; persona invention; quote mining; coercive incentives; re-identification; treating silence as satisfaction; and erasing dissent or negative cases.

RECOVERY AND RECONSIDERATION

Stop unsafe research, protect participants, restrict or delete data as authorized, preserve an audit trail, disclose limitations, and commission corrective research. Reconsider decisions when sampling, consent, context, or interpretation is challenged.

COMPLETION CRITERIA

Question, population, method, protection, evidence, analysis, findings, limitations, decisions, and acceptance authority are explicit. Remain CONTENT-COMPLETE pending reconciliation, hostile review, and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve protocol, consent, instruments, evidence, analysis, dissent, decisions, retention state, and next action. Hand off to UX, HCI, Experience, Service, Accessibility, Privacy, Governance, Product, and Testing.

RELATED CODEX SECTIONS

004-015 Strategy and stakeholder sections; 075 Human-Computer Interaction; 077-079 Experience, Service, and UX Strategy; 103 Accessibility Design; 247 Accessibility Testing; 249 Usability Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance files, Sections 001-079, and evidence conventions. Actual research questions, populations, consent regime, recruitment access, cultural context, and decision needs remain unresolved.

081

SECTION 081 - UX ARCHITECTURE

CODEX/03_SECTIONS/081_UX_ARCHITECTURE.txt

SHA-256 05ba286ebf015c9f627a7ebd3311ad4f258ac197a12bbc909286e70aa13a42bf

5,277 BYTES / PRESERVED SOURCE IDENTITY

SECTION 081 — UX Architecture

STATUS: CONTENT-COMPLETE

PURPOSE

Define the structural organization of user experience across journeys, capabilities, channels, states, patterns, content, accessibility, and operating contexts.

REQUIRED OUTCOME

- A scoped UX landscape connecting user outcomes to journeys, interaction structures, content, systems, services, and governance.
- Stable experience models, principles, patterns, evidence, controls, measures, and lifecycle handoffs.

WHY THIS MATTERS

Without an experience architecture, teams optimize isolated screens and create inconsistent states, terminology, accessibility, and recovery. Architecture makes cross-product experience coherence inspectable.

APPLICABILITY

Applies to products, platforms, services, suites, multi-channel ecosystems, design systems, and complex user or operator journeys.

NON-APPLICABILITY CONDITIONS

No material journey, pattern, content, state, accessibility, channel, or governance effect; or an unchanged approved UX architecture with impact recorded.

PREREQUISITES AND DEPENDENCIES

Strategy, research, journeys, information and application architecture, domain rules, design system, content, accessibility, privacy, security, operations, and decision authority.

REQUIRED INPUTS

Strategy, research, journeys, information and application architecture, domain rules, design system, content, accessibility, privacy, security, operations, and decision authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve experience principles, boundaries, populations, inclusion, consistency, tradeoffs, and acceptance. They decide exceptions and retain authority over user impact and commitments.

AI RESPONSIBILITIES AND BOUNDARIES

AI may map journeys, identify pattern drift, and propose structures. It must preserve evidence, context, and dissent and must not impose a universal user model or silently change standards.

APPLICABLE EXISTING AGENT ROLES

UX Strategist, Architect, Researcher, User Experience Reviewer, Interaction Designer, Content Designer, Analyst, Critic, Governance Reviewer, Accessibility specialist, Tester, Verifier, Implementer, Documentation Writer, and Memory Curator.

REQUIRED ACTIVITIES

Define experience scope, principles, journey and state taxonomy, channels, patterns, content relationships, accessibility, ownership, dependencies, exceptions, measurements, validation, and handoff.

HUMAN-AI INTERACTION REQUIREMENTS

Show journey, state, pattern, actor, authority, automation, evidence, accessibility, and recovery. AI suggestions remain proposed and do not bypass human review or design-system governance.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Connect journeys, states, information, interactions, components, services, content, identity, permissions, analytics, accessibility, localization, error, and lifecycle. Keep architecture separate from implementation.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show UX baseline, journeys, pattern versions, users, dependencies, evidence, exceptions, accessibility status, and controls to approve, correct, supersede, restrict, or reopen.

REQUIRED OUTPUTS AND ARTIFACTS

UX architecture charter; journey and state models; experience principles; pattern and channel map; ownership and dependency records; accessibility, content, evidence, decision, exception, and handoff package.

EVIDENCE REQUIREMENTS

Identify population, context, journey, version, source, method, result, uncertainty, and limitation. A pattern inventory or consistency score proves only bounded structural coverage.

VERIFICATION AND ACCEPTANCE TESTS

Verify journey completeness, state consistency, pattern reuse, accessibility, content, localization, error and recovery, cross-channel continuity, and impact of exceptions with representative users.

FAILURE STATES AND ANTI-PATTERNS

Forcing uniformity where context differs; pattern libraries without behavior; ignoring backstage systems; visual consistency over comprehension; inaccessible states; and architecture that cannot represent exception or recovery.

RECOVERY AND RECONSIDERATION

Quarantine harmful patterns, preserve prior versions and evidence, provide alternate path, and obtain owner review. Reopen when products, populations, journeys, platforms, policies, or evidence change.

COMPLETION CRITERIA

Scope, principles, journeys, states, patterns, ownership, accessibility, evidence, controls, recovery, open issues, and acceptance authority are explicit. Remain CONTENT-COMPLETE pending reconciliation, hostile review, and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve models, versions, rationale, exceptions, evidence, research, decisions, and next action. Hand off to Interaction, UI, Visual, Content, Design Systems, Application, Accessibility, Testing, and Operations.

RELATED CODEX SECTIONS

075 Human-Computer Interaction; 076 Interaction Design; 077 Experience Design; 082 UX Design; 083 UI Design; 097 Content Design; 103 Accessibility Design; 104 Design Systems.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance files, Sections 014-080, and evidence conventions. Actual product landscape, populations, channel constraints, standards, and acceptance thresholds remain unresolved.

082

SECTION 082 - UX DESIGN

CODEX/03_SECTIONS/082_UX_DESIGN.txt

SHA-256 22ce15df1f614d0d8647b4d307ae33be9a60868c7f6cd84f6b584627ed1228fe

5,051 BYTES / PRESERVED SOURCE IDENTITY

SECTION 082 — UX Design

STATUS: CONTENT-COMPLETE

PURPOSE

Define the design of usable, accessible, understandable, and recoverable experiences that meet authorized user and service outcomes.

REQUIRED OUTCOME

- A declared user, task, context, journey, interaction, content, accessibility, risk, and outcome scope.
- Tested designs and specifications with traceable evidence, decisions, limitations, and implementation handoff.

WHY THIS MATTERS

UX design translates evidence into behavior and presentation. It must account for the whole task, not only the attractive or nominal path.

APPLICABILITY

Applies to user, operator, admin, support, assistive, mobile, desktop, physical, conversational, and multimodal experiences.

NON-APPLICABILITY CONDITIONS

No material user, task, state, content, accessibility, permission, or recovery effect; or an unchanged approved UX design with impact recorded.

PREREQUISITES AND DEPENDENCIES

Research, requirements, journeys, information and interaction models, content, devices, accessibility, privacy, security, design system, implementation constraints, and acceptance authority.

REQUIRED INPUTS

Research, requirements, journeys, information and interaction models, content, devices, accessibility, privacy, security, design system, implementation constraints, and acceptance authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

User and product owners approve outcomes, populations, inclusion, burden, risk, content meaning, tradeoffs, and acceptance. They retain correction, appeal, and stop authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may generate alternatives, copy drafts, prototypes, and critique. It must preserve research context and uncertainty and must not claim usability, accessibility, consent, or user intent without evidence.

APPLICABLE EXISTING AGENT ROLES

UX Designer, User Experience Reviewer, Researcher, Interaction Designer, Visual Designer, Content Designer, Architect, Accessibility specialist, Tester, Verifier, Critic, Governance Reviewer, Implementer, and Documentation Writer.

REQUIRED ACTIVITIES

Frame goals; model tasks and states; design flows, content, feedback, permissions, and recovery; prototype; test with representative and assistive users; resolve findings; specify and hand off.

HUMAN-AI INTERACTION REQUIREMENTS

Show user, state, actor, automation, authority, consequence, uncertainty, and recovery. AI proposals remain drafts and must not bypass review, accessibility, privacy, or implementation checks.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Define behavior, structure, content, visual, accessibility, localization, data, identity, error, analytics, and lifecycle contracts. Ensure design intent survives implementation and degraded operation.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show design version, task, population, evidence, accessibility findings, open defects, implementation status, and controls to approve, correct, annotate, supersede, or recover.

REQUIRED OUTPUTS AND ARTIFACTS

UX requirements, journey and task flows, wireframes or prototypes, interaction and content specs, accessibility annotations, research findings, test results, decisions, and implementation handoff.

EVIDENCE REQUIREMENTS

Identify task, user, context, device, design version, method, participant coverage, result, and limitation. A prototype, heuristic review, or AI critique is bounded evidence.

VERIFICATION AND ACCEPTANCE TESTS

Verify task success, comprehension, accessibility, error prevention, recovery, content, permissions, privacy, latency, localization, and implementation fidelity.

FAILURE STATES AND ANTI-PATTERNS

Designing the happy path only; inaccessible critical actions; ambiguous state; hidden automation; decorative hierarchy that obscures task; and calling a heuristic review user validation.

RECOVERY AND RECONSIDERATION

Pause affected release, preserve findings and user evidence, provide alternate or manual path, correct design, and retest. Reopen when requirements, population, technology, content, or evidence changes.

COMPLETION CRITERIA

Goals, users, tasks, flows, states, content, accessibility, evidence, tests, risks, recovery, open findings, and acceptance authority are explicit. Remain CONTENT-COMPLETE pending reconciliation, hostile review, and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve design versions, prototypes, specifications, findings, decisions, evidence, exceptions, and next action. Hand off to UI, Visual, Content, Design Systems, Application, Accessibility, Testing, and Operations.

RELATED CODEX SECTIONS

075 Human-Computer Interaction; 076 Interaction Design; 081 UX Architecture; 083 UI Design; 097 Content Design; 103 Accessibility Design; 104 Design Systems; 249 Usability Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance files, Sections 015 and 075-081, and evidence conventions. Actual users, devices, accessibility standards, implementation constraints, and acceptance criteria remain unresolved.

083

SECTION 083 - UI DESIGN

CODEX/03_SECTIONS/083_UI_DESIGN.txt

SHA-256 5b322a55b6757ff0db5f2c926a254700147909da4b3dec97b216136bfaa06af3

5,105 BYTES / PRESERVED SOURCE IDENTITY

SECTION 083 — UI Design

STATUS: CONTENT-COMPLETE

PURPOSE

Define the design of interface surfaces, controls, layout, hierarchy, feedback, states, and visual affordances that let people complete authorized tasks.

REQUIRED OUTCOME

- A scoped UI system tied to users, tasks, states, content, interaction, accessibility, devices, and outcomes.
- Tested interface specifications with evidence, behavior, responsive rules, controls, and implementation handoff.

WHY THIS MATTERS

UI is the visible contract between people and system state. Visual clarity cannot compensate for incorrect behavior, inaccessible controls, or hidden authority.

APPLICABILITY

Applies to screens, dashboards, forms, controls, navigation, responsive surfaces, device displays, operator consoles, and system status views.

NON-APPLICABILITY CONDITIONS

No material surface, control, state, hierarchy, accessibility, content, or recovery effect; or an unchanged approved UI specification with impact recorded.

PREREQUISITES AND DEPENDENCIES

UX and interaction models, content, design system, user research, device constraints, accessibility, localization, data, identity, privacy, security, and acceptance authority.

REQUIRED INPUTS

UX and interaction models, content, design system, user research, device constraints, accessibility, localization, data, identity, privacy, security, and acceptance authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve meaning, hierarchy, inclusion, interaction authority, risk, content, and acceptance. They resolve visual or usability tradeoffs and retain correction and stop authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may generate layouts, variants, tokens, and critiques. It must preserve context, accessibility, and content meaning and must not claim comprehension or silently alter permission or hierarchy.

APPLICABLE EXISTING AGENT ROLES

UI Designer, UX Designer, Interaction Designer, Visual Designer, User Experience Reviewer, Researcher, Accessibility specialist, Architect, Tester, Verifier, Critic, Implementer, Governance Reviewer, and Documentation Writer.

REQUIRED ACTIVITIES

Define surfaces, tasks, states, hierarchy, components, content, responsive behavior, input, feedback, errors, accessibility, localization, and implementation specifications; prototype and test representative paths.

HUMAN-AI INTERACTION REQUIREMENTS

Show current state, action, actor, automation, consequence, permission, validation, and recovery. AI-generated UI remains proposed; critical controls retain accessible and non-AI alternatives.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Specify component behavior, state, tokens, responsive constraints, focus, input, content, data, permission, error, telemetry, localization, and degraded rendering. Keep visual design separate from security enforcement.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show UI version, surface, state coverage, device, accessibility, content, evidence, implementation status, defects, and controls to annotate, approve, correct, or restore.

REQUIRED OUTPUTS AND ARTIFACTS

UI specifications, component states, responsive and accessibility rules, prototypes, content mapping, visual assets, test results, evidence, decisions, and handoff package.

EVIDENCE REQUIREMENTS

Identify surface, device, viewport, user, state, version, method, result, and limitation. A screenshot or visual review proves only appearance in bounded conditions.

VERIFICATION AND ACCEPTANCE TESTS

Verify hierarchy, state feedback, responsive behavior, focus, keyboard and assistive access, contrast, content fit, localization, errors, permissions, loading, offline, and recovery.

FAILURE STATES AND ANTI-PATTERNS

Screenshot-driven design; color as the only status signal; clipped text; inaccessible focus; inconsistent state; decorative motion blocking task; and visual polish treated as usability evidence.

RECOVERY AND RECONSIDERATION

Disable or correct affected surface, preserve state and user intent, provide accessible fallback, and retest devices and assistive modes. Reopen when content, components, platform, population, or evidence changes.

COMPLETION CRITERIA

Surfaces, states, controls, hierarchy, accessibility, content, evidence, tests, recovery, open issues, and acceptance authority are explicit. Remain CONTENT-COMPLETE pending reconciliation, hostile review, and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve UI versions, states, tokens, assets, content, findings, decisions, evidence, and next action. Hand off to Visual, Interaction, UX, Content, Design Systems, Frontend, Accessibility, and Testing.

RELATED CODEX SECTIONS

076 Interaction Design; 082 UX Design; 084 Visual Design; 097 Content Design; 103 Accessibility Design; 104 Design Systems; 118 Frontend Engineering; 247 Accessibility Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance files, Sections 075-082, and evidence conventions. Actual device matrix, content, accessibility standards, component library, and acceptance thresholds remain unresolved.

084

SECTION 084 - VISUAL DESIGN

CODEX/03_SECTIONS/084_VISUAL_DESIGN.txt

SHA-256 9df5e6175cb40c718534b9b4bb5a557f46088bba5dc6dab55c2c642df55323dc

5,338 BYTES / PRESERVED SOURCE IDENTITY

SECTION 084 — Visual Design

STATUS: CONTENT-COMPLETE

PURPOSE

Define visual language, composition, typography, color, imagery, iconography, motion cues, and hierarchy in service of comprehension, identity, accessibility, and authorized outcomes.

REQUIRED OUTCOME

- A scoped visual system with audience, medium, content, brand, accessibility, cultural, technical, and lifecycle constraints.
- Reusable visual rules, assets, specifications, evidence, exceptions, and acceptance criteria.

WHY THIS MATTERS

Visual choices shape attention, interpretation, trust, and access. Aesthetic consistency that obscures status, hierarchy, or difference can create operational and exclusionary harm.

APPLICABILITY

Applies to digital interfaces, products, services, environments, campaigns, data displays, identities, and visual systems with material interpretation or access effects.

NON-APPLICABILITY CONDITIONS

No material visual meaning, hierarchy, accessibility, identity, content, or interaction effect; or an unchanged approved visual baseline with impact recorded.

PREREQUISITES AND DEPENDENCIES

Brand and domain rules, UI and UX models, content, audience research, accessibility, localization, device and medium constraints, legal rights, asset provenance, and acceptance authority.

REQUIRED INPUTS

Brand and domain rules, UI and UX models, content, audience research, accessibility, localization, device and medium constraints, legal rights, asset provenance, and acceptance authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve meaning, audience, representation, cultural and accessibility obligations, rights, identity, risk, and acceptance. They resolve conflicts and retain publication and correction authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may propose visual alternatives, generate drafts, and inspect consistency. It must disclose source and license limits, preserve accessibility and representation context, and not publish assets or infer taste as fact.

APPLICABLE EXISTING AGENT ROLES

Visual Designer, UI Designer, UX Designer, Graphic Designer, Researcher, User Experience Reviewer, Accessibility specialist, Content Designer, Critic, Governance Reviewer, Tester, Verifier, Implementer, and Documentation Writer.

REQUIRED ACTIVITIES

Define visual principles, hierarchy, type, color, imagery, iconography, spacing, data display, responsive behavior, accessibility, localization, asset rights, variants, and usage rules; prototype and validate.

HUMAN-AI INTERACTION REQUIREMENTS

Show hierarchy, status, contrast, actor, automation, consequence, accessibility, source, and alternative representation. AI-generated assets remain proposals and must not obscure controls or provenance.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use tokens, semantic color, readable typography, contrast, nonvisual cues, responsive constraints, asset metadata, localization, performance, and content integrity. Visual styling must not substitute for authorization or status truth.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show visual baseline, medium, audience, asset provenance, accessibility, content, contrast, open findings, and controls to approve, replace, annotate, withdraw, or restore.

REQUIRED OUTPUTS AND ARTIFACTS

Visual system and style guide, tokens, type and color rules, asset catalog and rights, compositions, accessibility annotations, responsive specifications, test evidence, decisions, and handoff.

EVIDENCE REQUIREMENTS

Identify asset, source, license, audience, medium, version, method, rendering context, result, and limitation. A mockup or brand review proves bounded appearance, not comprehension or access.

VERIFICATION AND ACCEPTANCE TESTS

Verify contrast, scaling, reflow, non-color cues, typography, localization, content fit, asset rights, assistive use, low bandwidth, device variation, and implementation fidelity.

FAILURE STATES AND ANTI-PATTERNS

Color-only status; inaccessible type; decorative hierarchy that misleads; unlicensed assets; cultural stereotypes; brand rules over user need; and visual consistency treated as comprehension.

RECOVERY AND RECONSIDERATION

Withdraw harmful or infringing assets, preserve provenance and prior version, provide accessible alternate representation, notify owners, and retest. Reopen when audience, medium, brand, content, rights, or standards change.

COMPLETION CRITERIA

Visual purpose, audience, assets, rules, accessibility, provenance, evidence, tests, recovery, exceptions, open risks, and acceptance are explicit. Remain CONTENT-COMPLETE pending reconciliation, hostile review, and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve rules, tokens, assets, licenses, versions, decisions, findings, evidence, exceptions, and next action. Hand off to UI, Graphic, Motion, Content, Design Systems, Accessibility, Localization, and Frontend.

RELATED CODEX SECTIONS

083 UI Design; 085 Graphic Design; 086 Motion Design; 097 Content Design; 101 Localization; 103 Accessibility Design; 104 Design Systems; 118 Frontend Engineering.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance files, Sections 075-083, and evidence conventions. Actual brand authority, assets, licenses, audiences, cultural context, media, and accessibility standards remain unresolved.

085

SECTION 085 - GRAPHIC DESIGN

CODEX/03_SECTIONS/085_GRAPHIC_DESIGN.txt

SHA-256 2784a7de3e52462d2f197cdaaf8b2f47bf66d8eeb6b3f5e3d673f9d5c541cb5b

4,403 BYTES / PRESERVED SOURCE IDENTITY

SECTION 085 — Graphic Design

STATUS: CONTENT-COMPLETE

PURPOSE

Define the design of visual compositions, symbols, illustrations, diagrams, and graphic assets that communicate an authorized message or support a task.

REQUIRED OUTCOME

- Scoped audience, message, medium, cultural context, accessibility, asset rights, production, and lifecycle.
- Approved or proposed graphic rules, assets, provenance, tests, evidence, and handoff.

WHY THIS MATTERS

Graphics can clarify or distort meaning, exclude audiences, or create rights and safety risk. Production quality is not comprehension.

APPLICABILITY

Applies to print, digital, instructional, brand, editorial, data, packaging, environmental, and product graphics.

NON-APPLICABILITY CONDITIONS

No material message, representation, accessibility, asset, rights, or task effect; or an unchanged approved graphic baseline with impact recorded.

PREREQUISITES AND DEPENDENCIES

Message, audience, content, domain, visual system, research, accessibility, localization, asset provenance, rights, production constraints, and acceptance authority.

REQUIRED INPUTS

Message, audience, content, domain, visual system, research, accessibility, localization, asset provenance, rights, production constraints, and acceptance authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve meaning, representation, rights, audience, cultural context, accessibility, publication, and acceptance.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft layouts and assets, but must preserve provenance, licensing, context, and uncertainty. It must not publish or claim audience comprehension without review.

APPLICABLE EXISTING AGENT ROLES

Graphic Designer, Visual Designer, UX Designer, Content Designer, Researcher, Accessibility specialist, Governance Reviewer, Critic, Tester, Verifier, Implementer, and Documentation Writer.

REQUIRED ACTIVITIES

Define message and audience; gather references; create alternatives; check hierarchy, representation, accessibility, rights, production, localization, and use context; test; document; hand off.

HUMAN-AI INTERACTION REQUIREMENTS

Show message, source, audience, actor, status, accessible alternative, and consequence. AI drafts remain proposed and must not obscure warnings or controls.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Control asset identity, vector or raster constraints, color, type, composition, export, metadata, localization, accessibility, rights, versioning, and archival.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show graphic baseline, asset source, license, audience, medium, accessibility, evidence, findings, and controls to approve, replace, withdraw, or restore.

REQUIRED OUTPUTS AND ARTIFACTS

Graphic brief, concepts, final or proposed assets, source and rights register, specifications, accessibility review, research, tests, decisions, and handoff.

EVIDENCE REQUIREMENTS

Identify asset, source, license, audience, medium, version, method, result, and limitation. A polished composition proves bounded appearance only.

VERIFICATION AND ACCEPTANCE TESTS

Verify message comprehension, contrast, scaling, nonvisual alternative, localization, rights, production fidelity, and representative audience response.

FAILURE STATES AND ANTI-PATTERNS

Unlicensed assets; stereotypes; decorative overload; inaccessible text in images; misleading scale or cropping; and publication without provenance.

RECOVERY AND RECONSIDERATION

Withdraw harmful or infringing work, preserve source and decision history, issue an accessible replacement, and retest affected contexts.

COMPLETION CRITERIA

Message, audience, assets, rights, accessibility, evidence, tests, recovery, open issues, and acceptance authority are explicit. Remain CONTENT-COMPLETE pending review and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve asset IDs, sources, rights, versions, findings, decisions, evidence, exceptions, and next action. Hand off to Visual, UI, Content, Localization, Accessibility, and Publishing.

RELATED CODEX SECTIONS

084 Visual Design; 097 Content Design; 101 Localization; 103 Accessibility Design; 223 Intellectual Property; 224 Licensing; 247 Accessibility Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract and shared governance files. Actual message, audience, assets, rights, cultural context, medium, and standards remain unresolved.

086

SECTION 086 - MOTION DESIGN

CODEX/03_SECTIONS/086_MOTION_DESIGN.txt

SHA-256 b0c79c4c782e76e4c8ee67b373215fbfd5f47361a15e8a88b786e3a73ce98aee

4,354 BYTES / PRESERVED SOURCE IDENTITY

SECTION 086 — Motion Design

STATUS: CONTENT-COMPLETE

PURPOSE

Define purposeful movement, transition, timing, feedback, and animation that communicate state and support comprehension without causing harm or blocking control.

REQUIRED OUTCOME

- Scoped motion purpose, audience, device, timing, accessibility, performance, reduced-motion, and lifecycle rules.
- Motion specifications, assets, evidence, tests, controls, and implementation handoff.

WHY THIS MATTERS

Motion can explain state or trigger vestibular, cognitive, performance, and safety problems. Delight is not evidence of benefit.

APPLICABILITY

Applies to interface animation, product behavior, data visualization, identity, marketing, games, physical displays, and feedback cues.

NON-APPLICABILITY CONDITIONS

No material motion, timing, state, accessibility, performance, or user-control effect; or an unchanged motion baseline with impact recorded.

PREREQUISITES AND DEPENDENCIES

Interaction and UI states, content, device performance, accessibility, reduced-motion, timing, research, media rights, and acceptance authority.

REQUIRED INPUTS

Interaction and UI states, content, device performance, accessibility, reduced-motion, timing, research, media rights, and acceptance authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve motion purpose, safety, accessibility, timing, content, publication, and acceptance.

AI RESPONSIBILITIES AND BOUNDARIES

AI may propose transitions or timing and analyze performance. It must expose uncertainty and accessibility implications and must not enable motion that overrides user control.

APPLICABLE EXISTING AGENT ROLES

Motion Designer, UI Designer, Interaction Designer, UX Reviewer, Accessibility specialist, Researcher, Tester, Verifier, Critic, Implementer, Governance Reviewer, and Documentation Writer.

REQUIRED ACTIVITIES

Define motion intent, triggers, duration, easing, hierarchy, feedback, interruption, reduced-motion, performance, fallback, rights, and tests; validate with representative and sensitive users.

HUMAN-AI INTERACTION REQUIREMENTS

Show what moved, why, duration, actor, consequence, pause or skip control, and reduced-motion alternative. AI proposals remain drafts.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Specify state-linked transitions, timing budgets, interruptibility, focus, performance, reduced-motion, offline fallback, and no-motion status alternatives.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show motion baseline, trigger, state, audience, device, accessibility, performance, evidence, and controls to pause, reduce, disable, or restore.

REQUIRED OUTPUTS AND ARTIFACTS

Motion specification, storyboards, timing sheets, assets, reduced-motion rules, performance and accessibility tests, decisions, evidence, and handoff.

EVIDENCE REQUIREMENTS

Identify motion, trigger, device, audience, version, method, result, and limitation. A video or designer review proves bounded appearance.

VERIFICATION AND ACCEPTANCE TESTS

Verify comprehension, timing, interruptibility, reduced-motion, assistive use, performance, focus, localization, loading, and degraded behavior.

FAILURE STATES AND ANTI-PATTERNS

Motion without purpose; seizure or vestibular risk; blocking transitions; no pause; hidden loading; motion as the only status; and performance measured on one device.

RECOVERY AND RECONSIDERATION

Disable or reduce motion, preserve state, provide static status, notify owners, and retest affected populations and devices before restoring.

COMPLETION CRITERIA

Purpose, timing, states, accessibility, controls, evidence, tests, recovery, open risks, and acceptance are explicit. Remain CONTENT-COMPLETE pending review and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve timing, assets, versions, findings, decisions, evidence, accessibility results, and next action. Hand off to UI, Interaction, Visual, Accessibility, Frontend, and Testing.

RELATED CODEX SECTIONS

075 Human-Computer Interaction; 076 Interaction Design; 083 UI Design; 084 Visual Design; 103 Accessibility Design; 247 Accessibility Testing; 323 Performance Management.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract and shared governance files. Actual motion sensitivity, devices, timing, content, performance, and accessibility thresholds remain unresolved.

087

SECTION 087 - SOUND DESIGN

CODEX/03_SECTIONS/087_SOUND_DESIGN.txt

SHA-256 dca5286e9772390f8080dc2937ac3f4360c657b92cd3b177504bd165bc7c5a8c

4,578 BYTES / PRESERVED SOURCE IDENTITY

SECTION 087 — Sound Design

STATUS: CONTENT-COMPLETE

PURPOSE

Define the intentional use of sound, silence, speech, acoustics, and audio feedback to communicate, orient, warn, and support an experience.

REQUIRED OUTCOME

- Scoped listener, environment, message, device, hearing access, cultural context, performance, and lifecycle.
- Audio specifications, assets, provenance, alternatives, tests, controls, evidence, and handoff.

WHY THIS MATTERS

Sound can convey urgency and state but can also exclude, startle, distract, expose private information, or create physical risk. Audio quality is not comprehension.

APPLICABILITY

Applies to interfaces, games, products, vehicles, environments, alerts, media, accessibility cues, and systems where audio affects action or understanding.

NON-APPLICABILITY CONDITIONS

No material audio message, alert, accessibility, privacy, safety, or interaction effect; or an unchanged sound baseline with impact recorded.

PREREQUISITES AND DEPENDENCIES

Interaction and content requirements, audience and environment, devices, acoustics, hearing access, localization, privacy, safety, rights, and acceptance authority.

REQUIRED INPUTS

Interaction and content requirements, audience and environment, devices, acoustics, hearing access, localization, privacy, safety, rights, and acceptance authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve meaning, urgency, audience, accessibility, privacy, rights, safety, publication, and acceptance.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft audio, classify recordings, and propose cues. It must preserve provenance, consent, context, and uncertainty and must not publish or infer comprehension without review.

APPLICABLE EXISTING AGENT ROLES

Sound Designer, Voice Designer, UX and Interaction Designers, Researcher, Accessibility specialist, Content Designer, Tester, Verifier, Critic, Governance Reviewer, Implementer, and Documentation Writer.

REQUIRED ACTIVITIES

Define audio purpose, hierarchy, timing, volume, spatialization, interruption, silence, captions or visual alternative, recording rights, environment, and tests.

HUMAN-AI INTERACTION REQUIREMENTS

Show cue meaning, urgency, actor, privacy, consequence, volume, caption or visual alternative, and user control. AI-generated audio remains proposed.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Specify audio identity, format, loudness, latency, mixing, priority, spatial behavior, accessibility alternative, localization, caching, privacy, and degraded operation.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show sound baseline, cue, environment, device, audience, accessibility, privacy, evidence, and controls to mute, caption, replace, or restore.

REQUIRED OUTPUTS AND ARTIFACTS

Audio brief, cue catalog, assets and rights, specifications, captions or alternatives, accessibility and environmental tests, evidence, decisions, and handoff.

EVIDENCE REQUIREMENTS

Identify recording, source, rights, listener, environment, device, version, method, result, and limitation. A studio sample or loudness metric proves bounded quality only.

VERIFICATION AND ACCEPTANCE TESTS

Verify meaning, audibility, non-audio alternative, privacy, startling risk, localization, latency, mixing, noisy environment, assistive use, and device variation.

FAILURE STATES AND ANTI-PATTERNS

Audio-only critical status; excessive loudness; unlicensed recordings; hidden privacy exposure; culturally misleading cues; and sound that cannot be paused or replaced.

RECOVERY AND RECONSIDERATION

Mute, caption, replace, or disable harmful cues, preserve state and evidence, communicate through alternate channel, and retest affected environments.

COMPLETION CRITERIA

Purpose, cues, audience, alternatives, rights, accessibility, evidence, tests, recovery, open risks, and acceptance are explicit. Remain CONTENT-COMPLETE pending review and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve assets, rights, cue IDs, versions, findings, decisions, evidence, alternatives, and next action. Hand off to Voice, UI, Interaction, Accessibility, Localization, Testing, and Operations.

RELATED CODEX SECTIONS

075 HCI; 076 Interaction Design; 088 Voice Interface Design; 089 Conversational Design; 103 Accessibility Design; 101 Localization; 142 Audio Engineering; 247 Accessibility Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract and shared governance files. Actual listeners, environments, devices, rights, cultural context, accessibility, and safety thresholds remain unresolved.

088

SECTION 088 - VOICE INTERFACE DESIGN

CODEX/03_SECTIONS/088_VOICE_INTERFACE_DESIGN.txt

SHA-256 4279e13f914386ea577ce5ae0b3596271dc527c306a359358fa9299e0fa40b24

5,041 BYTES / PRESERVED SOURCE IDENTITY

SECTION 088 — Voice Interface Design

STATUS: CONTENT-COMPLETE

PURPOSE

Define spoken interaction, input, recognition, prompts, turn-taking, feedback, confirmation, privacy, and recovery for voice-enabled systems.

REQUIRED OUTCOME

- A scoped speaker, listener, context, language, device, task, authority, privacy, and lifecycle model.
- Voice flows, prompts, intents, disambiguation, fallback, evidence, accessibility, testing, and acceptance records.

WHY THIS MATTERS

Voice is affected by noise, accent, language, privacy, interruption, and ambiguity. Fluent recognition does not prove correct intent or safe action.

APPLICABILITY

Applies to voice assistants, IVR, hands-free controls, accessibility tools, vehicles, devices, services, and conversational products with spoken input or output.

NON-APPLICABILITY CONDITIONS

No voice input, output, command, privacy, accessibility, or task effect; or an unchanged approved voice baseline with impact recorded.

PREREQUISITES AND DEPENDENCIES

Tasks, users, languages, accents, devices, acoustic context, speech and intent models, content, identity, permissions, privacy, security, fallback, and acceptance authority.

REQUIRED INPUTS

Tasks, users, languages, accents, devices, acoustic context, speech and intent models, content, identity, permissions, privacy, security, fallback, and acceptance authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve commands, permissions, disclosures, confirmations, recording, accessibility, language, risk, and acceptance. They retain stop, correction, appeal, and human-support authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft prompts, intents, tests, and summaries. It must preserve transcript, speaker, language, confidence, and privacy context; it must not infer intent or execute consequential actions from uncertain speech.

APPLICABLE EXISTING AGENT ROLES

Voice Interface Designer, Conversational Designer, UX Reviewer, Researcher, Accessibility specialist, Security and Privacy Auditors, Architect, Tester, Verifier, Critic, Governance Reviewer, Implementer, and Documentation Writer.

REQUIRED ACTIVITIES

Define tasks, invocation, turn-taking, prompts, recognition, intent, confirmation, interruption, errors, fallback, privacy, language, accessibility, and safe action; test with representative speakers and environments.

HUMAN-AI INTERACTION REQUIREMENTS

Show transcript or interpretation, confidence, actor, permission, consequence, confirmation, recording state, and recovery. AI proposals remain advisory; keyboard, visual, and human alternatives remain available.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Separate audio capture, recognition, intent, policy, confirmation, execution, response, logging, and recovery. Enforce identity and authorization independently of recognition confidence.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show voice session, speaker, language, device, transcript, confidence, permission, action queue, recording, evidence, and controls to cancel, mute, correct, escalate, or recover.

REQUIRED OUTPUTS AND ARTIFACTS

Voice architecture, intents and utterances, prompt and dialogue specs, language and accessibility coverage, privacy and permission model, test evidence, fallback, decisions, and handoff.

EVIDENCE REQUIREMENTS

Identify speaker population, language, acoustic context, device, model, version, transcript, method, result, uncertainty, and limitation. Recognition accuracy is not task or safety acceptance.

VERIFICATION AND ACCEPTANCE TESTS

Verify accents, noise, silence, interruption, ambiguity, spoofing, permissions, confirmation, privacy, accessibility, localization, fallback, latency, and cancellation.

FAILURE STATES AND ANTI-PATTERNS

Executing on low confidence; recording without clear control; no transcript or correction; language exclusion; voice-only critical action; and treating recognition score as intent truth.

RECOVERY AND RECONSIDERATION

Cancel or suspend action, preserve transcript and evidence, mute or revoke capture, use alternate input or human support, and reassess affected speakers before resuming.

COMPLETION CRITERIA

Tasks, speakers, languages, authority, privacy, dialogue, evidence, tests, fallback, recovery, open risks, and acceptance are explicit. Remain CONTENT-COMPLETE pending review and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve dialogue versions, prompts, models, transcripts, permissions, findings, decisions, evidence, and next action. Hand off to Conversational, Audio, Speech, Accessibility, Localization, Security, Privacy, and Testing.

RELATED CODEX SECTIONS

075 HCI; 087 Sound Design; 089 Conversational Design; 101 Localization; 103 Accessibility Design; 173 Speech Engineering; 186 Prompt Engineering; 247 Accessibility Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract and shared governance files. Actual languages, speaker populations, recording rules, speech models, device environments, and acceptance thresholds remain unresolved.

089

SECTION 089 - CONVERSATIONAL DESIGN

CODEX/03_SECTIONS/089_CONVERSATIONAL_DESIGN.txt

SHA-256 ac0b1dc460b7ee0ac3b95266a9876d2ca3b970f85509d58e20d7ce12b6df1e0a

4,700 BYTES / PRESERVED SOURCE IDENTITY

SECTION 089 — Conversational Design

STATUS: CONTENT-COMPLETE

PURPOSE

Define dialogue, turn-taking, intent, context, repair, escalation, and conversational content for systems that interact through language.

REQUIRED OUTCOME

- Scoped participants, goals, channels, languages, authority, privacy, states, fallback, and lifecycle.
- Dialogue models, content, evidence, tests, controls, and acceptance criteria.

WHY THIS MATTERS

Conversation can create false trust and hide system limits. A fluent response is not understanding, authorization, or a successful outcome.

APPLICABILITY

Applies to chat, assistants, agents, messaging, support, voice transcripts, and mixed human-machine dialogue.

NON-APPLICABILITY CONDITIONS

No material dialogue, intent, disclosure, action, accessibility, privacy, or recovery effect; or an unchanged approved dialogue baseline with impact recorded.

PREREQUISITES AND DEPENDENCIES

User research, tasks, domain language, policies, content, models, tools, identity, permissions, privacy, safety, escalation, and acceptance authority.

REQUIRED INPUTS

User research, tasks, domain language, policies, content, models, tools, identity, permissions, privacy, safety, escalation, and acceptance authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve goals, persona, disclosure, authority, tone, safety, privacy, escalation, and acceptance. They retain correction, stop, appeal, and human-support authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft dialogue and analyze transcripts. It must preserve context, uncertainty, consent, and provenance and must not invent policy, claim understanding, or execute beyond authority.

APPLICABLE EXISTING AGENT ROLES

Conversational Designer, Voice Interface Designer, UX Reviewer, Researcher, Content Designer, Security and Privacy Auditors, Governance Reviewer, Tester, Verifier, Critic, Implementer, and Documentation Writer.

REQUIRED ACTIVITIES

Model goals, turns, intents, context, confirmation, ambiguity, repair, error, handoff, privacy, language, accessibility, tool use, and recovery; prototype and test representative and adversarial conversations.

HUMAN-AI INTERACTION REQUIREMENTS

Show speaker, context, intent, confidence, tool, authority, consequence, recording, and recovery. AI-generated responses remain provisional; users can correct, exit, and reach humans.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Separate dialogue state, policy, retrieval, generation, tool execution, identity, audit, and human handoff. Enforce authority outside language interpretation and preserve terminal state.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show conversation identity, state, speaker, tool calls, permissions, evidence, uncertainty, open failures, and controls to stop, clear, correct, escalate, or recover.

REQUIRED OUTPUTS AND ARTIFACTS

Dialogue map, intents, prompts, content rules, state model, escalation paths, privacy and authority model, test transcripts, evidence, decisions, and handoff.

EVIDENCE REQUIREMENTS

Identify speaker, context, language, model, prompt, tools, version, transcript, method, result, uncertainty, and limitation. A successful chat is bounded evidence.

VERIFICATION AND ACCEPTANCE TESTS

Verify ambiguity, refusal, prompt injection, privacy, identity, permissions, tool errors, long turns, context loss, accessibility, localization, handoff, and recovery.

FAILURE STATES AND ANTI-PATTERNS

Anthropomorphic claims; endless loops; hidden tool actions; no exit; invented sources or policy; confident ambiguity; recording without control; and treating engagement as resolution.

RECOVERY AND RECONSIDERATION

Stop tool actions, preserve transcript and audit, revoke access, provide human or nonconversational fallback, and reassess affected users before resuming.

COMPLETION CRITERIA

Goals, participants, states, authority, content, evidence, tests, fallback, recovery, open risks, and acceptance are explicit. Remain CONTENT-COMPLETE pending review and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve dialogue versions, prompts, transcripts, tool and policy identities, decisions, evidence, incidents, and next action. Hand off to Voice, AI, Prompt, Security, Privacy, Content, Testing, and Operations.

RELATED CODEX SECTIONS

075 HCI; 088 Voice Interface Design; 097 Content Design; 178 Agent Systems Engineering; 186 Prompt Engineering; 188 Human-in-the-Loop Operations; 182 AI Safety; 249 Usability Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract and shared governance files. Actual audiences, languages, models, tools, authority, privacy, escalation, and acceptance thresholds remain unresolved.

090

SECTION 090 - SPATIAL INTERFACE DESIGN

CODEX/03_SECTIONS/090_SPATIAL_INTERFACE_DESIGN.txt

SHA-256 a7fa162737b7711c27283db898bd54582d67292f973a304186c262d6a499ac0e

4,686 BYTES / PRESERVED SOURCE IDENTITY

SECTION 090 — Spatial Interface Design

STATUS: CONTENT-COMPLETE

PURPOSE

Define interaction in physical or three-dimensional space, including placement, scale, orientation, movement, embodiment, context, and safety.

REQUIRED OUTCOME

- Scoped space, users, devices, environment, tasks, tracking, content, accessibility, privacy, and lifecycle.
- Spatial behavior, anchors, controls, evidence, tests, safety, and recovery records.

WHY THIS MATTERS

Spatial interfaces affect orientation, attention, physical movement, and shared environments. Immersion or novelty does not prove comprehension or safety.

APPLICABILITY

Applies to 3D interfaces, room-scale systems, tangible interaction, spatial audio, robotics interfaces, and mixed physical-digital environments.

NON-APPLICABILITY CONDITIONS

No spatial placement, tracking, movement, safety, accessibility, privacy, or user-control effect; or an unchanged approved spatial baseline with impact recorded.

PREREQUISITES AND DEPENDENCIES

User and environment research, device and tracking limits, task model, physical safety, accessibility, content, privacy, identity, performance, and acceptance authority.

REQUIRED INPUTS

User and environment research, device and tracking limits, task model, physical safety, accessibility, content, privacy, identity, performance, and acceptance authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve spatial purpose, safety zones, representation, accessibility, privacy, and acceptance. They retain stop, physical override, and incident authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may propose spatial layouts or analyze tracking. It must expose uncertainty and environmental limits and must not place content or command physical action without authority.

APPLICABLE EXISTING AGENT ROLES

Spatial Designer, UX and Interaction Designers, Architect, Researcher, Robotics and Safety specialists, Accessibility specialist, Tester, Verifier, Critic, Governance Reviewer, Implementer, and Documentation Writer.

REQUIRED ACTIVITIES

Define space, anchors, scale, orientation, gestures, devices, environment, shared ownership, accessibility, privacy, performance, safety, failure, and recovery; prototype and test in representative spaces.

HUMAN-AI INTERACTION REQUIREMENTS

Show spatial state, tracking confidence, actor, boundaries, automation, consequence, privacy, and safe exit. AI placements remain proposals and physical stop paths remain independent.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Define coordinate frames, anchors, occlusion, latency, calibration, collision and boundary handling, shared space, identity, data, accessibility, and degraded tracking behavior.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show space, device, user, tracking, boundaries, hazards, evidence, and controls to pause, clear, recalibrate, exit, stop, or recover.

REQUIRED OUTPUTS AND ARTIFACTS

Spatial model, interaction specification, environment and safety map, tracking and calibration record, prototypes, accessibility tests, evidence, decisions, and handoff.

EVIDENCE REQUIREMENTS

Identify space, device, tracking, calibration, user, task, version, method, result, and limitation. A virtual demo proves bounded behavior only.

VERIFICATION AND ACCEPTANCE TESTS

Verify tracking loss, latency, scale, accessibility, shared space, privacy, collision boundaries, fatigue, motion sensitivity, calibration, interruption, and safe exit.

FAILURE STATES AND ANTI-PATTERNS

Ignoring physical context; stale anchors; inaccessible gestures; hidden boundaries; tracking confidence treated as truth; and immersion that prevents stop or orientation.

RECOVERY AND RECONSIDERATION

Pause or remove spatial content, provide visual or non-spatial fallback, preserve state and evidence, recalibrate or isolate equipment, and retest the environment.

COMPLETION CRITERIA

Space, tasks, devices, tracking, safety, accessibility, evidence, tests, recovery, open risks, and acceptance are explicit. Remain CONTENT-COMPLETE pending review and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve maps, anchors, calibration, versions, findings, decisions, evidence, incidents, and next action. Hand off to AR, VR, XR, Robotics, HCI, Accessibility, Safety, and Testing.

RELATED CODEX SECTIONS

074 Robotics Engineering; 075 HCI; 090 Spatial Interface Design; 091 AR Design; 092 VR Design; 093 XR Design; 103 Accessibility Design; 247 Accessibility Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract and shared governance files. Actual spaces, devices, tracking, physical hazards, populations, privacy, and accessibility thresholds remain unresolved.

091

SECTION 091 - AR DESIGN

CODEX/03_SECTIONS/091_AR_DESIGN.txt

SHA-256 e3eea90a2f8947c93a48d88a6288213284537eb354bb91dbc46c78f453647cd9

4,857 BYTES / PRESERVED SOURCE IDENTITY

SECTION 091 — AR Design

STATUS: CONTENT-COMPLETE

PURPOSE

Define augmented-reality experiences that place digital information or controls in relation to the physical world while preserving comprehension, safety, privacy, and user control.

REQUIRED OUTCOME

- A scoped user, physical environment, device, tracking, content, task, interaction, safety, privacy, and lifecycle model.
- AR behavior, anchors, occlusion, accessibility, evidence, tests, fallback, and acceptance records.

WHY THIS MATTERS

AR can obscure real hazards, misalign information, expose surroundings, and overload attention. Alignment and novelty are not proof of safe use.

APPLICABILITY

Applies to headsets, mobile AR, industrial overlays, navigation, training, maintenance, medical, retail, and collaborative physical-digital experiences.

NON-APPLICABILITY CONDITIONS

No overlay, tracking, physical task, privacy, safety, accessibility, or user-control effect; or an unchanged approved AR baseline with impact recorded.

PREREQUISITES AND DEPENDENCIES

Environment and user research, device and tracking limits, spatial model, content, task, physical safety, privacy, identity, accessibility, performance, and acceptance authority.

REQUIRED INPUTS

Environment and user research, device and tracking limits, spatial model, content, task, physical safety, privacy, identity, accessibility, performance, and acceptance authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve overlay purpose, safety, privacy, representation, accessibility, physical boundaries, and acceptance. They retain emergency stop and physical control authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may generate overlays or analyze spatial context. It must disclose tracking and recognition uncertainty and must not place consequential guidance or command action without authority.

APPLICABLE EXISTING AGENT ROLES

AR Designer, Spatial Designer, UX and Interaction Designers, Robotics and Safety specialists, Researcher, Accessibility specialist, Security and Privacy Auditors, Tester, Verifier, Critic, Implementer, and Governance Reviewer.

REQUIRED ACTIVITIES

Define context, anchors, recognition, overlay, occlusion, field of view, input, warnings, accessibility, privacy, safety, failure, and recovery; prototype and test with representative environments and users.

HUMAN-AI INTERACTION REQUIREMENTS

Show physical context, tracking confidence, overlay source, actor, authority, hazard, uncertainty, and safe exit. AI-generated content remains proposed; visual and physical fallbacks remain available.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Define coordinate frames, anchors, calibration, latency, occlusion, depth, data capture, identity, permissions, safe rendering, and degraded or no-tracking behavior.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show environment, device, overlay, confidence, hazards, privacy, evidence, and controls to hide, pause, recalibrate, exit, stop, or recover.

REQUIRED OUTPUTS AND ARTIFACTS

AR concept and behavior model, environment map, content and anchor specifications, safety and privacy assessment, accessibility tests, prototypes, evidence, decisions, and handoff.

EVIDENCE REQUIREMENTS

Identify environment, device, tracking, content, user, task, version, method, result, uncertainty, and limitation. A lab overlay proves bounded conditions only.

VERIFICATION AND ACCEPTANCE TESTS

Verify alignment, latency, occlusion, tracking loss, physical hazard visibility, privacy, accessibility, motion sensitivity, input, shared space, fallback, and safe exit.

FAILURE STATES AND ANTI-PATTERNS

Overlay hiding hazards; stale or false anchors; attention capture; unannounced recording; inaccessible controls; and recognizing a physical object as permission to act.

RECOVERY AND RECONSIDERATION

Hide overlays, return control to physical environment, preserve evidence, stop related actions, notify users, and recalibrate or fall back before resuming.

COMPLETION CRITERIA

Environment, device, overlays, authority, safety, privacy, accessibility, evidence, tests, recovery, open risks, and acceptance are explicit. Remain CONTENT-COMPLETE pending review and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve maps, anchors, content, versions, calibration, findings, decisions, evidence, and next action. Hand off to Spatial, VR, XR, HCI, Robotics, Safety, Privacy, Accessibility, and Testing.

RELATED CODEX SECTIONS

090 Spatial Interface Design; 092 VR Design; 093 XR Design; 103 Accessibility Design; 182 AI Safety; 218 Privacy Engineering; 247 Accessibility Testing; 252 Safety Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract and shared governance files. Actual environments, devices, tracking, physical hazards, data capture, populations, and acceptance thresholds remain unresolved.

092

SECTION 092 - VR DESIGN

CODEX/03_SECTIONS/092_VR_DESIGN.txt

SHA-256 1535cb8861b7564102b5d0ab1913f1edbd58776b11769206fb99efc828c561de

4,810 BYTES / PRESERVED SOURCE IDENTITY

SECTION 092 — VR Design

STATUS: CONTENT-COMPLETE

PURPOSE

Define virtual-reality experiences and interactions that create an artificial spatial environment while preserving orientation, consent, accessibility, safety, privacy, and exit.

REQUIRED OUTCOME

- A scoped user, virtual environment, device, locomotion, embodiment, task, content, safety, privacy, and lifecycle model.
- VR interaction, comfort, accessibility, evidence, tests, fallback, and acceptance records.

WHY THIS MATTERS

VR can affect balance, perception, identity, motion comfort, and awareness of real surroundings. Immersion must never remove informed control or safe exit.

APPLICABILITY

Applies to headsets, room-scale systems, simulation, training, games, therapy, design review, social VR, and virtual workspaces.

NON-APPLICABILITY CONDITIONS

No immersive, locomotion, embodiment, physical, privacy, accessibility, or user-control effect; or an unchanged approved VR baseline with impact recorded.

PREREQUISITES AND DEPENDENCIES

User and environment research, device limits, virtual world model, locomotion, content, physical safety, accessibility, privacy, identity, performance, and acceptance authority.

REQUIRED INPUTS

User and environment research, device limits, virtual world model, locomotion, content, physical safety, accessibility, privacy, identity, performance, and acceptance authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve purpose, content, age or population bounds, comfort, consent, safety, privacy, accessibility, and acceptance. They retain immediate exit and physical supervision authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may generate worlds, avatars, and scenarios, but must disclose generation and uncertainty and must not manipulate, surveil, or create consequential effects without authority.

APPLICABLE EXISTING AGENT ROLES

VR Designer, Spatial Designer, UX and Interaction Designers, Researcher, Accessibility and Safety specialists, Privacy Auditor, Tester, Verifier, Critic, Implementer, Governance Reviewer, and Documentation Writer.

REQUIRED ACTIVITIES

Define world, embodiment, locomotion, interaction, comfort, boundary, content, consent, privacy, accessibility, interruption, safe exit, and recovery; prototype and test representative users and devices.

HUMAN-AI INTERACTION REQUIREMENTS

Show virtual and physical state, user identity, tracking, comfort, content source, consequence, privacy, and exit. AI-generated scenes remain proposed and supervision remains available.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Define coordinate and boundary systems, tracking, latency, locomotion, haptics, embodiment, data, identity, moderation, performance, comfort settings, and safe exit.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show session, user, device, world, tracking, comfort, boundaries, incidents, evidence, and controls to pause, recenter, mute, exit, supervise, or recover.

REQUIRED OUTPUTS AND ARTIFACTS

VR concept, world and interaction model, comfort and safety rules, content and avatar provenance, privacy and moderation model, tests, evidence, decisions, and handoff.

EVIDENCE REQUIREMENTS

Identify user, device, world, version, locomotion, environment, method, result, discomfort or safety signal, and limitation. A demo is bounded evidence.

VERIFICATION AND ACCEPTANCE TESTS

Verify tracking, latency, comfort, motion sickness, boundaries, safe exit, accessibility, privacy, harassment or abuse controls, content provenance, and device variation.

FAILURE STATES AND ANTI-PATTERNS

No exit; forced locomotion; hidden recording; deceptive embodiment; inaccessible supervision; blurred real-world hazards; and immersive content treated as harmless by default.

RECOVERY AND RECONSIDERATION

Pause and exit, preserve session evidence, provide physical supervision and alternate mode, disable harmful content, and review affected users before resuming.

COMPLETION CRITERIA

Purpose, world, users, devices, comfort, safety, privacy, accessibility, evidence, tests, recovery, open risks, and acceptance are explicit. Remain CONTENT-COMPLETE pending review and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve world, content, avatars, configurations, sessions, findings, incidents, decisions, evidence, and next action. Hand off to Spatial, AR, XR, HCI, Safety, Privacy, Accessibility, and Testing.

RELATED CODEX SECTIONS

090 Spatial Interface Design; 091 AR Design; 093 XR Design; 094 Game Design; 103 Accessibility Design; 182 AI Safety; 218 Privacy Engineering; 252 Safety Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract and shared governance files. Actual users, devices, content, physical environment, comfort, age, privacy, moderation, and acceptance thresholds remain unresolved.

093

SECTION 093 - XR DESIGN

CODEX/03_SECTIONS/093_XR_DESIGN.txt

SHA-256 3b89adf545489ee0daf694128debe8d55005f4a2b5fc8d68193fd95714369f1f

4,609 BYTES / PRESERVED SOURCE IDENTITY

SECTION 093 — XR Design

STATUS: CONTENT-COMPLETE

PURPOSE

Define extended-reality experiences spanning augmented, virtual, mixed, and emerging spatial modalities while preserving user agency, safety, accessibility, privacy, and continuity.

REQUIRED OUTCOME

- A bounded XR population, environment, device, modality, content, interaction, tracking, safety, privacy, and lifecycle scope.
- Cross-modal behavior, evidence, tests, fallback, controls, and acceptance records.

WHY THIS MATTERS

XR combines physical and synthetic context and can fail through tracking, embodiment, attention, identity, or environment mismatch. A compelling experience is not proof of safe or inclusive use.

APPLICABILITY

Applies to systems combining AR, VR, spatial, embodied, immersive, wearable, or collaborative interfaces.

NON-APPLICABILITY CONDITIONS

No modality, spatial, physical, privacy, accessibility, or user-control effect; or an unchanged approved XR baseline with impact recorded.

PREREQUISITES AND DEPENDENCIES

Modalities, users, environments, devices, spatial models, content, interaction, research, physical safety, privacy, identity, accessibility, performance, and authority.

REQUIRED INPUTS

Modalities, users, environments, devices, spatial models, content, interaction, research, physical safety, privacy, identity, accessibility, performance, and authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve modality, content, safety, consent, privacy, accessibility, representation, and acceptance and retain physical stop and exit authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may generate or adapt XR content and analyze context. It must disclose uncertainty and generation and must not manipulate users or act physically without authority.

APPLICABLE EXISTING AGENT ROLES

XR Designer, Spatial, AR, VR, UX, Interaction, Research, Safety, Privacy, Accessibility, Security, Testing, Verification, Governance, Implementation, and Documentation roles.

REQUIRED ACTIVITIES

Define modalities, transitions, environment, tracking, content, embodiment, input, accessibility, privacy, safety, performance, moderation, fallback, and recovery; prototype and test cross-device and cross-modal paths.

HUMAN-AI INTERACTION REQUIREMENTS

Show modality, physical and virtual state, tracking, actor, authority, content source, uncertainty, hazards, and exit. AI adaptations remain proposed and nonimmersive fallback remains available.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Define modality contracts, coordinate systems, identity, data, rendering, latency, calibration, boundaries, shared space, accessibility, and graceful degradation.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show XR session, modality, user, device, environment, hazards, evidence, and controls to pause, switch mode, exit, supervise, or recover.

REQUIRED OUTPUTS AND ARTIFACTS

XR architecture, modality and transition model, spatial and content specifications, safety, privacy, accessibility, moderation, tests, evidence, decisions, and handoff.

EVIDENCE REQUIREMENTS

Identify user, modality, device, environment, content, version, method, result, uncertainty, and limitation. A demo proves bounded conditions only.

VERIFICATION AND ACCEPTANCE TESTS

Verify transitions, tracking, latency, comfort, accessibility, privacy, safety, shared space, content provenance, moderation, fallback, and exit.

FAILURE STATES AND ANTI-PATTERNS

Treating immersion as safety; hiding mode changes; no physical awareness; inaccessible transitions; unbounded generated content; and one-device evidence.

RECOVERY AND RECONSIDERATION

Pause, exit, switch modality, preserve evidence, provide supervision, and isolate harmful content or equipment before resuming.

COMPLETION CRITERIA

Modalities, users, devices, safety, privacy, accessibility, evidence, tests, recovery, risks, and acceptance are explicit. Remain CONTENT-COMPLETE pending review and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve session and content identity, device and calibration, findings, incidents, decisions, evidence, and next action. Hand off to AR, VR, Spatial, HCI, Safety, Privacy, Accessibility, and Testing.

RELATED CODEX SECTIONS

090-092 Spatial, AR, and VR Design; 094 Game Design; 103 Accessibility Design; 182 AI Safety; 218 Privacy Engineering; 247 Accessibility Testing; 252 Safety Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract and shared governance files. Actual XR modalities, populations, devices, environments, content, safety, privacy, and acceptance thresholds remain unresolved.

094

SECTION 094 - GAME DESIGN

CODEX/03_SECTIONS/094_GAME_DESIGN.txt

SHA-256 e0701c551e456c0ec4674351aba378759e23158b1976fbe25766cec96af91ced

4,808 BYTES / PRESERVED SOURCE IDENTITY

SECTION 094 — Game Design

STATUS: CONTENT-COMPLETE

PURPOSE

Define rules, goals, systems, player agency, feedback, challenge, progression, content, safety, accessibility, and fairness for games and game-like experiences.

REQUIRED OUTCOME

- A declared player population, context, platform, mechanics, content, economy, social model, moderation, and lifecycle.
- Coherent rule and content specifications with evidence, tests, safeguards, and acceptance criteria.

WHY THIS MATTERS

Games shape behavior through rules, rewards, uncertainty, and social systems. Engagement or retention is not proof of wellbeing, fairness, or meaningful outcome.

APPLICABILITY

Applies to digital, physical, educational, serious, social, competitive, casino-like, and interactive game systems.

NON-APPLICABILITY CONDITIONS

No game rule, player behavior, content, reward, social, safety, accessibility, privacy, or monetization effect; or an unchanged approved game baseline with impact recorded.

PREREQUISITES AND DEPENDENCIES

Goals, player research, rules, mechanics, content, platform, economy, identity, privacy, safety, accessibility, moderation, monetization, analytics, and authority.

REQUIRED INPUTS

Goals, player research, rules, mechanics, content, platform, economy, identity, privacy, safety, accessibility, moderation, monetization, analytics, and authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve purpose, audience, content, age bounds, fairness, monetization, safety, accessibility, moderation, and acceptance. They retain pause, refund, appeal, and retirement authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may generate content, balance proposals, and tests. It must disclose generation and preserve player safety and privacy; it must not manipulate, target vulnerability, or change live rules without authority.

APPLICABLE EXISTING AGENT ROLES

Game Designer, Level Designer, Narrative Designer, UX Reviewer, Researcher, Content Designer, Safety and Privacy Auditors, Tester, Verifier, Critic, Governance Reviewer, Implementer, and Documentation Writer.

REQUIRED ACTIVITIES

Define goals, rules, mechanics, progression, feedback, balance, content, accessibility, social interaction, moderation, economy, safety, analytics, failure, and recovery; prototype and test.

HUMAN-AI INTERACTION REQUIREMENTS

Show rule, player state, automation, consequence, probability, content source, permissions, spending, moderation, and exit. AI content remains reviewable and player controls remain available.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Separate rules, state, content, economy, identity, matchmaking, moderation, telemetry, monetization, and lifecycle. Make randomness, consent, limits, and recovery inspectable.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show game and player state, content version, economy, moderation, evidence, risks, and controls to pause, restrict, report, refund, appeal, or retire.

REQUIRED OUTPUTS AND ARTIFACTS

Game design document, rules and economy model, player journeys, content and moderation plan, accessibility and safety assessment, tests, evidence, decisions, and handoff.

EVIDENCE REQUIREMENTS

Identify player population, platform, rules, version, context, method, result, uncertainty, and limitation. Engagement, playtime, or win rate is bounded evidence.

VERIFICATION AND ACCEPTANCE TESTS

Verify rules, balance, accessibility, privacy, spending limits, randomness, exploits, moderation, safety, recovery, platform compatibility, and representative player outcomes.

FAILURE STATES AND ANTI-PATTERNS

Dark patterns; targeting vulnerable players; opaque odds; pay-to-progress pressure; inaccessible controls; unmoderated abuse; and optimizing retention over wellbeing or fairness.

RECOVERY AND RECONSIDERATION

Pause or disable harmful mechanics, protect players, preserve records, refund or compensate where authorized, moderate, and review affected populations before resuming.

COMPLETION CRITERIA

Goals, rules, players, content, economy, safety, accessibility, evidence, tests, recovery, open risks, and acceptance are explicit. Remain CONTENT-COMPLETE pending review and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve rules, content, builds, economy, moderation, decisions, evidence, incidents, and next action. Hand off to Level, Narrative, UI, UX, Audio, Game Engineering, Safety, and Operations.

RELATED CODEX SECTIONS

075 HCI; 094 Game Design; 095 Level Design; 096 Narrative Design; 138 Game Engineering; 247 Accessibility Testing; 216 Abuse Prevention; 358 Commerce.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract and shared governance files. Actual players, platform, age, content, economy, moderation, monetization, safety, and acceptance requirements remain unresolved.

095

SECTION 095 - LEVEL DESIGN

CODEX/03_SECTIONS/095_LEVEL_DESIGN.txt

SHA-256 7c9827b67d1d56bbdae14347f05dca00b9556117103ce2ab80cb091ed37b5c1d

4,541 BYTES / PRESERVED SOURCE IDENTITY

SECTION 095 — Level Design

STATUS: CONTENT-COMPLETE

PURPOSE

Define spaces, encounters, pacing, navigation, challenge, affordances, resources, and progression that support an intended player or participant experience.

REQUIRED OUTCOME

- A scoped space, player, objective, mechanics, content, difficulty, accessibility, platform, and lifecycle model.
- Level layouts and rules with evidence, tests, safety, recovery, and acceptance criteria.

WHY THIS MATTERS

Space teaches rules and shapes behavior. Poor pacing, navigation, difficulty, or accessibility can make the intended experience impossible or unfair.

APPLICABILITY

Applies to game levels, simulations, training spaces, virtual worlds, educational environments, and interactive physical or spatial scenes.

NON-APPLICABILITY CONDITIONS

No spatial, navigation, challenge, content, accessibility, safety, or progression effect; or an unchanged approved level baseline with impact recorded.

PREREQUISITES AND DEPENDENCIES

Game or simulation rules, player research, objectives, mechanics, narrative, platform, accessibility, content, performance, safety, and acceptance authority.

REQUIRED INPUTS

Game or simulation rules, player research, objectives, mechanics, narrative, platform, accessibility, content, performance, safety, and acceptance authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve objectives, difficulty, audience, representation, safety, accessibility, progression, and acceptance.

AI RESPONSIBILITIES AND BOUNDARIES

AI may generate layouts and analyze telemetry, but must disclose generation and preserve player agency. It must not alter live difficulty or target vulnerabilities without authority.

APPLICABLE EXISTING AGENT ROLES

Level Designer, Game Designer, Narrative Designer, UX Reviewer, Researcher, Accessibility specialist, Tester, Verifier, Critic, Governance Reviewer, Implementer, and Documentation Writer.

REQUIRED ACTIVITIES

Define goals, geometry, navigation, affordances, encounters, pacing, resources, difficulty, checkpoints, accessibility, hazards, feedback, and recovery; prototype and test representative play.

HUMAN-AI INTERACTION REQUIREMENTS

Show objective, location, state, challenge, risk, automation, accessibility, and recovery. AI layouts remain proposals and player exit or assist controls remain visible.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Connect geometry, rules, navigation, triggers, state, assets, performance, checkpoints, telemetry, accessibility, and failure recovery. Avoid hidden progression dependencies.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show level version, player state, objective, difficulty, accessibility, performance, evidence, defects, and controls to pause, assist, skip, reset, or report.

REQUIRED OUTPUTS AND ARTIFACTS

Level brief, maps, encounter and pacing model, asset and trigger specification, accessibility notes, playtest evidence, decisions, defect register, and handoff.

EVIDENCE REQUIREMENTS

Identify player, level, platform, version, task, method, result, and limitation. Completion or average time is bounded evidence.

VERIFICATION AND ACCEPTANCE TESTS

Verify navigation, pacing, difficulty, checkpoints, exploits, accessibility, performance, ambiguity, safety, recovery, and representative player outcomes.

FAILURE STATES AND ANTI-PATTERNS

Unsignposted hazards; impossible navigation; difficulty spikes; inaccessible checkpoints; invisible dependencies; and telemetry optimized for friction rather than learning or enjoyment.

RECOVERY AND RECONSIDERATION

Offer assist, skip, reset, safe exit, and support; preserve progress and evidence; correct the level; and retest affected populations.

COMPLETION CRITERIA

Goals, space, mechanics, players, difficulty, accessibility, evidence, tests, recovery, risks, and acceptance are explicit. Remain CONTENT-COMPLETE pending review and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve maps, triggers, assets, versions, playtests, findings, decisions, evidence, and next action. Hand off to Game, Narrative, UX, Accessibility, Audio, Engineering, and Testing.

RELATED CODEX SECTIONS

094 Game Design; 096 Narrative Design; 090 Spatial Interface Design; 103 Accessibility Design; 138 Game Engineering; 247 Accessibility Testing; 239 Performance Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract and shared governance files. Actual platform, player population, mechanics, difficulty, accessibility, safety, and acceptance thresholds remain unresolved.

096

SECTION 096 - NARRATIVE DESIGN

CODEX/03_SECTIONS/096_NARRATIVE_DESIGN.txt

SHA-256 84a355e05ff569024ed132ac96774db5259825848db0836e12682240f4ea4cf2

4,876 BYTES / PRESERVED SOURCE IDENTITY

SECTION 096 — Narrative Design

STATUS: CONTENT-COMPLETE

PURPOSE

Define story, character, world, theme, dialogue, sequence, choice, consequence, and continuity in support of an intended experience or communication outcome.

REQUIRED OUTCOME

- A scoped audience, purpose, world, characters, content, medium, choice model, representation, sensitivity, and lifecycle.
- Coherent narrative structures with provenance, review, evidence, safety, and acceptance records.

WHY THIS MATTERS

Narrative shapes interpretation, identity, empathy, and behavior. Coherence or emotional impact does not excuse deception, exclusion, or unreviewed harm.

APPLICABILITY

Applies to games, interactive media, products, education, simulations, marketing, immersive environments, and conversational or branching narratives.

NON-APPLICABILITY CONDITIONS

No story, character, representation, choice, emotional, cultural, safety, or content effect; or an unchanged approved narrative baseline with impact recorded.

PREREQUISITES AND DEPENDENCIES

Purpose, audience research, world and domain context, character and content rules, medium, localization, sensitivity, accessibility, rights, and acceptance authority.

REQUIRED INPUTS

Purpose, audience research, world and domain context, character and content rules, medium, localization, sensitivity, accessibility, rights, and acceptance authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve theme, representation, audience, content boundaries, rights, sensitivity, and acceptance. They resolve value conflicts and retain publication and correction authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may brainstorm, draft, translate, and check continuity. It must disclose generated content and preserve source, cultural context, and uncertainty; it must not invent lived experience or publish sensitive content without review.

APPLICABLE EXISTING AGENT ROLES

Narrative Designer, Game and Level Designers, Content Designer, Researcher, Cultural or domain reviewers, UX Reviewer, Accessibility specialist, Critic, Governance Reviewer, Tester, Verifier, Implementer, and Documentation Writer.

REQUIRED ACTIVITIES

Define premise, audience, world, characters, arcs, dialogue, branching, consequence, pacing, representation, content warnings, accessibility, localization, rights, and review; test and revise.

HUMAN-AI INTERACTION REQUIREMENTS

Show source, author, character, fiction status, choice, consequence, sensitivity, and exit. AI-generated narrative remains proposed and users can skip, report, or reach support.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Maintain story graph, canon, state, branching, content metadata, provenance, localization, moderation, accessibility, and safe interruption. Separate fiction from factual claims.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show narrative version, audience, state, content warnings, representation review, evidence, open issues, and controls to approve, edit, withdraw, skip, or report.

REQUIRED OUTPUTS AND ARTIFACTS

Narrative bible, story graph, character and world records, dialogue, content warnings, rights and provenance, sensitivity and accessibility review, tests, decisions, and handoff.

EVIDENCE REQUIREMENTS

Identify source, author, audience, medium, version, cultural context, review method, result, and limitation. Emotional response or continuity score is bounded evidence.

VERIFICATION AND ACCEPTANCE TESTS

Verify continuity, choice consequences, representation, content warnings, accessibility, localization, age suitability, rights, moderation, and player or audience comprehension.

FAILURE STATES AND ANTI-PATTERNS

Stereotype; cultural appropriation; unmarked generation; coercive emotional manipulation; false factual framing; inaccessible dialogue; and branching that invalidates user choice.

RECOVERY AND RECONSIDERATION

Withdraw or revise harmful content, preserve source and decisions, provide warnings and alternate path, notify owners, and retest affected audiences.

COMPLETION CRITERIA

Purpose, audience, world, content, representation, rights, evidence, tests, recovery, risks, and acceptance are explicit. Remain CONTENT-COMPLETE pending review and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve story graph, canon, sources, versions, reviews, decisions, evidence, warnings, and next action. Hand off to Game, Level, Content, Localization, UX, Accessibility, Legal, and Testing.

RELATED CODEX SECTIONS

094 Game Design; 095 Level Design; 096 Narrative Design; 097 Content Design; 101 Localization; 103 Accessibility Design; 223 Intellectual Property; 249 Usability Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract and shared governance files. Actual audience, cultural context, content boundaries, rights, localization, age, and acceptance requirements remain unresolved.

097

SECTION 097 - CONTENT DESIGN

CODEX/03_SECTIONS/097_CONTENT_DESIGN.txt

SHA-256 0e01fe15705417c4ea8b08ee752d053d73caa09df544c25e296d1aba8dadeced

4,778 BYTES / PRESERVED SOURCE IDENTITY

SECTION 097 — Content Design

STATUS: CONTENT-COMPLETE

PURPOSE

Define content that helps a declared audience understand, decide, act, recover, and maintain an accurate mental model of a product, service, or system.

REQUIRED OUTCOME

- Scoped audience, purpose, context, channel, language, content type, accessibility, and lifecycle.
- Content models, terminology, drafts, evidence, review, publication, correction, and retirement controls.

WHY THIS MATTERS

Content is part of behavior and can create legal, safety, privacy, and trust consequences. Clear prose is not automatically accurate or inclusive.

APPLICABILITY

Applies to interface, help, instructional, transactional, error, notification, onboarding, support, and operational content.

NON-APPLICABILITY CONDITIONS

No material meaning, instruction, disclosure, action, accessibility, localization, or recovery effect; or an unchanged approved content baseline with impact recorded.

PREREQUISITES AND DEPENDENCIES

Information architecture, user research, requirements, domain terms, policies, product behavior, accessibility, localization, legal review, source evidence, and authority.

REQUIRED INPUTS

Information architecture, user research, requirements, domain terms, policies, product behavior, accessibility, localization, legal review, source evidence, and authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve meaning, claims, audience, tone, disclosure, rights, safety, accessibility, and publication. They retain correction, withdrawal, and acceptance authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft, transform, translate, and critique content. It must preserve source, context, uncertainty, and rights and must not invent facts, cite nonexistent sources, or publish consequential text.

APPLICABLE EXISTING AGENT ROLES

Content Designer, Technical Writer, Copywriter, UX Reviewer, Researcher, Domain Expert, Accessibility specialist, Legal and Governance Reviewers, Tester, Verifier, Implementer, and Documentation Writer.

REQUIRED ACTIVITIES

Define audience and task; gather sources; model terms and states; draft; check accuracy, accessibility, tone, localization, rights, and behavior; test comprehension; publish, monitor, correct, and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show source, status, audience, actor, state, consequence, uncertainty, and correction route. AI content remains draft until reviewed and users retain non-AI support.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Connect content to product states, terminology, permissions, data, localization, versioning, search, accessibility, audit, and retirement. Keep content changes traceable to behavior changes.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show content version, owner, source, audience, locale, status, evidence, approvals, findings, and controls to edit, withdraw, supersede, or restore.

REQUIRED OUTPUTS AND ARTIFACTS

Content model, glossary, drafts, source and rights record, state and message inventory, accessibility and localization notes, reviews, tests, evidence, decisions, and handoff.

EVIDENCE REQUIREMENTS

Identify source, claim, audience, context, version, reviewer, method, result, and limitation. Readability or engagement is bounded evidence.

VERIFICATION AND ACCEPTANCE TESTS

Verify accuracy, task comprehension, state alignment, accessibility, localization, legal and privacy disclosure, error recovery, source traceability, and implementation fidelity.

FAILURE STATES AND ANTI-PATTERNS

Invented claims; vague errors; manipulative consent; untranslated critical content; inaccessible tables; stale help; and content that promises behavior the system does not perform.

RECOVERY AND RECONSIDERATION

Withdraw or correct content, preserve prior evidence, notify affected users, provide human support, and retest behavior and locales before republishing.

COMPLETION CRITERIA

Audience, purpose, claims, sources, states, accessibility, evidence, tests, correction, recovery, open risks, and acceptance are explicit. Remain CONTENT-COMPLETE pending review and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve content IDs, sources, versions, locales, reviews, decisions, evidence, corrections, and next action. Hand off to Content Strategy, UX, UI, Technical Writing, Localization, Legal, and Testing.

RELATED CODEX SECTIONS

030 Information Architecture; 097 Content Design; 098 Content Strategy; 099 Technical Writing; 100 Copywriting; 101 Localization; 103 Accessibility Design; 249 Usability Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract and shared governance files. Actual audiences, source authorities, claims, languages, legal review, content lifecycle, and acceptance thresholds remain unresolved.

098

SECTION 098 - CONTENT STRATEGY

CODEX/03_SECTIONS/098_CONTENT_STRATEGY.txt

SHA-256 aa2f609d13fa9db37b70f8e45992025851f2ac75dce36880f03148145535ee00

4,636 BYTES / PRESERVED SOURCE IDENTITY

SECTION 098 — Content Strategy

STATUS: CONTENT-COMPLETE

PURPOSE

Define the strategic purpose, portfolio, governance, investment, measurement, and lifecycle of content across audiences, channels, products, and services.

REQUIRED OUTCOME

- A declared content mission, audiences, channels, outcomes, governance, resources, risks, and horizon.
- Prioritized content system, ownership, standards, measures, evidence, decisions, and retirement model.

WHY THIS MATTERS

Content strategy prevents fragmented publication, contradictory claims, inaccessible information, and unmanaged maintenance burden.

APPLICABILITY

Applies when content is a material product, service, operational, marketing, instructional, support, or records capability.

NON-APPLICABILITY CONDITIONS

No material content outcome, audience, channel, claim, governance, resource, or lifecycle effect; or an unchanged approved strategy with impact recorded.

PREREQUISITES AND DEPENDENCIES

Mission, product and service strategy, information architecture, research, content inventory, channel data, policies, legal, accessibility, localization, staffing, and decision authority.

REQUIRED INPUTS

Mission, product and service strategy, information architecture, research, content inventory, channel data, policies, legal, accessibility, localization, staffing, and decision authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans set content purpose, priorities, claims, audiences, resources, risk, governance, and acceptance. They approve commitments and resolve conflicting stakeholder needs.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inventory content, synthesize evidence, and propose priorities. It must expose evidence quality and opportunity cost and must not set editorial policy or publish claims.

APPLICABLE EXISTING AGENT ROLES

Content Strategist, Content Designer, Technical Writer, Copywriter, UX Strategist, Researcher, Analyst, Governance Reviewer, Legal, Accessibility, Localization, Verifier, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Define mission, audiences, inventory, lifecycle, channels, taxonomy, ownership, standards, workflow, measures, risks, resourcing, governance, and review cadence.

HUMAN-AI INTERACTION REQUIREMENTS

Show content objective, audience, source, owner, status, evidence, risk, and decision. AI recommendations remain advisory and cannot bypass publication or records controls.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Connect strategy to information, product, service, workflow, localization, accessibility, legal, records, search, analytics, and retirement architecture.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show portfolio, gaps, freshness, owners, evidence, measures, risks, approvals, and controls to prioritize, pause, correct, withdraw, or retire.

REQUIRED OUTPUTS AND ARTIFACTS

Content strategy, inventory, audience map, governance model, lifecycle, standards, measurement plan, resource and risk register, decisions, and handoff.

EVIDENCE REQUIREMENTS

Identify source, audience, channel, period, method, result, uncertainty, and limitation. Traffic, reach, or engagement is bounded evidence.

VERIFICATION AND ACCEPTANCE TESTS

Verify strategic traceability, audience inclusion, accessibility, claim accuracy, freshness, channel fit, governance, resource feasibility, and retirement capacity.

FAILURE STATES AND ANTI-PATTERNS

Publishing without owners; optimizing volume; treating traffic as value; ignoring maintenance; contradictory channels; inaccessible archives; and no retirement path.

RECOVERY AND RECONSIDERATION

Pause unsupported publication, preserve decisions, correct or withdraw content, assign owners, commission research, and rebaseline affected channels.

COMPLETION CRITERIA

Mission, audiences, portfolio, governance, resources, evidence, measures, risks, recovery, open gaps, and acceptance authority are explicit. Remain CONTENT-COMPLETE pending review and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve strategy, inventory, standards, decisions, evidence, owners, dependencies, and next action. Hand off to Information, Content, UX, Technical Writing, Localization, Records, and Operations.

RELATED CODEX SECTIONS

030 Information Architecture; 054 Content Architecture; 097 Content Design; 099 Technical Writing; 100 Copywriting; 101 Localization; 225 Records Management; 342 Knowledge Management.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract and shared governance files. Actual content mission, audiences, channels, resources, records rules, legal constraints, and acceptance measures remain unresolved.

099

SECTION 099 - TECHNICAL WRITING

CODEX/03_SECTIONS/099_TECHNICAL_WRITING.txt

SHA-256 f1fdb6092e6258d6539dd70621a9d7a2359ef843c5993d73fc1a98d427e3dd47

4,877 BYTES / PRESERVED SOURCE IDENTITY

SECTION 099 — Technical Writing

STATUS: CONTENT-COMPLETE

PURPOSE

Define accurate, task-oriented technical documentation for building, using, operating, troubleshooting, securing, maintaining, and retiring a system.

REQUIRED OUTCOME

- A declared reader, task, version, environment, prerequisite, authority, format, and lifecycle scope.
- Verified documentation with source traceability, examples, warnings, accessibility, localization, and recovery guidance.

WHY THIS MATTERS

Technical documentation can become an operational control. Incorrect or stale instructions can cause outage, unsafe action, data loss, or unsupported commitments.

APPLICABILITY

Applies to developer, API, operator, administrator, user, support, installation, migration, security, and retirement documentation.

NON-APPLICABILITY CONDITIONS

No material instruction, configuration, API, task, safety, security, accessibility, or recovery effect; or an unchanged approved document with impact recorded.

PREREQUISITES AND DEPENDENCIES

Implementation and runtime evidence, requirements, interfaces, procedures, versions, users, accessibility, localization, security, support, records, and technical authority.

REQUIRED INPUTS

Implementation and runtime evidence, requirements, interfaces, procedures, versions, users, accessibility, localization, security, support, records, and technical authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve technical truth, warnings, supported scope, risks, publication, and acceptance. Owners retain correction, withdrawal, and escalation authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft, transform, index, and test documentation. It must preserve source, version, environment, commands, and uncertainty and must not invent behavior or claim execution it did not perform.

APPLICABLE EXISTING AGENT ROLES

Technical Writer, Documentation Writer, Researcher, Implementer, Tester, Verifier, Architect, Operations, Security Auditor, Governance Reviewer, Critic, Accessibility specialist, Localization, and Memory Curator.

REQUIRED ACTIVITIES

Identify readers and tasks; gather authoritative sources; draft procedures and examples; test commands; add warnings and recovery; review accessibility and localization; publish, monitor, correct, and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show document state, version, source, environment, command consequence, authority, uncertainty, and correction route. AI drafts remain untrusted until technical review.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Link docs to versions, APIs, configs, permissions, logs, support, search, localization, accessibility, records, and retirement. Examples must be reproducible or labeled illustrative.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show document owner, version, compatibility, freshness, source, tests, warnings, open issues, and controls to approve, correct, withdraw, supersede, or restore.

REQUIRED OUTPUTS AND ARTIFACTS

Documentation plan, information model, task guides, reference, examples, troubleshooting, security and recovery notes, source and review records, tests, decisions, and handoff.

EVIDENCE REQUIREMENTS

Identify source, artifact, version, environment, reader, method, test result, date, and limitation. A rendered page or grammar review proves bounded quality only.

VERIFICATION AND ACCEPTANCE TESTS

Execute representative commands, verify outputs and permissions, test version compatibility, accessibility, localization, error recovery, security warnings, and independent reproduction.

FAILURE STATES AND ANTI-PATTERNS

Stale commands; invented flags; examples that leak secrets; missing prerequisites; no recovery; version ambiguity; inaccessible code blocks; and documentation treated as tested because it renders.

RECOVERY AND RECONSIDERATION

Withdraw unsafe guidance, preserve source and prior version, publish correction, notify affected users, and retest dependent procedures before restoring status.

COMPLETION CRITERIA

Reader, task, version, source, evidence, tests, warnings, recovery, lifecycle, open risks, and acceptance are explicit. Remain CONTENT-COMPLETE pending review and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve docs, sources, versions, test outputs, reviews, decisions, corrections, and next action. Hand off to Content, Application, Software, Operations, Security, Support, Localization, and Records.

RELATED CODEX SECTIONS

019 Requirements Engineering; 030 Information Architecture; 097 Content Design; 101 Localization; 225 Records Management; 341 Documentation Engineering; 283 Systems Administration; 333 Technical Support.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract and shared governance files. Actual readers, source authority, supported versions, environments, records rules, and acceptance standards remain unresolved.

100

SECTION 100 - COPYWRITING

CODEX/03_SECTIONS/100_COPYWRITING.txt

SHA-256 c73b9018e1a5be87d47784e7896604299fe3c2b816bab2832bf5d7d130b0da92

4,662 BYTES / PRESERVED SOURCE IDENTITY

SECTION 100 — Copywriting

STATUS: CONTENT-COMPLETE

PURPOSE

Define persuasive, clear, and responsible language for an approved audience, offer, identity, or action without misleading, coercing, or hiding material conditions.

REQUIRED OUTCOME

- A declared audience, objective, claim, channel, tone, evidence, rights, accessibility, localization, and lifecycle scope.
- Copy briefs, drafts, approvals, source records, tests, corrections, and publication controls.

WHY THIS MATTERS

Copy can create commitments, shape consent, and influence vulnerable audiences. Persuasion must not substitute for truth, comprehension, or fair choice.

APPLICABILITY

Applies to marketing, product, campaign, transactional, onboarding, notification, fundraising, public, and brand copy.

NON-APPLICABILITY CONDITIONS

No material claim, offer, disclosure, consent, audience, accessibility, localization, or action effect; or an unchanged approved copy baseline with impact recorded.

PREREQUISITES AND DEPENDENCIES

Objective, audience research, offer, product behavior, substantiation, legal and policy constraints, brand, accessibility, localization, privacy, and approval authority.

REQUIRED INPUTS

Objective, audience research, offer, product behavior, substantiation, legal and policy constraints, brand, accessibility, localization, privacy, and approval authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve claims, audience, fairness, disclosure, consent, rights, risk, publication, and acceptance. They retain correction, withdrawal, and complaint authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may generate variants and analyze performance. It must preserve substantiation, audience context, and uncertainty and must not invent testimonials, scarcity, guarantees, or targeting rationale.

APPLICABLE EXISTING AGENT ROLES

Copywriter, Content Designer, UX Writer, Researcher, Product Owner, Legal and Governance Reviewers, Accessibility specialist, Localization, Tester, Verifier, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Define objective and audience; gather substantiation; draft and review claims; check accessibility, localization, fairness, consent, rights, and behavior; test comprehension; publish, monitor, correct, and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show claim source, audience, actor, offer, material condition, consent, automation, and correction. AI variants remain drafts and critical disclosures cannot be hidden.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Connect copy to product state, pricing, policy, identity, localization, accessibility, analytics, records, and legal approval. Separate persuasion from authorization.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show copy version, claim source, audience, locale, approval, evidence, performance limits, complaints, and controls to approve, correct, withdraw, or restore.

REQUIRED OUTPUTS AND ARTIFACTS

Copy brief, claims and substantiation register, variants, content and legal review, accessibility and localization review, test evidence, decisions, and handoff.

EVIDENCE REQUIREMENTS

Identify claim, source, audience, channel, version, method, result, limitation, and reviewer. Conversion or engagement is bounded evidence.

VERIFICATION AND ACCEPTANCE TESTS

Verify factual claims, material conditions, comprehension, accessibility, localization, consent, fairness, behavior alignment, rights, and implementation fidelity.

FAILURE STATES AND ANTI-PATTERNS

Dark patterns; false urgency; hidden fees; fake social proof; inaccessible disclosures; unsubstantiated superiority; and optimizing conversion over informed choice.

RECOVERY AND RECONSIDERATION

Pause publication, correct or withdraw claims, notify affected audiences, preserve evidence, offer remedy where authorized, and retest before reuse.

COMPLETION CRITERIA

Audience, objective, claims, substantiation, disclosures, evidence, tests, recovery, risks, and acceptance are explicit. Remain CONTENT-COMPLETE pending review and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve briefs, claims, sources, variants, approvals, locales, decisions, complaints, evidence, and next action. Hand off to Content, Product, Legal, Localization, Accessibility, and Marketing.

RELATED CODEX SECTIONS

004 Mission Strategy; 097 Content Design; 098 Content Strategy; 101 Localization; 103 Accessibility Design; 219 Legal; 222 Ethics; 249 Usability Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract and shared governance files. Actual offers, claims, audiences, substantiation, legal regimes, languages, and acceptance measures remain unresolved.

101

SECTION 101 - LOCALIZATION

CODEX/03_SECTIONS/101_LOCALIZATION.txt

SHA-256 c5bd1173fda9768c1843ab15b4c5b7ddf110d28f4723454502c8de2848045ef9

5,347 BYTES / PRESERVED SOURCE IDENTITY

SECTION 101 — Localization

STATUS: CONTENT-COMPLETE

PURPOSE

Define adaptation of content, interaction, data, behavior, and support for a declared locale while preserving meaning, functionality, accessibility, legal fit, and cultural respect.

REQUIRED OUTCOME

- A scoped source, locales, audiences, cultural context, content, formats, legal, accessibility, workflow, and lifecycle model.
- Traceable translations, locale rules, review, tests, fallback, evidence, and acceptance records.

WHY THIS MATTERS

Literal translation can break meaning, layout, dates, numbers, safety, humor, consent, or legal disclosure. Locale coverage is not cultural validity.

APPLICABILITY

Applies to products, services, content, interfaces, support, data, voice, games, documentation, and operations used across locales.

NON-APPLICABILITY CONDITIONS

No locale, translation, formatting, cultural, accessibility, legal, or audience effect; or an unchanged approved localization baseline with impact recorded.

PREREQUISITES AND DEPENDENCIES

Source content, locale requirements, terminology, translators, cultural research, legal and policy rules, format constraints, accessibility, testing, and authority.

REQUIRED INPUTS

Humans approve locale scope, meaning, cultural fit, legal disclosures, accessibility, publication, and acceptance. They resolve untranslatable or sensitive content.

HUMAN RESPONSIBILITIES AND AUTHORITY

AI may translate or identify variants, but must preserve source, context, uncertainty, and human review. It must not invent cultural authority or publish high-consequence translation alone.

AI RESPONSIBILITIES AND BOUNDARIES

AI may prepare locale-specific drafts, detect inconsistent terminology, flag truncation or cultural risk, and assemble traceable comparison and test evidence from approved source material. It must preserve source intent, locale scope, formatting rules, accessibility requirements, legal meaning, uncertainty, and reviewer attribution; distinguish translation from adaptation; and identify content that requires qualified cultural, legal, domain, or lived-experience review. It must not invent local authority, infer user identity or culture, silently replace approved terminology, expose sensitive source content, or publish, withdraw, or materially alter localized content without the required human decision and recorded evidence.

APPLICABLE EXISTING AGENT ROLES

Localization Specialist, Translator, Content Designer, UX Reviewer, Researcher, Cultural or domain Expert, Accessibility specialist, Tester, Verifier, Governance Reviewer, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Define locales and priority; inventory source; establish glossary; translate and adapt; review culturally and legally; test layout, data, voice, accessibility, and behavior; publish, monitor, correct, and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show source and locale, translation status, reviewer, uncertainty, actor, consequence, fallback, and correction. AI translations remain proposed until qualified review.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Support locale metadata, plural and gender rules, scripts, direction, dates, numbers, units, sorting, search, content, identity, accessibility, caching, and fallback without silently mixing locales.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show locale, source version, translation, reviewer, freshness, evidence, missing strings, defects, and controls to approve, correct, fall back, or withdraw.

REQUIRED OUTPUTS AND ARTIFACTS

Localization plan, locale and terminology register, translated assets, cultural and legal review, accessibility and layout tests, evidence, decisions, and handoff.

EVIDENCE REQUIREMENTS

Identify source, locale, translator or model, context, version, method, result, uncertainty, and limitation. Back-translation or fluency is bounded evidence.

VERIFICATION AND ACCEPTANCE TESTS

Verify meaning, terminology, pluralization, direction, dates, numbers, layout, accessibility, legal disclosure, cultural fit, fallback, and representative locale behavior.

FAILURE STATES AND ANTI-PATTERNS

Machine translation without review; English fallback for critical content; truncated scripts; cultural stereotypes; wrong units; and treating fluency as correctness.

RECOVERY AND RECONSIDERATION

Use approved fallback, remove harmful translation, preserve source and review history, notify owners, commission qualified review, and retest affected locales.

COMPLETION CRITERIA

Locales, source, meaning, review, accessibility, evidence, tests, fallback, recovery, risks, and acceptance are explicit. Remain CONTENT-COMPLETE pending review and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve source and locale versions, glossaries, translators, reviews, decisions, evidence, defects, and next action. Hand off to Internationalization, Content, UX, Accessibility, Testing, and Operations.

RELATED CODEX SECTIONS

097 Content Design; 099 Technical Writing; 101 Localization; 102 Internationalization; 103 Accessibility Design; 250 Localization Testing; 248 Compatibility Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract and shared governance files. Actual locales, source authority, cultural reviewers, legal regimes, scripts, and acceptance thresholds remain unresolved.

102

SECTION 102 - INTERNATIONALIZATION

CODEX/03_SECTIONS/102_INTERNATIONALIZATION.txt

SHA-256 1e36350903afeec6c633c72b5aee077dea285e469c66055c89a632541187ac97

5,275 BYTES / PRESERVED SOURCE IDENTITY

SECTION 102 — Internationalization

STATUS: CONTENT-COMPLETE

PURPOSE

Define product and system foundations that allow locale, language, culture, legal, accessibility, and regional variation without unsafe rewrites or hidden assumptions.

REQUIRED OUTCOME

- A scoped locale model, language, data, formatting, content, architecture, workflow, testing, ownership, and lifecycle baseline.
- Explicit variation points, defaults, fallback, evidence, and acceptance criteria.

WHY THIS MATTERS

Internationalization failures surface late as broken layouts, incorrect calculations, exclusion, or legal error. Early variation modeling reduces systemic rework and hidden regional assumptions.

APPLICABILITY

Applies to products, platforms, data, content, APIs, services, operations, and systems expected to support more than one locale or cultural context.

NON-APPLICABILITY CONDITIONS

No locale, language, format, culture, legal, accessibility, or regional behavior effect; or an unchanged approved foundation with impact recorded.

PREREQUISITES AND DEPENDENCIES

Target locales, language and script requirements, data and format rules, content, architecture, legal, privacy, accessibility, testing, deployment, and authority.

REQUIRED INPUTS

Humans approve supported locales, defaults, regional obligations, data boundaries, accessibility, legal fit, tradeoffs, and acceptance.

HUMAN RESPONSIBILITIES AND AUTHORITY

AI may identify hard-coded assumptions and generate locale tests. It must preserve evidence and uncertainty and must not select regional policy or infer cultural meaning without authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inspect code, content, schemas, and tests for locale assumptions; propose variation points; generate representative fixtures; and report unsupported scripts, formats, parsing paths, or fallback behavior. It must preserve locale and regional scope, data meaning, privacy boundaries, accessibility, precision, timezone, currency, and uncertainty, and must separate detection from implementation and acceptance. It must not silently change global defaults, normalize or reinterpret user data, select legal or regional policy, treat one locale as proof of general support, or merge architecture changes without authorized review and recorded test evidence.

APPLICABLE EXISTING AGENT ROLES

Architect, Analyst, Localization Specialist, Technical Writer, Content Designer, UX Reviewer, Tester, Verifier, Accessibility specialist, Security and Privacy Auditors, Governance Reviewer, Implementer, and Documentation Writer.

REQUIRED ACTIVITIES

Inventory locale assumptions; define variation points; externalize strings and formats; support scripts, direction, plural, units, sorting, data, time, identity, accessibility; test locales; document and hand off.

HUMAN-AI INTERACTION REQUIREMENTS

Show locale, source, default, fallback, format, actor, data boundary, uncertainty, and user control. AI findings remain recommendations pending qualified review.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use locale-aware schemas, formatting, collation, parsing, resource loading, content keys, test fixtures, timezone, currency, accessibility, and deployment without global mutable assumptions.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show supported locales, coverage, defaults, fallback, format rules, defects, evidence, dependencies, and controls to approve, restrict, switch, or recover.

REQUIRED OUTPUTS AND ARTIFACTS

Internationalization requirements, locale model, resource and format contracts, test matrix, hard-coded assumption inventory, evidence, decisions, and handoff.

EVIDENCE REQUIREMENTS

Identify locale, platform, data, format, version, test method, result, uncertainty, and limitation. Passing one locale proves bounded support only.

VERIFICATION AND ACCEPTANCE TESTS

Verify scripts, direction, plural, dates, numbers, units, sorting, timezones, currency, parsing, accessibility, fallback, data integrity, and locale switching.

FAILURE STATES AND ANTI-PATTERNS

Hard-coded strings; locale as decoration; incorrect numeric or date parsing; global defaults; untestable formatting; and falling back to a language that hides critical meaning.

RECOVERY AND RECONSIDERATION

Restrict unsupported locale, preserve data, use approved fallback, correct formatting, notify owners, and retest affected locales before expansion.

COMPLETION CRITERIA

Locale model, variation, data, formats, accessibility, tests, evidence, fallback, recovery, risks, and acceptance are explicit. Remain CONTENT-COMPLETE pending review and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve locale rules, resources, fixtures, decisions, defects, evidence, and next action. Hand off to Localization, Content, Data, Application, UI, Accessibility, Testing, and Operations.

RELATED CODEX SECTIONS

030 Information Architecture; 031 Data Architecture; 042 API Architecture; 097 Content Design; 101 Localization; 103 Accessibility Design; 248 Compatibility Testing; 250 Localization Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract and shared governance files. Actual locale matrix, supported scripts, regional data rules, legal obligations, and acceptance thresholds remain unresolved.

103

SECTION 103 - ACCESSIBILITY DESIGN

CODEX/03_SECTIONS/103_ACCESSIBILITY_DESIGN.txt

SHA-256 d3711668ee9a1ab7e40776a3338ae8ad1e96213d5bb66824a29a05220d117511

5,386 BYTES / PRESERVED SOURCE IDENTITY

SECTION 103 — Accessibility Design

STATUS: CONTENT-COMPLETE

PURPOSE

Define inclusive experiences, interfaces, content, environments, and controls that people with varied abilities can perceive, understand, operate, and recover from.

REQUIRED OUTCOME

- A scoped population, task, context, technology, access need, standard, risk, and lifecycle model.
- Accessibility requirements, alternatives, evidence, tests, exceptions, and owner controls.

WHY THIS MATTERS

Accessibility is an outcome and a rights concern, not a final visual check. Exclusion often occurs in error, recovery, timing, content, and support paths.

APPLICABILITY

Applies to products, services, content, physical or digital interfaces, devices, documents, environments, and owner or operator controls.

NON-APPLICABILITY CONDITIONS

No material access, task, content, control, assistive, timing, or recovery effect; or an unchanged approved accessibility baseline with impact recorded.

PREREQUISITES AND DEPENDENCIES

Users and disability research, requirements, interaction and content models, target technology, applicable standards, devices, assistive tools, privacy, and acceptance authority.

REQUIRED INPUTS

Humans approve inclusion goals, standards, risk, accommodations, exceptions, resources, and acceptance. They must include affected people and retain correction, appeal, and stop authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

AI may identify barriers and generate test ideas. It must not infer disability, claim accessibility from heuristics, or replace lived-experience review and qualified testing.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inspect interfaces, content, documents, and workflows for potential barriers; map findings to approved standards; propose equivalent alternatives; and generate bounded test cases for keyboard, assistive technology, sensory, cognitive, timing, and recovery paths. It must preserve user agency, privacy, dignity, access context, uncertainty, and the distinction between automated evidence and lived experience, and must identify where qualified accessibility review is required. It must not infer a person's disability, expose sensitive access information, declare conformance from heuristics alone, remove an accommodation, or make an inaccessible path the only path without authorized human review and recorded acceptance.

APPLICABLE EXISTING AGENT ROLES

Accessibility Designer, User Experience Reviewer, Researcher, Interaction, UI, Content, and Visual Designers, Tester, Verifier, Governance Reviewer, Critic, Implementer, and Documentation Writer.

REQUIRED ACTIVITIES

Define access needs and tasks; map barriers; set standards; design alternatives; test with assistive technologies and affected users; record exceptions; remediate; and retest.

HUMAN-AI INTERACTION REQUIREMENTS

Show access mode, user control, state, actor, automation, alternative, consequence, and recovery. AI suggestions remain advisory; accessible non-AI paths remain first-class.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Support semantic structure, keyboard and alternative input, focus, contrast, text resizing, captions, audio description, timing, error, localization, privacy, and equivalent outcomes.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show population, standard, coverage, assistive technology, findings, exceptions, evidence, and controls to remediate, provide accommodation, restrict, or reopen.

REQUIRED OUTPUTS AND ARTIFACTS

Accessibility requirements, user and task records, standard mapping, design annotations, alternatives, test evidence, exception and accommodation records, decisions, and handoff.

EVIDENCE REQUIREMENTS

Identify population, access mode, technology, standard, task, method, result, limitation, and date. Automated scans and conformance claims prove bounded coverage only.

VERIFICATION AND ACCEPTANCE TESTS

Verify keyboard, screen reader, magnification, contrast, captions, timing, input, focus, language, error, recovery, cognitive load, physical access, and representative lived experience.

FAILURE STATES AND ANTI-PATTERNS

Treating a scan as acceptance; inaccessible recovery; color-only meaning; inaccessible authentication; accommodations hidden; and claiming compliance without version, scope, or evidence.

RECOVERY AND RECONSIDERATION

Provide equivalent path or accommodation, preserve user progress, remediate and communicate barriers, record exceptions, and retest before removing restrictions.

COMPLETION CRITERIA

Population, tasks, standards, alternatives, evidence, tests, exceptions, recovery, risks, and acceptance authority are explicit. Remain CONTENT-COMPLETE pending review and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve standards, findings, user evidence, assistive matrix, exceptions, decisions, evidence, and next action. Hand off to HCI, UX, UI, Content, Localization, Testing, Operations, and Governance.

RELATED CODEX SECTIONS

053 Accessibility Architecture; 075 HCI; 076 Interaction Design; 083 UI Design; 097 Content Design; 247 Accessibility Testing; 248 Compatibility Testing; 249 Usability Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance files, and accessibility evidence conventions. Actual standards, populations, technologies, accommodations, jurisdiction, and acceptance thresholds remain unresolved.

104

SECTION 104 - DESIGN SYSTEMS

CODEX/03_SECTIONS/104_DESIGN_SYSTEMS.txt

SHA-256 029dec5e4929c917a45334ad4bc3f95c292e0457bb20354622288459fffb04a9

5,245 BYTES / PRESERVED SOURCE IDENTITY

SECTION 104 — Design Systems

STATUS: CONTENT-COMPLETE

PURPOSE

Define reusable design tokens, components, patterns, content, behavior, accessibility, governance, and release controls that support coherent experiences without suppressing justified variation.

REQUIRED OUTCOME

- A scoped system, consumers, primitives, patterns, ownership, versioning, contribution, accessibility, evidence, and lifecycle model.
- Governed components and documentation with tests, exceptions, migration, and acceptance records.

WHY THIS MATTERS

Design systems reduce duplication but can spread a defect at scale. Reuse must preserve context, accessibility, behavior, and user outcomes.

APPLICABILITY

Applies to shared UI, interaction, content, visual, motion, voice, code, token, and service patterns.

NON-APPLICABILITY CONDITIONS

No shared component, token, pattern, accessibility, behavior, release, or consumer effect; or an unchanged approved system baseline with impact recorded.

PREREQUISITES AND DEPENDENCIES

UX and UI architecture, component consumers, requirements, accessibility, content, code, versioning, testing, contribution, support, and authority.

REQUIRED INPUTS

Humans own system purpose, standards, accessibility, versioning, exceptions, contribution, support, and acceptance. They decide when reuse harms context and retain deprecation authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

AI may inventory drift, generate variants, and suggest tokens or tests. It must preserve source and context and must not merge, release, or enforce patterns without governance.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inventory components, tokens, variants, consumers, drift, and test gaps; propose reusable patterns; and generate documentation, migration options, and bounded regression evidence from approved system sources. It must preserve component context, version, consumer impact, accessibility, content behavior, compatibility, ownership, and uncertainty, and must distinguish a suggestion from an approved system rule or release. It must not impose reuse where task outcomes or access needs require variation, silently change shared defaults, bypass contribution or review controls, merge or release breaking changes, deprecate consumer paths, or claim system health from snapshot or component-level evidence alone.

APPLICABLE EXISTING AGENT ROLES

Design Systems Designer, UX, UI, Interaction, Visual, Content, and Accessibility Designers, Architect, Implementer, Tester, Verifier, Governance Reviewer, Critic, Documentation Writer, and Memory Curator.

REQUIRED ACTIVITIES

Define principles and scope; inventory primitives and consumers; specify tokens and states; establish contribution, review, versioning, accessibility, tests, migration, deprecation, and support.

HUMAN-AI INTERACTION REQUIREMENTS

Show component version, state, consumer, actor, authority, accessibility, evidence, exception, and migration. AI suggestions remain proposals.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Represent tokens, components, behavior, content, accessibility, code, docs, dependencies, tests, release, compatibility, telemetry, and retirement as versioned artifacts.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show system health, versions, consumers, defects, accessibility, evidence, changes, and controls to approve, override, migrate, deprecate, or restore.

REQUIRED OUTPUTS AND ARTIFACTS

Design-system charter, principles, tokens, component specs and code, accessibility and behavior tests, contribution and release workflow, consumer map, decisions, and handoff.

EVIDENCE REQUIREMENTS

Identify component, version, consumer, platform, state, test, method, result, and limitation. Component existence or snapshot coverage proves bounded reuse only.

VERIFICATION AND ACCEPTANCE TESTS

Verify states, accessibility, responsive behavior, content fit, compatibility, version migration, consumer impact, documentation, performance, and regression.

FAILURE STATES AND ANTI-PATTERNS

One-size-fits-all components; hidden breaking changes; inaccessible defaults; token drift; documentation without code parity; and reuse prioritized over task outcome.

RECOVERY AND RECONSIDERATION

Quarantine defective component, preserve consumer versions, publish migration or fallback, notify owners, and retest all affected consumers before broad release.

COMPLETION CRITERIA

Scope, ownership, components, versions, accessibility, evidence, tests, migration, recovery, open risks, and acceptance are explicit. Remain CONTENT-COMPLETE pending review and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve system versions, components, consumers, dependencies, tests, decisions, evidence, exceptions, and next action. Hand off to UI, UX, Frontend, Content, Accessibility, Testing, and Release.

RELATED CODEX SECTIONS

081 UX Architecture; 083 UI Design; 084 Visual Design; 097 Content Design; 103 Accessibility Design; 118 Frontend Engineering; 258 Configuration Management; 296 Feature Management.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract and shared governance files. Actual consumers, platforms, standards, component scope, contribution model, and acceptance thresholds remain unresolved.

105

SECTION 105 - PROTOTYPING

CODEX/03_SECTIONS/105_PROTOTYPING.txt

SHA-256 834f6ea90bafe8be0666e7e1b30fbb4388ca3ab7c8c515f44b5a39b82d1cd117

21,999 BYTES / PRESERVED SOURCE IDENTITY

SECTION 105 — Prototyping

STATUS: CONTENT-COMPLETE

PURPOSE

Define the governed use of prototypes to learn about a proposed product, service, process, interaction, architecture, technology, or operating concept before treating it as an approved design or implementation.

Prototyping creates bounded representations for inquiry. A prototype may be a sketch, storyboard, role-play, paper flow, clickable interface, service simulation, physical mock-up, data sample, technical spike, throwaway implementation, model experiment, or controlled pilot. It is evidence about selected questions under declared conditions, not evidence that the eventual system is complete, secure, accessible, reliable, lawful, adopted, or effective.

REQUIRED OUTCOME

- A declared prototype question, learning objective, scope, audience, fidelity, time horizon, owner, authority, and stop condition.
- A prototype plan that identifies what is represented, what is intentionally omitted, what is simulated, what is real, and what claims the artifact cannot support.
- A traceable relationship from the prototype to requirements, user needs, processes, functions, architecture, risks, decisions, and next actions.
- A usable prototype artifact or experiment that allows the intended question to be examined under stated conditions.
- Evidence from observation, inspection, measurement, technical execution, or user interaction that is bounded by method, sample, environment, and limitations.
- A decision record stating what was learned, what remains unknown, which assumptions changed, whether the prototype should be discarded, revised, promoted, or stopped, and who has authority for that decision.

WHY THIS MATTERS

Prototypes make uncertainty visible while change is still relatively inexpensive. They expose misunderstanding, workflow gaps, technical constraints, accessibility barriers, unsafe assumptions, integration friction, and user burden that are easy to miss in prose or diagrams.

They also create a recurring source of false confidence. A polished mock-up can hide missing data, permissions, error states, support work, and operational cost. A technical spike can prove that a narrow path runs without proving security, resilience, maintainability, or user value. A prototype must therefore preserve the boundary between learning evidence and acceptance evidence.

APPLICABILITY

- A proposed outcome, requirement, interaction, workflow, service, architecture, component, integration, model, or operational concept contains material uncertainty.
- Stakeholders need to compare alternatives, expose assumptions, or inspect a future state before committing to implementation.
- A change affects users, operators, maintainers, support, accessibility, safety, privacy, security, data, reliability, cost, or lifecycle obligations.
- A technical or service risk can be reduced through a bounded experiment, simulation, spike, mock, pilot, or rehearsal.
- AI, automation, tools, data, external providers, or human review are proposed for a consequential workflow.

NON-APPLICABILITY CONDITIONS

- The work is a purely explanatory sketch with no claim beyond communication and no material change to requirements, decisions, user expectations, data, or authority.

- An approved and current prototype or design evidence package answers the exact question for the same scope, population, environment, version, and decision, and impact analysis shows no material change.
- The work is routine implementation of an already accepted design and introduces no new uncertainty or user, safety, security, privacy, accessibility, operational, or lifecycle question.
- A live experiment would expose real people, restricted data, production state, or external commitments beyond the approved boundary. In that case the prototype is not exempt; it must be narrowed, simulated, or separately authorized.

A non-applicability decision must identify the controlling artifact, exact question, evidence date, owner, assumptions, and reopening trigger. Visual simplicity, short duration, or disposable code is not by itself sufficient reason to omit governance.

PREREQUISITES AND DEPENDENCIES

- Sections 001-020 provide objective, stakeholder, user, domain, feasibility, risk, requirements, systems, verification, validation, transition, and recovery context.
- Sections 021-104 provide the applicable enterprise, solution, systems, domain, business, capability, process, functional, operational, information, data, application, user-experience, design, accessibility, and design-system boundaries.
- The prototype owner, learning question, intended audience, decision authority, allowed environment, data boundary, and consequence limit are identified.
- Applicable safety, privacy, security, accessibility, legal, records, licensing, supplier, research, and ethical controls are known or explicitly unresolved.

REQUIRED INPUTS

- Authorized problem or opportunity statement, intended outcome, non-goals, scope, affected parties, population, environment, jurisdiction, and lifecycle horizon.
- Requirements, user research, domain rules, process and functional models, architecture constraints, risks, acceptance criteria, and open decisions.
- Prototype hypothesis, learning questions, alternatives, assumptions, success signals, disconfirming signals, fidelity level, and time limit.
- Representative or synthetic data, test accounts, interfaces, dependencies, tools, content, devices, facilities, accessibility needs, and operational constraints.
- Experiment protocol, consent or participation arrangements, privacy and security controls, logging plan, evidence plan, and disposal or retention plan.

HUMAN RESPONSIBILITIES AND AUTHORITY

The authorized human owner retains authority over the prototype question, protected outcomes, participant treatment, risk tolerance, data and environment access, external commitments, interpretation of evidence, promotion, abandonment, and final decision.

- Confirm the question, scope, fidelity, audience, non-goals, consequence boundary, and decision date.
- Approve participant involvement, data use, environment, tools, suppliers, incentives, accessibility accommodations, and safety controls.
- Decide whether evidence is sufficient to revise requirements, continue learning, authorize a pilot, begin implementation, or stop.
- Resolve conflicts involving user welfare, rights, privacy, security, safety, accessibility, cost, schedule, feasibility, and organizational commitments.
- Ensure users, operators, domain experts, maintainers, support, security, privacy, safety, accessibility, legal, and governance specialists participate when consequence requires.
- Retain the ability to pause, narrow, quarantine, destroy, revoke, withdraw, or reject a prototype and its claims.

AI RESPONSIBILITIES AND BOUNDARIES

AI may help frame hypotheses, generate alternative concepts, create bounded mock data, draft prototype code or content, simulate scenarios, compare observations, identify missing states, and summarize evidence within authorized scope.

AI must label generated, simulated, observed, measured, inferred, and human-supplied material separately; preserve prototype identity, version, environment, data, prompt, tool, and limitation; expose uncertainty and disconfirming evidence; and stop when participant welfare, authority, privacy, security, safety, accessibility, or evidence integrity is unresolved.

AI must not present a prototype as a product, fabricate user feedback, recruit or manipulate participants without authorization, use restricted or personal data by inference, silently connect to production, approve a pilot, accept risk, or convert visual polish, model output, code execution, or engagement into proof of value.

APPLICABLE EXISTING AGENT ROLES

- Architect: frames prototype boundaries, fidelity, dependencies, system implications, and promotion risks.
- Researcher: gathers source, domain, user, technical, operational, and participant evidence.
- Analyst: compares hypotheses, alternatives, observations, feasibility, and uncertainty.
- Critic: challenges false confidence, omitted states, hidden work, weak evidence, and premature promotion.
- Security Auditor: examines data, identity, access, isolation, secrets, abuse, and disposal.
- User Experience Reviewer: examines comprehension, burden, accessibility, dignity, error, correction, and recovery.
- Governance Reviewer: checks authority, consent, policy, records, claims, and decision boundaries.
- Planner: sequences questions, experiments, review gates, and next actions.
- Coordinator: synchronizes participants, owners, dependencies, and handoffs.
- Implementer: builds the bounded artifact without silently expanding its purpose or authority.
- Refactorer: improves disposable or exploratory structure without changing the stated learning question.
- Tester: exercises intended, boundary, failure, accessibility, security, and recovery scenarios.
- Verifier: independently checks the prototype, method, evidence, and claims.
- Documentation Writer: records the protocol, artifact, observations, decisions, limitations, and disposal.
- Memory Curator: preserves versions, evidence, dissent, decisions, and continuity.
- Optimizer: improves learning efficiency without suppressing negative results or protected outcomes.
- Red Team: attempts authorized misuse, prompt manipulation, data leakage, bypass, failure, and misleading interpretation.
- Decision Arbiter Advisory: compares evidence-backed options under authorized criteria without deciding for the owner.

Role selection remains proportional to consequence. Existing role contracts remain controlling and are not redefined here.

REQUIRED ACTIVITIES

1. State the learning question, hypothesis, intended decision, owner, audience, scope, non-goals, fidelity, environment, and time limit.
2. Identify the requirements, user needs, domain rules, process steps, functions, architecture constraints, and risks the prototype is intended to examine.
3. Choose the smallest representation or experiment that can produce useful evidence without hiding material behavior.
4. Define what is real, simulated, stubbed, sampled, synthetic, omitted, manually operated, or unsafe to expose.
5. Define participant, operator, data, privacy, security, accessibility, safety, licensing, supplier, and records controls.
6. Build or configure the prototype with stable identity, version, provenance, environment, dependencies, and disposal conditions.
7. Include relevant normal, boundary, denied, delayed, error, degraded, correction, recovery, and exit states at the chosen fidelity.
8. Conduct authorized observation, inspection, measurement, technical execution, role-play, pilot, or comparison using a documented protocol.
9. Capture direct observations, participant feedback, task results, technical signals, failures, dissent, surprises, and unanswered questions without embellishment.
10. Analyze evidence against the hypothesis and separate prototype behavior from expected future behavior.

11. Review accessibility, privacy, security, safety, user burden, operational work, support, maintainability, and lifecycle implications.
12. Red-team the prototype for misleading affordances, hidden authority, data leakage, unsafe defaults, prompt or input manipulation, and false promotion.
13. Decide whether to discard, revise, repeat, transfer, pilot, implement, or stop, with owner authority and evidence limits recorded.
14. Preserve or dispose of the artifact, data, credentials, logs, participant records, and environment according to the approved lifecycle.

HUMAN-AI INTERACTION REQUIREMENTS

- Show whether each prototype element is a human decision, generated suggestion, simulation, stub, live dependency, observed behavior, or proposed future behavior.
- Let the owner inspect and correct the question, assumptions, data, participant treatment, scope, fidelity, interpretation, and decision without losing prior versions or dissent.
- Require human review before a prototype touches real users, personal or restricted data, production systems, external commitments, or consequential decisions.
- Make limitations, omitted states, manual work, synthetic data, model behavior, provider dependencies, and evidence gaps visible at the point of interpretation.
- Preserve non-AI, manual, accessible, stop, correction, complaint, and recovery paths during prototype sessions.
- Do not allow a conversational interface to bypass consent, authorization, isolation, data minimization, audit, or disposal controls.

ARCHITECTURE AND SYSTEM IMPLICATIONS

- Represent prototype identity, question, hypothesis, version, fidelity, scope, environment, dependencies, data, participants, observations, decisions, and lifecycle as attributable artifacts.
- Isolate prototype code, data, credentials, networks, providers, and state from production unless an explicit authority approves a controlled integration.
- Keep prototype definition separate from implementation, pilot, deployed system, user account, runtime state, and acceptance baseline.
- Make simulation, stubbing, manual intervention, synthetic data, disabled controls, and omitted behavior explicit in the artifact and evidence.
- Preserve traceability from question to requirement, prototype element, observation, decision, implementation implication, test, and acceptance criterion.
- Ensure prototype disposal, data deletion, credential revocation, log handling, and provider cleanup are testable.
- Treat AI models, prompts, tools, providers, and generated assets as versioned dependencies with evaluation, monitoring, substitution, and revocation paths.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

- Show the prototype name, question, owner, version, fidelity, environment, audience, status, time limit, and evidence freshness.
- Show what is live, simulated, stubbed, manual, omitted, generated, restricted, unsafe, or awaiting approval.
- Let the owner inspect assumptions, participants, data, dependencies, observations, failures, dissent, decisions, and open questions.
- Provide authorized pause, narrow, revoke, quarantine, reset, dispose, export, approve, reject, repeat, and handoff controls.
- Explain what the prototype does and does not demonstrate, which evidence supports each claim, and what would change the decision.
- Preserve status, consent, complaint, evidence, accessibility, correction, and recovery controls when nonessential tools or models fail.

REQUIRED OUTPUTS AND ARTIFACTS

- Prototype charter with learning question, hypothesis, owner, scope, fidelity, audience, environment, non-goals, and stop condition.

- Prototype plan, protocol, scenario set, participant or operator plan, risk controls, data plan, and disposal plan.
- Versioned prototype artifact, configuration, source or construction record, dependencies, test data, and environment description.
- Observation, measurement, feedback, failure, dissent, assumption, and decision records.
- Traceability from prototype questions and elements to requirements, user needs, process, function, architecture, risk, and acceptance.
- Limitation and non-claim statement distinguishing learning evidence from implementation, readiness, effectiveness, or acceptance evidence.
- Promotion, revision, abandonment, pilot, implementation, or stop decision with owner, authority, unresolved gaps, and next action.
- Continuity and disposal record covering artifact, data, credentials, logs, participants, retention, deletion, and handoff.

EVIDENCE REQUIREMENTS

- Each claim identifies the prototype, version, question, scope, environment, population, method, date, actor, source, result, and limitation.
- A mock-up, demo, successful execution, benchmark, simulation, interview, survey, or pilot proves only the bounded question it was designed to examine.
- Participant evidence identifies recruitment, consent or authorization, task, context, accessibility conditions, sample, protocol, dissent, and limitations.
- Technical evidence identifies configuration, dependencies, data, tools, provider or model, workload, instrumentation, failures, and reproducibility.
- Generated or synthetic evidence identifies origin, transformation, model or tool, review, and the reason it cannot be treated as observed user or production evidence.
- Failed, contradictory, stale, partial, inaccessible, missing, or negative evidence remains visible and constrains promotion.

VERIFICATION AND ACCEPTANCE TESTS

- Reproduce the prototype from its charter, version, configuration, data, environment, and protocol.
- Verify every stated learning question has an observation or result, and every unanswered question is recorded as unresolved.
- Exercise intended, boundary, denied, error, degraded, correction, recovery, accessibility, and disposal scenarios appropriate to the fidelity.
- Confirm that simulated, stubbed, manual, generated, and omitted behavior is correctly labeled and cannot be mistaken for live behavior.
- Attempt unauthorized data access, production connection, credential reuse, prompt manipulation, participant disclosure, and artifact export.
- Verify participant, user, and operator can stop, withdraw, correct, complain, request explanation, and recover from a failed or harmful session.
- Compare competing alternatives using the same declared protocol and disclose sample, measurement, and interpretation limits.
- Independently reproduce the decision package and confirm that promotion does not exceed the evidence.

FAILURE STATES AND ANTI-PATTERNS

- Treating a polished interface, clickable flow, technical spike, pilot, or demo as proof of product readiness or user value.
- Testing only the happy path while hiding denial, error, accessibility, support, recovery, manual work, or operational cost.
- Fabricating, selectively reporting, or averaging away negative feedback, dissent, failed runs, or affected-party burden.
- Using production data, credentials, users, or external systems without explicit scope and controls.
- Allowing AI to invent participant reactions, citations, metrics, requirements, or evidence.
- Expanding a throwaway artifact into a production dependency without security, privacy, accessibility, reliability, support, and lifecycle review.

- Optimizing prototype completion or engagement while weakening learning quality, participant welfare, or owner control.

RECOVERY AND RECONSIDERATION

- On unsafe behavior, privacy or security exposure, misleading claims, participant harm, evidence loss, authority conflict, or scope drift, stop the session and preserve relevant evidence.
- Identify the blocker, affected people, safe work that may continue, containment owner, authority, communication, and review condition.
- Revoke credentials, isolate systems, delete or quarantine data, reset state, withdraw materials, notify affected parties, and obtain specialist review as required.
- Re-run only after the question, controls, protocol, environment, and evidence plan are corrected and authorized.
- Reopen the prototype decision when new user evidence, technical evidence, policy, domain meaning, architecture, population, or lifecycle conditions invalidate the conclusion.

COMPLETION CRITERIA

- The question, hypothesis, scope, fidelity, owner, authority, environment, controls, evidence method, and stop condition are explicit.
- The artifact distinguishes real, simulated, generated, stubbed, manual, omitted, and observed behavior.
- Relevant user, operator, accessibility, privacy, security, safety, operational, and recovery conditions are addressed or justified.
- Results, failures, dissent, limitations, decision, unresolved questions, and next action are recorded.
- Disposal, retention, credential, data, participant, and handoff obligations are complete or explicitly blocked.
- The section is marked CONTENT-COMPLETE while source reconciliation, hostile review, and owner acceptance remain separate activities.

HANDOFF AND CONTINUITY REQUIREMENTS

- Preserve prototype identity, question, hypothesis, version, fidelity, scope, environment, dependencies, data, protocol, evidence, decisions, dissent, and limitations.
- Hand off only the evidence-supported implications to Requirements, User Research, User Validation, Architecture, Design, Engineering, Testing, Operations, Security, Privacy, Accessibility, and Governance work.
- Record what must not be carried forward, what requires revalidation, and which assumptions are still provisional.
- State the next safe action, blocked action, authority, evidence request, retention or disposal obligation, and exact continuation point.
- Do not claim implementation readiness, user validation, operational readiness, or owner acceptance from the prototype alone.

RELATED CODEX SECTIONS

- 015 User Research; 016 Domain Analysis; 017 Feasibility Analysis; 018 Risk Management; 019 Requirements Engineering; 020 Systems Engineering.
- 021-104 Enterprise through Design Systems, especially Process, Functional, Information, Data, Application, User Experience, Interaction, Accessibility, and Design Architecture.
- 231-257 Testing through Validation; 247 Accessibility Testing; 249 Usability Testing; 253 Model Testing; 254 Data Testing.
- 369-372 Experimentation through Feedback Management; 402-406 Post-Implementation Review through System Retirement.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

- This draft uses the Canonical Section Contract, shared governance files, Sections 001-104, Owner Cockpit Requirements, Status and Evidence Lexicon, and the Remaining-Section Integration Roadmap.
- No universal prototype fidelity scale, research protocol, participant incentive rule, statistical threshold, technology stack, or promotion gate applies to every future context.

- Actual domain authorities, participant populations, consent requirements, accessibility accommodations, data permissions, technical environments, and decision thresholds remain unresolved until supplied and verified.

106

SECTION 106 - USER VALIDATION

CODEX/03_SECTIONS/106_USER_VALIDATION.txt

SHA-256 1bad1e012067455a232ac6ce544eeb0e60a2230088e00a1bafd0a3e89a0de6c9

22,547 BYTES / PRESERVED SOURCE IDENTITY

SECTION 106 — User Validation

STATUS: CONTENT-COMPLETE

PURPOSE

Define the governed process for determining whether a proposed or implemented product, service, system, process, interaction, content, or operating change supports the intended users and affected people under declared real or representative conditions.

User Validation examines outcomes, not merely opinions or task completion. It tests whether the right problem is being addressed, whether people can understand and use the result, whether the result is accessible and safe, whether it fits real contexts and constraints, and whether users can recover, correct, complain, or decline. Validation evidence is bounded by population, context, method, version, and time; it does not automatically establish universal satisfaction, adoption, compliance, or operational effectiveness.

REQUIRED OUTCOME

- A declared validation question, intended user and affected-party outcome, population, context, version, owner, authority, and decision boundary.
- A participant and evidence plan that addresses representativeness, accessibility, safety, privacy, consent or authorization, bias, and limitations.
- Reproducible tasks, scenarios, instruments, observation methods, and acceptance criteria appropriate to the decision.
- Evidence showing what users and affected people could understand, accomplish, experience, challenge, correct, or recover from under stated conditions.
- Analysis that separates observed behavior, participant report, interpretation, recommendation, and owner decision.
- A decision record identifying validated outcomes, failed or unresolved outcomes, affected groups, required changes, residual risk, and the authority to accept, defer, narrow, reject, or reopen.

WHY THIS MATTERS

A system can meet its written requirements and still fail the people who must live with it. Users may be unable to discover a control, understand a status, complete a task, recover from an error, access content, challenge a decision, or trust what the system is claiming. People excluded from the study may carry the greatest burden.

User validation creates disciplined evidence about real-world fit while preserving uncertainty. It prevents the organization from substituting stakeholder preference, internal completion, model confidence, survey volume, or a small enthusiastic sample for demonstrated user and affected-party outcomes.

APPLICABILITY

- A change may affect user goals, comprehension, workflow, accessibility, safety, privacy, security, dignity, rights, trust, burden, or recovery.
- A product, service, system, process, content, interaction, AI behavior, policy, or operating model is proposed for a defined population or context.
- Requirements or acceptance depend on user outcomes, task performance, comprehension, accessibility, satisfaction, confidence, correction, appeal, or harm reduction.
- Different populations, environments, devices, languages, abilities, roles, or operating conditions may experience materially different results.
- AI, automation, recommendation, ranking, generation, or decision support may change what users see, decide, or can control.

NON-APPLICABILITY CONDITIONS

- The change has no material effect on user or affected-party outcomes, access, comprehension, burden, privacy, safety, security, accessibility, control, or recovery.

- An approved validation package covers the exact population, context, version, outcome, and change, and impact analysis shows no material effect.
- The work is an internal technical activity with no user-facing or affected-party claim and no change to user data, expectations, rights, or access.
- The activity is only a research or discovery input and makes no acceptance or effectiveness claim; it must still be labeled and its limitations preserved.

A non-applicability decision must identify the controlling evidence, exact population and context, version, owner, evidence date, assumptions, and reopening trigger. A small sample, familiar workflow, or internal approval is not by itself sufficient reason to omit validation.

PREREQUISITES AND DEPENDENCIES

- Sections 001-020 provide objectives, stakeholders, users, domain, feasibility, risk, requirements, systems, verification, validation, transition, and recovery context.
- Sections 021-104 provide applicable architecture, process, functional, operational, information, data, application, human-computer interaction, experience, accessibility, content, and design-system boundaries.
- The target population, affected non-users, intended outcome, decision authority, version, test environment, and consequence limits are explicit.
- Privacy, security, safety, accessibility, research ethics, legal, records, language, compensation, and safeguarding controls are known or marked unresolved.

REQUIRED INPUTS

- Authorized validation question, intended outcome, user jobs, affected parties, population, context, jurisdiction, environment, version, and lifecycle stage.
- Requirements, acceptance criteria, prototypes or implementation, domain rules, process and functional models, architecture, risks, policies, and known constraints.
- Participant segments, recruitment criteria, accessibility needs, language, devices, assistive technology, skills, support conditions, and exclusion risks.
- Tasks, scenarios, stimuli, data, accounts, instrumentation, consent or authorization materials, moderator guidance, and safety or escalation procedures.
- Analysis plan, success and failure signals, disconfirming evidence, sampling limits, privacy plan, retention and disposal plan, and decision threshold.

HUMAN RESPONSIBILITIES AND AUTHORITY

The authorized human owner retains authority over the outcome being validated, participant treatment, population definition, evidence sufficiency, risk acceptance, policy interpretation, design or release decision, and commitments made to users or affected people.

- Confirm the validation question, intended outcome, population, context, version, non-goals, and decision boundary.
- Approve recruitment, consent or authorization, accessibility accommodations, incentives, data use, recording, safeguarding, and escalation.
- Decide whether the evidence supports change, continued testing, limited release, full acceptance, deferral, rejection, or retirement.
- Resolve conflicts involving user welfare, rights, safety, privacy, security, accessibility, dignity, cost, schedule, feasibility, and organizational commitments.
- Ensure representative users, affected non-users, operators, domain experts, support, security, privacy, safety, accessibility, legal, and governance specialists participate as consequence requires.
- Retain the ability to pause, narrow, withdraw, compensate, correct, quarantine, reject, or reopen validation.

AI RESPONSIBILITIES AND BOUNDARIES

AI may help identify user segments, draft tasks, generate accessible variants, summarize authorized sessions, code observations, compare patterns, identify missing outcomes, and propose follow-up research within the approved boundary.

AI must preserve participant identity protections, consent or authorization, population, context, version, method, source, uncertainty, dissent, and limitations; distinguish direct evidence from interpretation; and escalate when participant welfare, rights, privacy, security, safety, accessibility, or evidence integrity is unresolved.

AI must not fabricate users, consent, reactions, quotes, success rates, accessibility outcomes, representative coverage, or acceptance. It must not manipulate participants, infer sensitive attributes beyond authorization, expose session data, decide that users are satisfied, or replace accountable human interpretation and acceptance.

APPLICABLE EXISTING AGENT ROLES

- Researcher: designs and conducts evidence gathering with users and affected people.
- User Experience Reviewer: examines usability, accessibility, comprehension, burden, dignity, trust, and recovery.
- Analyst: evaluates outcomes, patterns, segments, tradeoffs, uncertainty, and disconfirming evidence.
- Observer: records behavior, context, workarounds, hesitation, errors, and environmental conditions.
- Architect: traces findings into process, function, information, data, application, and system implications.
- Governance Reviewer: checks authority, consent, participant treatment, claims, privacy, records, and acceptance.
- Security Auditor: examines disclosure, identity, access, abuse, and secure recovery.
- Critic: challenges sample bias, leading instruments, false consensus, omitted populations, and weak causation.
- Planner: sequences recruitment, sessions, analysis, remediation, re-test, and decision gates.
- Coordinator: synchronizes participants, owners, specialists, logistics, and handoffs.
- Implementer: makes authorized changes arising from validated findings without changing the question silently.
- Tester: converts validated outcomes into reproducible acceptance, regression, accessibility, and recovery tests.
- Verifier: independently checks methods, records, findings, traceability, and claims.
- Documentation Writer: maintains protocols, instruments, findings, decisions, limitations, and participant-facing records.
- Memory Curator: preserves versions, evidence, consent boundaries, dissent, decisions, and continuity.
- Optimizer: improves outcome measures without gaming, excluding, or burdening participants.
- Red Team: probes manipulation, exclusion, privacy leakage, unsafe behavior, and denial or recovery failure.
- Decision Arbiter Advisory: compares alternatives under authorized user-outcome criteria without deciding for the owner.

Role selection remains proportional to consequence. Existing role contracts remain controlling and are not redefined here.

REQUIRED ACTIVITIES

1. Define the outcome, validation question, decision owner, population, context, version, scope, non-goals, and consequence boundary.
2. Identify users, affected non-users, operators, support staff, maintainers, vulnerable groups, and stakeholders who may experience different outcomes.
3. Translate requirements and intended outcomes into tasks, scenarios, observable behaviors, reported experiences, and acceptance criteria.
4. Select a method proportionate to the question, such as moderated study, field observation, accessibility evaluation, diary, pilot, survey, benchmark, support analysis, or operational evidence review.
5. Define recruitment, sampling, inclusion, exclusion, compensation, consent or authorization, safeguarding, privacy, recording, and withdrawal procedures.
6. Prepare representative tasks, content, accounts, data, devices, assistive technology, language, environment, and support conditions.
7. Conduct sessions or measurement without leading participants or hiding relevant failures, alternatives, burden, or uncertainty.
8. Record direct observations, task results, participant reports, context, workarounds, errors, emotional or safety signals, dissent, and non-completion.
9. Analyze results by population, context, device, language, ability, role, environment, and other material factors rather than averaging away differences.
10. Evaluate comprehension, completion, outcome quality, accessibility, burden, trust, privacy, safety, security, control, correction, appeal, and recovery.
11. Compare results with declared criteria and identify confounders, missing evidence, alternative explanations, and disconfirming cases.

12. Obtain specialist review for accessibility, privacy, security, safety, legal, research ethics, domain, and affected-party concerns as applicable.
13. Translate findings into requirements, design, process, architecture, implementation, support, training, policy, or research changes with traceability.
14. Re-test changed behavior and record the owner decision, residual risk, validity limit, handoff, and next safe action.

HUMAN-AI INTERACTION REQUIREMENTS

- Show whether each result is direct observation, participant report, instrument output, automated measure, AI interpretation, or owner decision.
- Let the owner and qualified reviewers inspect and correct population, task, context, coding, interpretation, acceptance criteria, and exclusions without losing history or dissent.
- Require human review before recording quotes, publishing findings, accepting risk, changing rights or access, or making commitments to users.
- Make participant burden, accessibility conditions, non-completion, harm signals, missing groups, uncertainty, and contradictory evidence visible.
- Preserve participant stop, withdrawal, complaint, correction, explanation, appeal, support, and recovery paths during AI-assisted sessions.
- Do not use an AI moderator, synthesizer, classifier, or recommender to bypass consent, safeguarding, privacy, authority, audit, or human review.

ARCHITECTURE AND SYSTEM IMPLICATIONS

- Represent validation question, outcome, population, context, version, method, session or measurement, evidence, finding, decision, and lifecycle as attributable records.
- Separate participant identity and protected data from analysis identifiers, findings, published summaries, and decision records.
- Preserve traceability from user need and requirement through task, observation, finding, remediation, re-test, acceptance, and operational monitoring.
- Make population, context, language, ability, device, environment, support, and version dimensions queryable so material differences are not erased.
- Support consent, withdrawal, redaction, access restriction, retention, deletion, correction, appeal, and secure export of validation records.
- Ensure instruments, prototypes, builds, AI providers, prompts, content, data, and configurations are versioned and reproducible.
- Instrument product behavior sufficiently to compare validation findings with later operational evidence without treating the comparison as automatic proof.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

- Show validation question, intended outcome, population, context, version, owner, authority, method, status, evidence freshness, and validity period.
- Show who participated, which groups were missing or underrepresented, what tasks were completed or failed, and where conditions differed.
- Let owners inspect raw or protected evidence according to permission, coded findings, dissent, limitations, risks, decisions, and open questions.
- Provide authorized pause, stop, withdraw, redact, restrict, correct, reopen, approve, reject, and handoff controls.
- Explain what was observed, inferred, measured, reported, unresolved, and accepted, including the evidence that would change the conclusion.
- Provide participants and users understandable status, complaint, correction, privacy, accessibility, appeal, and recovery controls appropriate to the activity.

REQUIRED OUTPUTS AND ARTIFACTS

- User Validation charter with question, intended outcome, owner, population, context, version, scope, authority, and decision boundary.

- Participant, recruitment, consent or authorization, accessibility, safeguarding, privacy, recording, compensation, and disposal plan.
- Protocol, tasks, scenarios, instruments, stimuli, environment, data, account, moderator, observer, and escalation records.
- Session, observation, measurement, participant feedback, non-completion, failure, harm, dissent, and limitation records.
- Analysis by material population, context, language, ability, device, role, environment, and version dimensions.
- Traceability from user needs and requirements to tasks, findings, changes, re-tests, acceptance, and operational monitoring.
- Remediation, decision, residual-risk, unresolved-question, and reopening records.
- Participant-facing summary or notification where required, plus secure continuity, retention, and deletion package.

EVIDENCE REQUIREMENTS

- Each finding identifies question, population, context, version, method, task, date, source, observer or instrument, result, uncertainty, and limitation.
- A quote, survey response, task success, completion time, benchmark, support ticket, rating, or session proves only its bounded claim.
- Representativeness claims identify recruitment, sample, coverage, exclusions, non-response, accessibility conditions, and limits.
- Accessibility evidence identifies users or evaluators, assistive technology, environment, task, standard or criterion, result, and limitation.
- AI-assisted analysis identifies model, provider, version, prompt or method, input boundary, reviewer, transformation, result, and limitation.
- Negative, contradictory, partial, missing, stale, withdrawn, inaccessible, or harmed-participant evidence remains visible and constrains acceptance.

VERIFICATION AND ACCEPTANCE TESTS

- Reproduce the validation from the exact protocol, version, tasks, instruments, population criteria, environment, and evidence package.
- Verify the stated outcome is measured directly or through a justified proxy and that proxy limits are explicit.
- Test normal, boundary, denied, error, degraded, accessibility, multilingual, privacy, safety, correction, appeal, and recovery journeys as applicable.
- Confirm that participants can understand the task, decline or stop, request help, report harm, correct a record, and recover from failure.
- Compare outcomes across relevant populations and contexts; investigate material disparities rather than hiding them in aggregate results.
- Introduce misleading content, missing information, stale state, unauthorized access, model failure, provider failure, and inaccessible controls and verify safe handling.
- Re-test every material remediation and verify the new evidence is tied to the changed version and same decision boundary.
- Independently review whether the owner decision exceeds what the evidence supports.

FAILURE STATES AND ANTI-PATTERNS

- Treating satisfaction, preference, a small positive sample, task completion, engagement, or internal approval as universal validation.
- Recruiting only convenient or enthusiastic participants and describing the result as representative.
- Leading participants, suppressing dissent, discarding non-completion, or averaging away material disparities.
- Testing a prototype or staged flow while claiming the production experience, support burden, accessibility, privacy, or recovery is validated.
- Fabricating quotes, users, consent, metrics, accessibility outcomes, or causal conclusions with AI or any other tool.
- Exposing participant data, recording without authority, ignoring withdrawal, or making complaint and correction impossible.
- Shipping a validated interaction without tracing required changes to architecture, implementation, operations, support, training, and monitoring.

RECOVERY AND RECONSIDERATION

- On participant harm, privacy or security exposure, unsafe behavior, accessibility exclusion, invalid method, sample failure, evidence loss, or scope drift, stop the activity and protect affected people.
- Identify the blocker, affected groups, containment owner, authority, communication, evidence preservation, and safe work that may continue.
- Withdraw or correct misleading findings, notify affected participants when required, restrict records, revoke access, and obtain specialist review.
- Reframe recruitment, tasks, instruments, version, environment, or analysis before repeating the activity.
- Reopen validation when the population, context, requirement, design, policy, model, data, environment, risk, or lifecycle stage changes materially.

COMPLETION CRITERIA

- Question, outcome, population, context, version, method, owner, authority, safeguards, criteria, and evidence limits are explicit.
- Representative tasks and material user, affected-party, accessibility, safety, privacy, security, control, and recovery conditions are addressed or justified.
- Findings include positive, negative, contradictory, missing, and unresolved evidence, with material differences preserved.
- Remediation, re-test, owner decision, residual risk, participant obligations, retention, deletion, and handoff are complete or explicitly blocked.
- The section is marked CONTENT-COMPLETE while source reconciliation, hostile review, and owner acceptance remain separate activities.

HANDOFF AND CONTINUITY REQUIREMENTS

- Preserve validation identity, question, outcome, population, context, version, protocol, participant protections, evidence, findings, dissent, decisions, limitations, and open gaps.
- Hand off evidence-supported changes to Requirements, User Research, Prototyping, UX, Interaction, Accessibility, Content, Process, Functional, Information, Data, Application, Engineering, Testing, Operations, Security, Privacy, and Governance work.
- Record which findings are local, which require architectural change, which require policy or training change, and which require future revalidation.
- State the next safe action, blocked action, authority, evidence request, participant or records obligation, validity period, and exact continuation point.
- Do not claim universal user acceptance, adoption, safety, compliance, or operational effectiveness from this section alone.

RELATED CODEX SECTIONS

- 014 Stakeholder Management; 015 User Research; 016 Domain Analysis; 017 Feasibility Analysis; 018 Risk Management; 019 Requirements Engineering; 020 Systems Engineering.
- 021-106 Architecture, User Experience, Design, Prototyping, and User Validation sections, especially Process, Functional, Information, Data, Application, HCI, Interaction, UX, Accessibility, Content, and Design Systems.
- 231-257 Testing through Validation; 247 Accessibility Testing; 249 Usability Testing; 251 Compliance Testing; 252 Safety Testing; 253 Model Testing.
- 331 Operational Readiness; 334 Customer Support; 369-372 Experimentation through Feedback Management; 402-406 Post-Implementation Review through System Retirement.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

- This draft uses the Canonical Section Contract, shared governance files, Sections 001-105, Owner Cockpit Requirements, Status and Evidence Lexicon, and the Remaining-Section Integration Roadmap.
- No universal participant recruitment standard, sample-size threshold, research ethics regime, accessibility test method, outcome metric, or acceptance threshold applies to every future context.

- Actual populations, affected non-users, consent and safeguarding requirements, language and accessibility needs, jurisdictional duties, evidence thresholds, and owner decisions remain unresolved until supplied and verified.